The Micro and Macro Implications of Multidimensional Skill Uncertainty*

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October 21, 2026

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Abstract

What are the consequences of multidimensional skill uncertainty for workers' wages and aggregate output? I develop and estimate a general equilibrium dynamic Roy model in which workers have imperfect information about their multidimensional skills and accumulate task-specific human capital. Estimated on Portuguese administrative data, the model rationalizes key patterns of occupational mobility, with learning about comparative advantage playing a central role in occupational reallocation among young and poorly matched workers. Removing information frictions raises aggregate output by 5.1%, primarily through better skill allocation across occupations, while generating the largest wage gains early in workers' careers and among the most mismatched. A feasible information treatment about comparative advantage across tasks recovers about 19% of this output loss. Finally, job transformation induced by Large Language Model adoption raises output while widening the output gap between the imperfect- and perfect-information economies along the transition path and in the new long run.

Keywords: Multidimensional Skills, Mismatch, Information Frictions, Learning, Occupational Mobility, Human Capital, Misallocation, Wage Inequality, Dynamic Roy Model, Task-Biased Technological Change

*I am indebted to Basile Grassi and Thomas Le Barbanchon for constant guidance and invaluable support, and to Jerome Adda and Luigi Bocola for many insightful conversations. I also thank Francesco Armillei, Adrien Bilal, Alexander Boca, Vasco Carvalho, Geraud Desazars, Rafael Dix-Carneiro, Jan Eeckhout, Pamela Giustinelli, Cristina Gualdani, John Grigsby, Simon Jäger, Francis Kramarz, Danial Lashkari, Arnaud Maurel, Romuald Meango, Andreas I. Mueller, Theodore Papageorgiou, Elena Pastorino, Nicola Pavoni, Richard Rogerson, Aysegül Sahin, Karthik Sastry, Isaac Sorkin, Wilbur Townsend and Gianluca Violante, as well as seminar participants at Bocconi, Princeton Junior Macroeconomics Workshop, Stanford Macro Lunch, Stanford Labor/Public Economics Workshop, 1st Aarhus Workshop on Labor Markets, EWMES 2025, CEPR Paris Symposium 2025, BeNA Labor Economics Workshop 2025, 2nd Boston College PhD Conference, Theories and Methods in Macro 2026, NASMES 2026, XXIX Workshop on Dynamic Macroeconomics, CEF 2026, the NBER Summer Institute 2026 (The Micro and Macro Perspectives of the Aggregate Labor Market), EEA/ESEM 2026, EALE 2026 and the TRAILS Conference on Occupational Mismatch and Skill Shortage for helpful comments. All errors are my own.

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1 Introduction

Workers do not possess a single one-dimensional skill but instead hold a portfolio of skills whose payoffs depend on the job that employs them. These latent multidimensional skills help explain a wide range of labor-market outcomes, from earnings trajectories and occupational choice (Heckman et al. 2006; Adda and Dustmann 2023) to who wins and who loses from occupational reallocation when task-biased technological shocks hit the economy (Freund and Mann 2025). Especially early in their careers, however, workers may know their skills only imperfectly and therefore direct their search toward jobs on the basis of perceived comparative advantage. Sorting on beliefs generates mismatch: workers end up in jobs that do not reflect their true comparative advantage. A large literature proposes a process of experimentation and learning-by-doing, in which workers gradually discover their abilities in different jobs while accumulating imperfectly transferable experience (Miller 1984; Antonovics and Golan 2012; Papageorgiou 2014; Pastorino 2024). Mismatch along different skill dimensions predicts corrective occupational moves with implications for wages and for human capital (Guvenen et al. 2020), which accumulates along these dimensions according to the task content of jobs (Gathmann and Schönberg 2010). Learning, imperfectly transferable human capital, and other non-wage-related factors interact to create strong path dependence in occupational choice that prevents a substantial fraction of workers from ever sorting into their ideal career path. Quantifying the importance of these components in driving occupational choice and mobility is crucial for understanding the consequences of early-career mismatch generated by information frictions about skills and for designing effective policies to mitigate it.

In this paper, I develop a general equilibrium dynamic Roy model with multidimensional skill learning to assess and quantify the consequences of imperfect information about skills for individual labor-market outcomes and aggregate output. In the model, workers begin their careers with incomplete information about their multidimensional skills and gradually refine their beliefs through noisy wage signals. At the same time, they accumulate task-specific experience depending on which tasks they perform on the job. The productivity of skills and experience in performing different tasks differs across occupations. Mismatch therefore arises endogenously. Two forces drive occupational sorting: learning about unobserved multidimensional skills and the accumulation of task-specific human capital. These forces, together with utility switching costs and compensating differentials, shape both the timing and the direction of occupational transitions over the life cycle.

I estimate the model on Portuguese administrative data using a two-stage estimation procedure that allows me to control for the dynamic selection induced by belief sorting. I show that the estimated model replicates important empirical patterns of occupational mobility, including the facts first documented by Groes et al. (2015) using Danish administrative data. As in Denmark, occupational mobility in Portugal is U-shaped and directional: within an occupation, low- and high-wage workers are more

likely to switch occupations in the next period; high-wage workers move to occupations that pay more on average, while low-wage workers move down to occupations that pay less. I add a new dimension to these facts: the change in task content when workers switch occupations. I find that, conditional on switching occupations, the change in task content is also U-shaped: relative to the origin occupation, both low- and high-wage workers within an occupation make larger moves in task content than the median worker in the same occupation. On average, those at the top move up the occupational ladder to more cognitive occupations, while those at the bottom move to more manual occupations, which are on lower rungs of the ladder. I use the model to rationalize these patterns and decompose the drivers of these different occupational moves. I find that while experience accumulation mainly induces upward mobility for high-wage workers within an occupation, learning about comparative advantage is an important driver of downward mobility of poorly matched workers: low-wage workers move because they learn that they are mismatched in their current occupation and that their set of skills is more productive elsewhere, while high-wage workers move because the accumulated experience in certain tasks is more productive in other occupations. Occupation switches driven by learning are therefore wage-improving, as workers move to more productive matches.

If workers knew their skills perfectly, they could sort into their most productive occupation right away and earn higher wages from the start of their careers. I use the estimated model to quantify the aggregate and individual costs of information frictions about skills. I simulate a counterfactual stationary equilibrium in which workers know their true skills from the beginning of their careers. I find that aggregate output in this perfect-information economy would be 5.1% higher than in the baseline economy with information frictions. Individual wage gains are largest at labor-market entry, where they are on average 10.5%, and fall to roughly zero after thirty years as the gap between the average wage level in the two economies closes. Almost all of the early aggregate gain comes from better sorting on skills; some workers also gain because sorting correctly from the start lets them accumulate more human capital along the right skill dimensions. These wage gains are concentrated at the bottom of the match-quality distribution within occupations, because these workers are the most mismatched and are those who, in the baseline economy, switch occupations because of learning. These mismatched workers are more likely to be specialists, that is, workers with a strong comparative advantage in one skill dimension, and therefore gain the most from sorting correctly, while generalists lose little from skill uncertainty.

Recovering the full output and wage losses may be infeasible in practice, but I can use the estimated model to assess how well existing career guidance policies reduce misallocation. I study a realistic information intervention guided by experimental evidence. This policy is an information treatment about comparative advantage in skills, which has been studied and implemented by Kiss et al. (2023). The treatment assesses young jobseekers' skills with a computerized assessment and provides them with information about their performance in the tests relative to other candidates taking the

same assessment. In my model, I simulate this intervention as a noisy signal about skills and calibrate the variance of the signal to match the average treatment effect in Kiss et al. (2023), so that the share of workers whose beliefs about their comparative advantage match their true comparative advantage rises by 13.7 percentage points. Delivered at labor-market entry, the intervention raises aggregate output by 0.97% and increases average wages by about 2.9% at entry, with the gains concentrated in the first decade and among the workers who are more mismatched in the baseline economy. The output and wage gains are highest when the assessment is delivered before labor-market entry, when workers have not yet accumulated task-specific experience: the same assessment delivered two years after entering the labor market produces gains that are only a little over half of those obtained when the signal is provided before workers start their careers. The effectiveness of career guidance declines quickly over the career.

The model also allows me to study how information frictions interact with standard frictions considered in the job-mobility literature. I represent these frictions as a fixed utility cost paid by workers when switching occupations. This reduced-form cost may capture search costs or broader mobility frictions that are not explicitly modeled. For example, when employment opportunities in an occupation are geographically concentrated, switching occupations may also require costly geographic relocation. Moreover, when switching occupation, workers may have to spend time unemployed while reallocating across labor markets, as in the island model of Lucas and Prescott (1974). I show that reducing this switching cost always reduces the misallocation induced by information frictions, echoing results from Papageorgiou (2014). However, decreasing this utility cost has non-monotonic effects on the level of aggregate output: small reductions in the switching cost increase the output level, but very large reductions reduce it. This non-monotonic effect is driven by two countervailing forces: on the one hand, small reductions in the switching cost allow workers to reduce their mismatch once they learn in which tasks they are most productive; on the other hand, large reductions create excessive churning that destroys accumulated experience in tasks, lowering workers' productivity. The excessive churning is driven by idiosyncratic taste shocks that make workers jump from one job to another. An economy in which a very experienced engineer decides to become a bartender because of her taste will have relatively lower aggregate output.

Finally, I study how artificial intelligence affects the misallocation induced by skill uncertainty. I focus on the consequences of Large Language Model (LLM) adoption, which changes the importance of specific tasks within occupations. The modeling of this task-biased technology shock is similar in spirit to the one recently studied by Freund and Mann (2025) and Althoff and Reichardt (2026). Following the task-exposure framework of Eloundou et al. (2024), LLMs increase the importance of manual tasks while reducing that of cognitive tasks. In the model, this implies that time spent on cognitive tasks decreases in proportion to each occupation's exposure to LLMs, while time spent on manual tasks increases. This implies that cognitive experience is accu-

mulated at a lower rate, and the returns to cognitive skills and experience decrease. Conversely, more time spent on manual tasks implies that experience along the manual dimension is accumulated faster, and the returns to manual skills and experience increase. The shock is modeled as an unanticipated MIT shock, and I solve for the transition path assuming that workers have perfect foresight over prices but still have imperfect information about their skills. I find that LLM adoption raises aggregate output by 2.1%. On impact, this task-biased reallocation toward manual tasks increases output by 1.0%. Gradually, as new cohorts enter the economy, output increases toward its long-run stationary equilibrium level. Output increases because new cohorts can immediately sort differently, based on the new task importance in the economy, compared to experienced workers who were already locked into certain career paths when the task-based reallocation shock hit the economy. I also show that in this long-run equilibrium, sorting on skills becomes more important, and so does learning, as cognitive experience, which was particularly important in driving wage growth, is accumulated more slowly and pays less. The wage gains from removing information frictions in the post-AI economy, compared to the pre-AI economy, are highest for high-ability workers with a manual comparative advantage. These workers are mainly highly educated individuals who enter the labor market believing that they have a comparative advantage in cognitive tasks and who benefit from reallocation toward more mixed occupations.

Related Literature. This paper contributes to four strands of the literature. The first strand studies the sources of wage growth and the misallocation of human capital (Hsieh et al. 2019). A large empirical literature documents the importance of job and occupational mobility for wage growth (Topel and Ward 1992; Neal 1999; Kambourov and Manovskii 2009; Pavan 2011; Adda and Dustmann 2023). I focus on learning about multidimensional skills and task-specific human capital accumulation as the two forces that drive reallocation to more productive matches and the wage dynamics it generates. Two recent papers document the role of information frictions in generating occupational mismatches. Goller et al. (2026) link standardized aptitude tests to administrative records in Switzerland and document that imperfect self-knowledge at labor-market entry predicts misaligned occupational aspirations, higher realized skill mismatch, and a higher probability of dropout. Kiss et al. (2023) show, in two field experiments with young jobseekers, that giving workers their assessment results redirects search toward jobs that reward the skills they score well on and substantially raises earnings, consistent with mismatch driven by imperfect information. I simulate the same intervention in my estimated model to trace its general-equilibrium and life-cycle consequences.

Closest to this paper is a strand that treats this misallocation as multidimensional, arising from a mismatch between a worker's several skills and the skills her occupation rewards (Guvenen et al. 2020; Lise and Postel-Vinay 2020; Baley et al. 2022). Guvenen et al. (2020) derive a multidimensional mismatch measure from a model of

Bayesian learning about workers' abilities, but do not estimate such a model. Lise and Postel-Vinay (2020) estimate a structural on-the-job search model with multidimensional skills, but workers know their skills, so mismatch is mainly driven by search frictions. Baley et al. (2022) let workers learn about unobserved skills, but each job uses a single skill dimension and their focus is on business-cycle dynamics rather than life-cycle sorting. In my model, learning is à la Jovanovic (1979) and Miller (1984): workers forecast a vector of time-invariant task-specific skills, and a worker is "mismatched" when, given current beliefs, she chooses an occupation that is not the one in which she is truly most productive. The approach does not require a measure of the supply of skills, such as test scores; I instead recover the distribution of unobserved skills directly from wages, using the approach of Bunting et al. (2024). Estimating the model allows me to quantify the micro and macro implications of early-career skill mismatch generated by imperfect information and to understand how much policies can recover.

The second studies learning, job assignment, and wage growth. In the terminology of Pastorino (2024), learning affects wages both directly—pay responds to beliefs about ability because it is an equilibrium outcome of what employers and workers know[1]—and indirectly, as workers and firms learn about comparative advantage and reallocate workers to jobs where their skills are used more productively. Pastorino (2024) finds that this indirect, assignment channel can be the dominant force shaping wage growth and dispersion: the direct belief-to-wage effects are relatively small, while learning substantially improves the sorting of workers to jobs, which then drives wage dynamics[2]. Motivated by this, and by the occupational mobility facts in Groes et al. (2015), I focus on the assignment channel and study how learning about multidimensional unobserved skills shapes occupational mobility and wages over the life cycle. A related structural literature studies occupational sorting when workers learn about occupation-specific abilities (James 2012; Arcidiacono et al. 2025; Papageorgiou 2014; de Melo and Papageorgiou 2025); Papageorgiou (2014) in particular shows how search frictions amplify the mismatch that learning generates. In all of these, ability is a single latent factor or one scalar per occupation, so the models are silent about which skill dimension underlies mismatch. Antonovics and Golan (2012) show that occupational choice and wage growth in the NLSY are consistent with workers learning about their abilities on the job; their framework, however, abstracts from on-the-job human capital accumulation and non-wage amenities, both of which are central here.

[1]Much of the employer-learning literature emphasizes this direct channel, among others Harris and Holmström (1982), Farber and Gibbons (1996), Altonji and Pierret (2001), Lange (2007), and Gibbons and Katz (1991). A closely related strand develops frameworks that incorporate both wage-setting effects and assignment dynamics, such as theories that integrate wage and promotion dynamics inside firms in Gibbons and Waldman (1999*b*) and subsequent extensions (e.g. Gibbons and Waldman 2006, and the synthesis in Gibbons and Waldman 1999*a*), as well as sectoral and occupational sorting frameworks in which learning affects wages both directly and through mobility (e.g. Gibbons et al. 2005).

[2]More recently, de Paula et al. (2025) build on Pastorino (2024) with a model of heterogeneous workers and firms, monopsony power, and symmetric learning, and find that standard static methods underestimate the importance of sorting because they ignore the dynamic effects of learning and human capital accumulation on future matching.

This paper departs from this literature by embedding learning in a task-based model of occupational choice. I show empirically that mobility across tasks is informative about different types of occupational switches, and that these switches have distinct implications for wages and human capital accumulation. The model also enriches occupational choice by allowing for compensating differentials, a utility switching cost, and preference heterogeneity, which improves its quantitative fit and helps isolate the separate roles of learning and human capital accumulation in generating job mobility.

Third, the paper relates to work on task-specific human capital (Gibbons and Waldman 2004; Poletaev and Robinson 2008; Gathmann and Schönberg 2010). The model in this paper is similar in spirit to the task-based frameworks developed by Gathmann and Schönberg (2010) to study the transferability of human capital across occupations, and Autor and Handel (2013), who propose a Roy (1951) model where workers sort based on comparative advantage across tasks. In my framework, as in these task-based approaches, tasks are bundled within occupations and workers' skills map into output through occupation-specific task loadings. However, unlike these papers, I focus on how workers' occupational allocation is shaped by imperfect information about multidimensional skills. The task-based approach is also the workhorse for studying technological change: routine-biased technical change and job polarization (Autor et al. 2003, 2006; Autor and Dorn 2013; Acemoglu and Autor 2011). More recently, Freund and Mann (2025) develop a general-equilibrium task-based model to study the effects of job transformation induced by Large Language Models (Eloundou et al. 2024), in which automating a task raises the importance of those that remain, and estimate the resulting individual wage effects from a distribution of task-specific skills. In their model, however, as in related general-equilibrium dynamic Roy models of technological change (Dvorkin and Monge-Naranjo 2019; Althoff and Reichardt 2026), workers know their skills exactly. In this paper, I study how job transformation affects mismatch driven by information frictions about skills, and how skill uncertainty affects occupational reallocation in response to this technological shock.

Finally, I contribute to the estimation of Roy-type models (Roy 1951) to study the effects of macroeconomic shocks when there are labor mobility frictions. Building on the identification results of Heckman and Honore (1990), dynamic Roy frameworks have been applied to sectoral switching costs (Lee and Wolpin 2006), labor-market responses to trade liberalization (Dix-Carneiro 2014; Traiberman 2019), and learning about occupation-specific ability (James 2011; Hincapié 2020; Arcidiacono et al. 2025). I contribute by introducing a two-stage estimation strategy that accommodates multidimensional unobserved heterogeneity, building on the nonparametric identification results of Bunting et al. (2024) for what they call the pure learning model[3].

[3] As I describe later in Section 4, identification in this class of models requires fewer restrictions than a standard Roy model. Moreover, parts of the functional form can accommodate heterogeneity observed by workers but unobserved by the econometrician; while this complicates identification and estimation in general, the estimation routine can incorporate such heterogeneity by extending the expectation-maximization (EM) algorithm for finite mixtures. To retain tractability for the dynamic programming problem and to ease interpretation, I focus on the simpler pure learning model.

The rest of the paper is organized as follows. Section 2 describes the data and documents key facts on occupational mobility. Section 3 presents the model, and Section 4 discusses identification and estimation. In Section 5 I use the estimated model to decompose the drivers of occupational mobility. Section 6 quantifies the costs of skill uncertainty and evaluates information and mobility policies, and Section 7 studies the AI shock. Section 8 concludes.

2 Data and Empirical Facts

2.1 Data

The main data source of this paper is *Quadros de Pessoal*, a matched employer–employee dataset created by the Portuguese Ministry of Labor in the 1980s. It includes Portuguese firms with at least one employee, excluding self-employed workers. My sample covers 1995–2019. The main wage measure is a "straight-time real hourly wage," constructed following Card et al. (2016)[4]. Nominal wages are deflated to 1995 euros using the Portuguese CPI.

I use two related samples throughout the paper, each serving a different purpose. Appendix A details how I construct the two samples and compares them. The large sample is designed to maximize the number of observations and representativeness for descriptive facts and baseline comparisons. It applies standard cleaning criteria to the raw data and is used to document the overarching empirical patterns of occupational mobility at the 3-digit occupation level. The estimation sample is a cohort panel of labor-market entrants, constructed by applying further restrictions to the large sample. It tracks the complete early-career trajectories of young workers entering the labor market, which I need to identify and estimate the learning and human-capital parameters of the structural model.

To measure the task content of occupations, I use two classifications. The first focuses on broad task categories, cognitive and manual skills[5]; the second classification is ESCO, the European Skills, Competences, Qualifications and Occupations dataset, which links occupations to specific skills across European labor markets. In both classifications, each occupation j receives weights ω_j^k over tasks/skills k that sum to one, $\sum_k \omega_j^k = 1$. These measures let me compare workers within occupations and track how task content changes when workers switch occupations. Throughout the paper, I interpret skills and tasks as two sides of the same object: workers supply a set of skills, each of which maps into a specific task, so that skills refer to the supply side of the labor market and tasks to its demand side.

[4]This measure is computed as the sum of the worker's base wage and regular salary supplements (e.g., monetary meal allowances) and is then divided by the worker's normal hours, defined as the contractual monthly hours fixed by the individual contract. The measure is defined as "straight-time" in the sense that both wage and hours exclude overtime.

[5]Detailed construction of these categories and the classification mapping over time is provided in Appendix A.

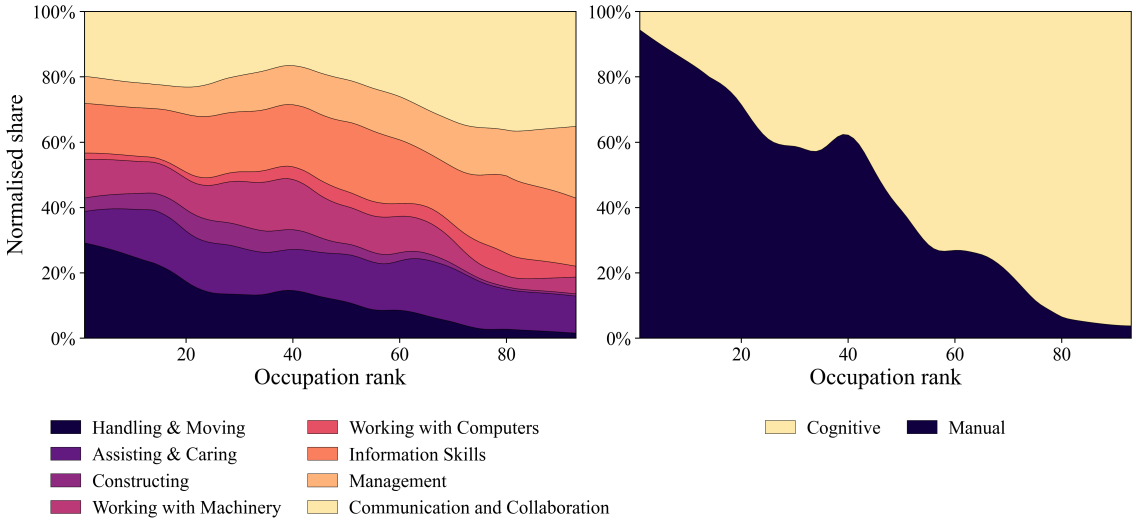


Figure 1: Task composition by occupation average wage rank. Occupations are ranked by mean hourly wage within each year (horizontal axis: occupation rank, from the lowest-paying occupation on the left to the highest-paying on the right; vertical axis: skill share in percent). *Left panel*: Each occupation's ESCO skill vector is normalized to shares that sum to 100%. *Right panel*: Cognitive and Manual task shares based on the cognitive/manual task classification. Both panels show smoothed average skill shares at each wage percentile. The figure is produced using the large sample data.

Figure 1 shows how task content varies across the occupational wage distribution under two complementary classifications. The left panel uses the eight ESCO skills, while the right panel employs the cognitive/manual task classification.

Two facts stand out. First, although a few occupations require a single skill, at every point of the wage distribution the average occupation bundles several skills. Second, there is a clear shift toward more cognitive skills as wages rise. In the left panel, the shares of lighter-colored layers—communication & collaboration, management skills, information skills, and working with computers—increase with occupational wage rank. In contrast, the darker layers associated with more manual content—handling and moving, working with machinery, and constructing—decline with occupational wage rank, while assisting and caring remains roughly constant. The right panel confirms this pattern more starkly: cognitive tasks rise with occupational wage rank while manual tasks decline.

Taken together, the figure points to a vertical job ladder in tasks: higher-paying occupations tilt increasingly toward cognitive tasks, and lower-paying occupations toward manual tasks. As I show below, workers move both up and down this ladder, and the direction of the move has different implications for their wages.

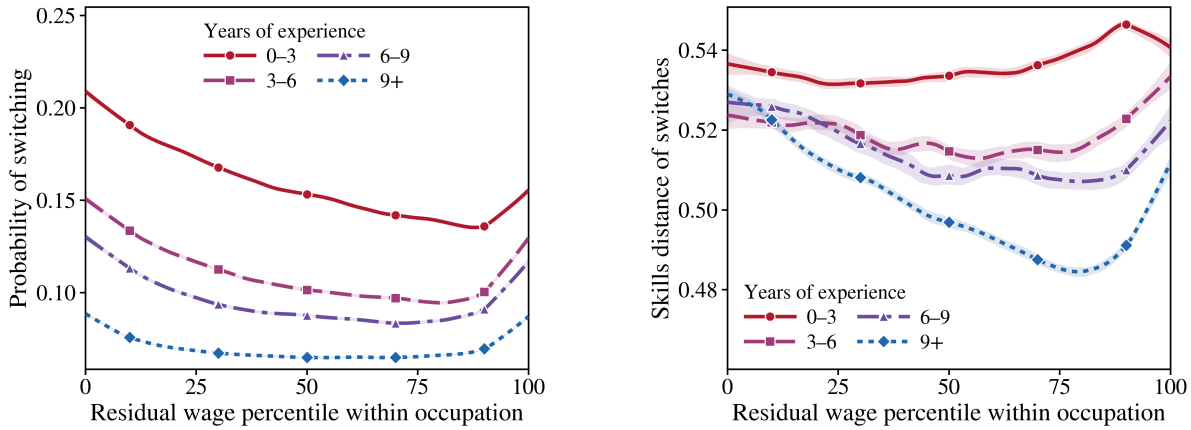
2.2 Occupational Mobility, Directionality in Tasks and Wages

I now document three facts on occupational mobility. They extend the evidence of Groes et al. (2015) for Denmark in two directions: I study the task content of occupational moves as a function of the worker's position within the origin occupation, and I document the wage changes that accompany these moves.

Fact 1: Mobility and the change in task content are both U-shaped. Because the administrative data cover nearly all workers in each occupation, I can characterize how mobility varies across the within-occupation wage distribution. As Groes et al. (2015) show, this coverage is essential to uncover the heterogeneous nature of occupational transitions. To isolate the component of within-occupation wage variation driven by unobserved fit, I residualize wages from observable characteristics. I estimate:

$$w_{it} = p_j + \iota_t + f_1(\text{tenure}_{it}) + f_2(\text{experience}_{it}) + \gamma_F \text{Female}_i + \gamma_e \text{Educ}_i + m_{ij} + v_{it} \quad (1)$$

where w_{it} is the log hourly wage, p_j are occupation fixed effects (the log occupation prices), ι_t are time fixed effects, $f_1(\cdot)$ and $f_2(\cdot)$ are cubic polynomials in occupational tenure and total labor-market experience, m_{ij} is the unobserved worker-occupation match quality, and v_{it} is mean-zero noise. The estimated residual is $\widehat{r}_{it} = \widehat{m}_{ij} + \widehat{v}_{it}$. Since v_{it} is mean-zero, the percentile rank of $\widehat{r}_{it}$ within occupation j is positively correlated with m_{ij}: on average, workers with higher match quality will rank higher in the residual distribution. Therefore the "within-occupation residual percentile" used throughout this section is a noisy but informative proxy for the worker's position in the match quality distribution within occupation. Appendix A shows that all patterns documented here are robust to using raw hourly wages instead of residualized wages.



(a) Probability of Switching by Experience

(b) Skills Distance for Occupation Switchers

Figure 2: *Panel (a)*: Probability of switching occupations between t and $t+1$ (vertical axis) as a function of residualized within-occupation wage percentile (horizontal axis), for different experience groups. *Panel (b)*: Skills distance between origin and destination occupations conditional on switching (vertical axis) by residualized within-occupation wage percentile (horizontal axis), computed using ESCO skill measures.

Figure 2(a)[6] shows that workers at both ends of the within-occupation wage distribution switch at higher rates, resembling the "U-shape" documented by Groes et al. (2015) for Denmark, and that mobility falls sharply with experience, from roughly 15%

[6]All figures for Facts 1 and 2 use non-parametric LOWESS regression with bandwidth 0.2. 95% confidence intervals are computed using 100 bootstrap iterations. In many figures the confidence bands are barely visible due to the precision of the estimates.

for workers with 0–3 years of experience to below 10% after 6 years.

Occupational transitions are not random. Following Gathmann and Schönberg (2010), I define skills distance as

$$d(j, j') = 1 - \text{Sim}(j, j'), \qquad \text{Sim}(j, j') = \frac{\boldsymbol{\omega}_j^\top \boldsymbol{\omega}_{j'}}{\|\boldsymbol{\omega}_j\|_2 \, \|\boldsymbol{\omega}_{j'}\|_2}, \tag{2}$$

where $\boldsymbol{\omega}_j$ collects the skill weights of occupation j, and $\text{Sim}(j, j') \in [0, 1]$ equals 1 for identical and 0 for orthogonal skill mixes.[7] The mean skills distance among actual occupation switchers is 0.52, with a standard deviation of 0.22. The most common observed switch, between laborers and building frame workers, has $d = 0.21$[8]. As in Gathmann and Schönberg (2010), I find that workers switch to occupations that are substantially closer in task space than a random mobility benchmark would predict (Figure 24 in the Appendix).

Figure 2(b) shows that the task distance between origin and destination occupations is related to the worker's match quality in the origin occupation. Workers at both ends of the wage distribution within occupation make larger moves in the task space, conditional on switching occupation. Task distance also falls with experience. This pattern is hard to reconcile with undirected search: as Guvenen et al. (2020) argue, if mismatch arose from search frictions alone, the change in mismatch upon switching would be independent of mismatch in the origin occupation. The relationship between match quality and the size of the task move is instead consistent with directed sorting, where workers learn about their comparative advantage and reallocate toward better-fitting occupations.

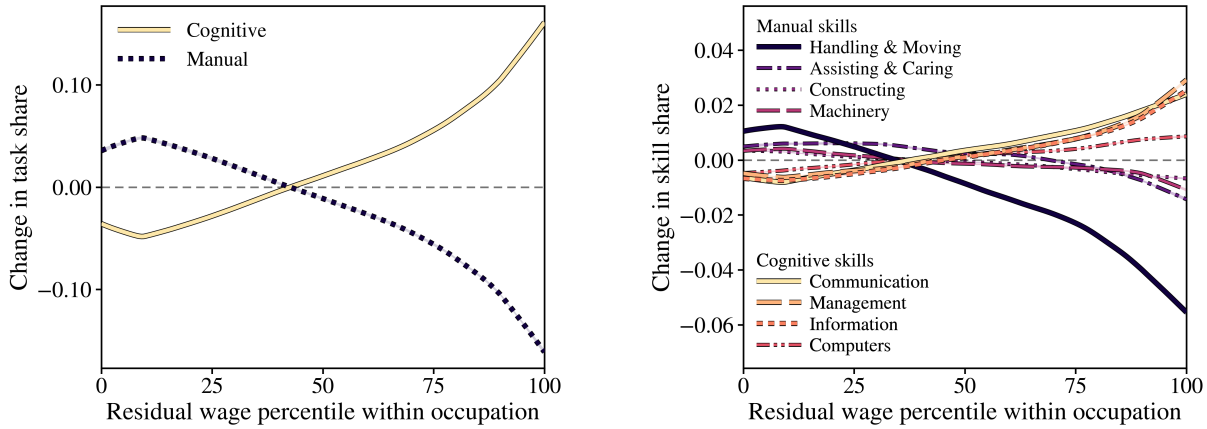
Appendix Figure 22 splits the patterns in Figure 2(a) into switches within and across firms. The probability of switching occupation within the same firm differs from the probability of switching across firms at both ends of the within-occupation wage distribution, especially at the top, where the increase in mobility comes mainly from switches within the same firm. The directionality patterns documented next, which characterize the nature of the move, are instead consistent within and across firms (Appendix Figure 23).

Fact 2: Mobility is directional in the task space. Occupational mobility in Portugal is also directional, in the terminology of Groes et al. (2015). Well-matched workers within an occupation are more likely to move to an occupation with a higher average wage compared to their current occupation, while poorly matched workers move to an

[7] Skills distance is computed using the ESCO classification at different levels of aggregation: 8 broad skill dimensions (Matrix 1.3), 74 intermediate dimensions (Matrix 2.3), and the full 296-skill classification (Matrix 3.3). The figure in the main text uses the 296-skill classification. For interpretability, when displaying changes in skill content upon switching (Figure 3), I use the 8 aggregated ESCO skill dimensions. In addition, I use the Cognitive–Manual classification of Mihaylov and Tijdens (2019), which will serve as the task space for the structural estimation.

[8] See Appendix Table 2 for detailed summary statistics on the skills distance distribution, including the most common switches by education group.

occupation with a lower average wage (Appendix Figure 21). However, when moving, poorly matched workers improve their relative position in the destination occupation, while well-matched workers worsen their relative position.



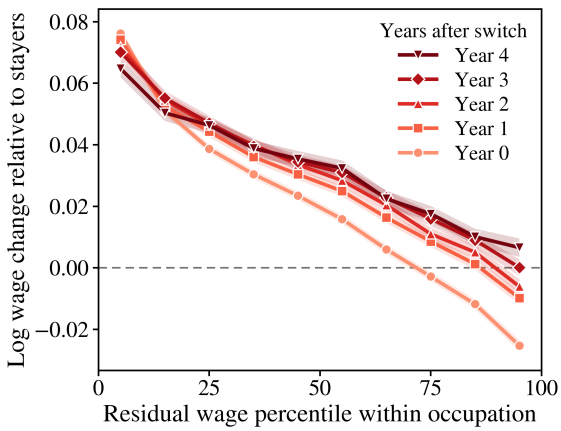
(a) Change in Cognitive and Manual Task Shares

(b) Change in ESCO Skill Shares

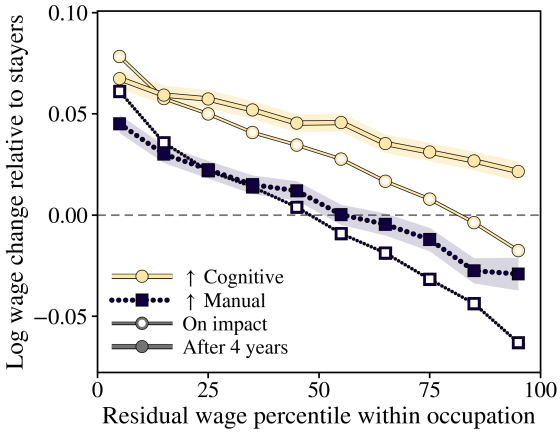
Figure 3: *Panel (a)*: Change in cognitive and manual task shares (vertical axis) by residualized within-occupation wage percentile (horizontal axis) for occupational movers. Change is $\Delta\text{Task-Share}_p^k = \text{LOWESS}_p\left(\omega_{j'(i)}^k - \omega_{j(i)}^k\right)$, the smoothed difference in task intensities between destination $j'(i)$ and origin $j(i)$ occupations. *Panel (b)*: Analogous changes based on the ESCO skill classification. By construction, given that $\sum_k \omega_j^k = 1$, the smoothed changes in shares at each percentile sum to 0.

Figure 3 examines the task content of occupational transitions. Workers at different positions in the match quality distribution move to occupations with different task content: high match quality workers move toward more cognitive occupations, while low match quality workers move toward more manual ones. The ESCO classification (Panel (b)) shows a similar pattern at a finer level: well-matched workers move to occupations requiring more communication, collaboration, creativity, and computer skills, while poorly matched workers move to occupations requiring more handling, moving, or caring skills.

Fact 3: Switching raises the wages of poorly matched workers. Figure 4 documents that workers realize wage gains upon switching occupations, relative to workers who start from the same position in the same occupation and stay (see Appendix A.1 for details). These gains vary systematically with a worker's initial position within their occupation: those in the bottom quintile gain 6 log points relative to stayers upon switching and 6 after four years, while workers in the top decile initially suffer wage cuts of 3 log points and close the gap by year four. Workers in the bottom quintile gain whatever the direction of the move in the task space, while workers at the top gain, after four years, only when they move toward more cognitive occupations. This pattern indicates that switching is especially beneficial for workers poorly matched to their initial occupation. The initial wage losses of high earners suggest either backloaded compensation or non-pecuniary benefits that influence job choice (Sorkin 2018).



(a) Wage change of switchers relative to stayers, by initial wage percentile



(b) Wage change of switchers relative to stayers, by initial wage percentile and task content of the switch

Figure 4: Wage changes upon occupational switches, relative to stayers in the same origin occupation, year, and decile of the initial wage percentile. Horizontal axis: residual wage percentile within the origin occupation in year $t-1$; vertical axis: cumulative change in the log real wage of switchers relative to stayers. *Panel (a)*: Years 0–4 after the switch, where Year 0 is the first year in the new occupation. *Panel (b)*: switches toward more cognitive occupations (light yellow, solid) and more manual occupations (dark blue, dotted), on impact (thin lines and open markers, Year 0) and four years after the switch (thick lines and filled markers). Shaded areas are 95% confidence bands clustered by worker.

2.3 Discussion of the Empirical Facts

Taken together, Facts 1 and 2 show that workers at both ends of the match-quality distribution are more likely to switch occupation than the median worker in the same origin occupation, and that they change tasks more when they switch. Facts 2 and 3 are instead informative about the nature of the switch: workers with low match quality improve their relative position in the destination occupation and gain relative to stayers, while workers with high match quality worsen their relative position and initially suffer wage cuts. Switches from the bottom and from the top of the distribution are therefore likely to be driven by different forces.

The fact that, conditional on moving, the size of the change in task content depends on origin match quality is a distinctive feature of the data and points to directed sorting rather than undirected search. The change in wages upon switching is also informative about the nature of the occupational move. The mobility of workers with low within-occupation match quality is consistent with learning about comparative advantage in tasks: these workers learn that they are mismatched in their current occupation and that they might be more productive elsewhere, so they are more likely to switch and, when they do, they change the task content of their job more. At the upper end of within-occupation match quality, switchers experience a wage cut on average, but their wages increase gradually afterwards. This suggests the presence of compensating differentials, which could take the form of backloaded wage compensation or of other occupation-specific amenities more closely linked to cognitive tasks. These findings call for a theory that accounts for the relationship between match quality, mobility, and

the task content of transitions, while allowing for more general compensating differentials in the form of dynamic returns or non-wage amenities. The model in the next section incorporates learning about multidimensional skills, task-specific human capital, and compensating differentials to quantify the consequences of this reallocation mechanism for wages over the life cycle.

3 Structural Model

The economy is populated by a continuum of workers and firms. Time is discrete and in each period t firms post piece-rate wage contracts in occupations $j \in \mathcal{J}$ and workers choose an occupation j in which to work.

3.1 Occupations, Tasks and Human Capital

Workers supply multidimensional human capital that is used to perform different tasks $k \in \mathcal{K}$, with $K \equiv |\mathcal{K}|$, within each occupation $j \in \mathcal{J}$. Worker i is endowed with a vector of task-specific skills $\boldsymbol{\theta}_i = (\theta_{ik})_{k\in\mathcal{K}}$ and accumulates task-specific experience $\boldsymbol{\tau}_{it} = (\tau_{ikt})_{k\in\mathcal{K}}$ while performing tasks on the job. The output of a specific task k in a given occupation j at time t is given by:

$$Y_{ijkt} = \ell_{jk} \exp\big(a_{jk}\,\theta_{ik}\big)\, H_{jk}(\tau_{ikt}), \tag{3}$$

where ℓ_{jk} is the share of time allocated to task k in occupation j, θ_{ik} is worker i's latent skill in task k, a_{jk} is the semi-elasticity of task-k output with respect to that skill, and task-specific experience enters productivity through a reduced-form learning-by-doing function $H_{jk}(\tau)$: a worker who spends more time on task k accumulates task-specific know-how that is productive wherever the task is used, with returns that may differ across occupations. The function is concave in logs, which imposes diminishing returns to learning-by-doing.[9] Tasks aggregate within each occupation j according to a Cobb-Douglas technology with task weights ω_{jk}:

$$Y_{ijt} = \exp(\varepsilon_{ijt}) \prod_{k\in\mathcal{K}} \left(\frac{Y_{ijkt}}{\omega_{jk}}\right)^{\omega_{jk}}, \tag{4}$$

where ω_{jk} governs how intensively occupation j uses task k, with $\sum_{k\in\mathcal{K}} \omega_{jk} = 1$, and $\exp(\varepsilon_{ijt})$ is an idiosyncratic worker-occupation-time productivity shock with $\varepsilon_{ijt} \sim F_j$ i.i.d. across workers and time. Following Althoff and Reichardt (2026), I assume that the firm chooses the static allocation of worker time across tasks optimally under the symmetric belief $(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it})$ it shares with the worker, and that this allocation is non-

[9] In the estimation, I parameterize the log returns to be a stepwise function: $h_{jk}(\tau) = \sum_{b\in\mathcal{B}} \delta_{jk}^{(b)} \mathbf{1}\{\tau \in b\}$, where $\mathcal{B}$ is a set of tenure bins. In this way, I make sure that I identify the wage progression along different task dimensions in each occupation flexibly as pinned down by the data.

contractible. Under the Cobb-Douglas aggregation, Appendix B.1 shows that the optimum is $\ell^*_{jk} = \omega_{jk}$. Task allocation is therefore independent of worker skills and experience, and depends only on the task importance ω_{jk} of the occupation. Substituting this into (4) yields

$$Y_{ijt} = \underbrace{\exp(\boldsymbol{\lambda}_j'\boldsymbol{\theta}_i)}_{\substack{\text{unobserved} \\ \text{skills bundle}}} \quad \underbrace{H_j(\boldsymbol{\tau}_{it})}_{\substack{\text{human-capital} \\ \text{bundle}}} \quad \underbrace{\exp(\varepsilon_{ijt})}_{\substack{\text{idiosyncratic} \\ \text{shock}}}, \tag{5}$$

where the effective loadings at occupation j on task k are

$$\lambda_{jk} \;=\; \omega_{jk}\, a_{jk}, \tag{6}$$

and $\boldsymbol{\omega}_j = (\omega_{jk})_{k\in\mathcal{K}}$ collects the occupation's task weights. The occupation-level human-capital bundle aggregates the task-level functions via the Cobb-Douglas weights:

$$H_j(\boldsymbol{\tau}_{it}) \;\equiv\; \prod_{k\in\mathcal{K}} H_{jk}(\tau_{ikt})^{\omega_{jk}}. \tag{7}$$

Taking logs, I define

$$h_j(\boldsymbol{\tau}_{it}) \;\equiv\; \log H_j(\boldsymbol{\tau}_{it}) \;=\; \sum_{k\in\mathcal{K}} \omega_{jk}\, h_{jk}(\tau_{ikt}), \tag{8}$$

the additive log-form of the human-capital bundle. Taking logs of (5) then yields the log-output expression

$$\log Y_{ijt} \;=\; \boldsymbol{\lambda}_j'\boldsymbol{\theta}_i \;+\; h_j(\boldsymbol{\tau}_{it}) \;+\; \varepsilon_{ijt}. \tag{9}$$

The occupation-specific parameters $\boldsymbol{\lambda}_j$, $h_{jk}(\cdot)$ and $\boldsymbol{\omega}_j$ jointly determine how worker human capital maps into occupation j output. They govern the relative importance of skills and experience across tasks and the concavity of the returns to experience, and, through the multiplicative structure, the complementarity among these components. This specification implies that worker skills are bundled in the sense that, once matched to an occupation, a worker must perform all tasks, unless $\omega_{jk} = 0$ or the loadings are zero for some task k.

Finally, task-specific experience evolves deterministically. A worker in occupation j spends a share ω_{jk} of her time on task k, so her task-specific tenure evolves linearly:

$$\boldsymbol{\tau}_{it+1} = \boldsymbol{\tau}_{it} + \boldsymbol{\omega}_j. \tag{10}$$

3.2 Labor Demand and Wage Equation

In each period t there is a continuum of competitive firms operating in each occupation j. Firms take the occupation output price in levels P_{jt} as given and compete for workers by posting output-contingent piece-rate wage contracts.[10] Firms in occupation j choose

[10]This output-contingent contracting environment differs from the standard employer-learning literature in which wages (or wage offers) depend directly on employers' beliefs about worker productivity,

employment across worker types to maximize profits

$$\max_{\{n_{jt}(\boldsymbol{\theta},\boldsymbol{\tau}_t)\}} \Pi_{jt} = P_{jt} Y_{jt} - \int W_{jt}(\boldsymbol{\theta},\boldsymbol{\tau}_t)\, n_{jt}(\boldsymbol{\theta},\boldsymbol{\tau}_t)\, dG_t(\boldsymbol{\theta},\boldsymbol{\tau}_t), \tag{11}$$

subject to the occupation production function

$$Y_{jt} = \int n_{jt}(\boldsymbol{\theta},\boldsymbol{\tau}_t)\, Y(\boldsymbol{\theta},\boldsymbol{\tau}_t)\, dG_t(\boldsymbol{\theta},\boldsymbol{\tau}_t), \tag{12}$$

where $G_t(\boldsymbol{\theta},\boldsymbol{\tau}_t)$ is the cross-sectional distribution of worker types at time t. Here $Y(\boldsymbol{\theta},\boldsymbol{\tau}_t)$ denotes the worker-occupation-specific output implied by the task technology in equation (4), which already embeds the idiosyncratic productivity shock $\exp(\varepsilon_{ijt})$. Under piece-rate contracts and perfect competition, firms post contracts until the wage paid equals realized worker-specific revenue, that is, the value of realized marginal product

$$W_{ijt} = P_{jt}\, Y_{ijt}. \tag{13}$$

Taking logs yields the log wage equation

$$w_{ijt} \equiv \log W_{ijt} = p_{jt} + \boldsymbol{\lambda}_j' \boldsymbol{\theta}_i + h_j(\boldsymbol{\tau}_{it}) + \varepsilon_{ijt}, \tag{14}$$

where $p_{jt} \equiv \log P_{jt}$. Log wages are determined by four components: (i) the log occupation price p_{jt}; (ii) the worker's latent skill bundle $\boldsymbol{\lambda}_j' \boldsymbol{\theta}_i$, where the occupation-specific loadings $\boldsymbol{\lambda}_j$ determine how task-specific skills map into output in occupation j; (iii) the log human-capital bundle $h_j(\boldsymbol{\tau}_{it})$, capturing the returns from accumulated task-specific experience; and (iv) the idiosyncratic productivity shock ε_{ijt}.

3.3 Learning and Information Structure

Workers belong to an observed education group $g \in \mathcal{G}$. Workers learn about their latent skills $\boldsymbol{\theta}_i$, which at labor-market entry are distributed within group g as

$$\boldsymbol{\theta}_i \mid g \sim \mathcal{N}(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g). \tag{15}$$

Workers are Bayesian and at the beginning of period t hold a subjective belief density $\tilde{f}_{it}(\boldsymbol{\theta})$ over $\boldsymbol{\theta}_i$ given their information set $\mathcal{I}_{it}$, which contains group membership and the history of past occupational choices, wages, and experience. I assume rational initial

so that observed wages reflect employers' posterior expectations (e.g. Farber and Gibbons 1996; Gibbons et al. 2005; Lange 2007; Pastorino 2024). In my framework, competitive piece-rates directly link wages to realized output; learning affects outcomes primarily through occupational sorting and continuation values rather than through belief-based wage-setting. Similar output-contingent wage contracts are used in Groes et al. (2015) and are consistent with empirical approaches that combine workers' choices with realized outcomes, as in Bunting et al. (2024). This wage contracting can also be viewed as a special case of the general wage contract in Camargo et al. (2022), $W_{ijt} = c_{ijt} + b_{ijt} Y_{ijt} = (1 - b_{ijt})\, \mathbb{E}\big[Y_{ijt} \mid I_{ijt}\big] + b_{ijt} Y_{ijt}$, with $b_{ijt} = 1$, so that wages depend one-for-one on realized output (up to the occupation-specific output price scaling in my model) rather than on posterior expectations.

beliefs in the sense that the initial prior coincides with the true group distribution,

$$\tilde{f}_{i0}(\boldsymbol{\theta}) = f_{\theta|g(i)}(\boldsymbol{\theta}) = \phi\Big(\boldsymbol{\theta}; \boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)}\Big), \tag{16}$$

where $\phi(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the multivariate normal density.

After choosing occupation j at time t and observing the wage-implied signal

$$s_{ijt} \equiv w_{ijt} - p_{jt} - h_j(\boldsymbol{\tau}_{it}) = \boldsymbol{\lambda}_j' \boldsymbol{\theta}_i + \varepsilon_{ijt}, \tag{17}$$

beliefs update according to Bayes' rule

$$\tilde{f}_{it+1}(\boldsymbol{\theta}) = \frac{f_{s|\theta}\big(s_{ijt} \mid \boldsymbol{\theta}\big)\, \tilde{f}_{it}(\boldsymbol{\theta})}{\int f_{s|\theta}\big(s_{ijt} \mid \boldsymbol{\theta}'\big)\, \tilde{f}_{it}(\boldsymbol{\theta}')\, d\boldsymbol{\theta}'}. \tag{18}$$

I assume that $\varepsilon_{ijt} \sim \mathcal{N}(0, \varsigma_{\varepsilon j}^2)$. Since the prior at $t = 0$ is multivariate normal and the signal in (17) is linear Gaussian, Bayes' rule implies that posteriors remain multivariate normal at all dates. Therefore the relevant beliefs can be summarized by a mean vector $\tilde{\boldsymbol{\theta}}_{it}$ and a variance-covariance matrix $\tilde{\boldsymbol{\Sigma}}_{it}$ such that

$$\boldsymbol{\theta}_i \mid \mathcal{I}_{it} \sim \mathcal{N}\big(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it}\big), \qquad \tilde{\boldsymbol{\theta}}_{i0} = \boldsymbol{\mu}_{g(i)}, \quad \tilde{\boldsymbol{\Sigma}}_{i0} = \boldsymbol{\Sigma}_{g(i)}. \tag{19}$$

Under normality, Bayes updating can be written in the Kalman filter form

$$\tilde{\boldsymbol{\theta}}_{it+1} = \tilde{\boldsymbol{\theta}}_{it} + \boldsymbol{K}_{ijt}\left(s_{ijt} - \boldsymbol{\lambda}_j' \tilde{\boldsymbol{\theta}}_{it}\right), \tag{20}$$

$$\tilde{\boldsymbol{\Sigma}}_{it+1} = \left(\boldsymbol{I}_K - \boldsymbol{K}_{ijt} \boldsymbol{\lambda}_j'\right) \tilde{\boldsymbol{\Sigma}}_{it}, \tag{21}$$

where $\boldsymbol{K}_{ijt}$ is the Kalman gain defined by

$$\boldsymbol{K}_{ijt} \equiv \tilde{\boldsymbol{\Sigma}}_{it} \boldsymbol{\lambda}_j \left(\boldsymbol{\lambda}_j' \tilde{\boldsymbol{\Sigma}}_{it} \boldsymbol{\lambda}_j + \varsigma_{\varepsilon j}^2\right)^{-1}. \tag{22}$$

Because $\tilde{\boldsymbol{\Sigma}}_{it}$ may have nonzero off-diagonal elements, observing s_{ijt} updates beliefs about all skill dimensions even if an occupation loads heavily on only a subset of skills.

3.4 Occupational Choice

Let the worker state at the beginning of period t be

$$\boldsymbol{x}_{it} \equiv \big(g(i),\, \tilde{\boldsymbol{\theta}}_{it},\, \tilde{\boldsymbol{\Sigma}}_{it},\, \boldsymbol{\tau}_{it},\, d_{it-1},\, \{p_{js}\}_{j \in \mathcal{J}, s \geq t}\big), \tag{23}$$

where $(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it})$ are beliefs about latent skills, $\boldsymbol{\tau}_{it}$ is task experience, and d_{it-1} is the last period's occupation. The sequence $\{p_{js}\}_{s \geq t}$ denotes log occupation prices from period t onward. I assume workers have perfect foresight over the future path of occupational prices and take $\{p_{js}\}$ as given. Figure 5 below summarizes the within-period timing of

the worker's choice and the resulting wage realization.

In each period t worker i chooses an occupation $j \in \mathcal{J}$ before observing the current wage shock. I assume log utility over wages, $u(W) = \log W$, and a switching cost $\mathcal{C}(j, d_{it-1})$ that is paid whenever the destination occupation j differs from the origin occupation d_{it-1}. The realized current-period utility from choosing j is

$$U_{ijt}(\boldsymbol{x}_{it}) = \log W_{ijt} + \eta_{jg(i)} - \mathcal{C}\left(j, d_{it-1}\right) + \epsilon_{ijt}, \tag{24}$$

where W_{ijt} is the wage paid in levels, $\eta_{jg(i)}$ is a non-pecuniary compensating differential specific to occupation j and the worker's education group $g(i)$, and ϵ_{ijt} is i.i.d. Extreme Value Type I with scale ϱ.[11] The switching cost is incurred only when the worker changes occupation, and I parameterize it as a single utility constant:

$$\mathcal{C}\left(j, d_{it-1}\right) = \kappa_0 \, \mathbf{1}_{\{j \neq d_{it-1}\}}. \tag{25}$$

The cost captures, in reduced form, all the non-wage related factors involved in changing occupation, for example search and retraining effort or the psychological cost of changing the type of job one does. A parsimonious cost lets me understand the roles of learning about task-specific skills and of task-specific experience in generating the task-content patterns of occupational mobility. Because the wage equation already makes moves costly through the task-specific human capital that a switcher leaves behind, κ_0 absorbs only the residual, non-pecuniary component of the cost of moving. Under log utility and the wage equation (14), expected utility is linear in expected log productivity:

$$\mathbb{E}_{\boldsymbol{\theta}_i, \varepsilon_{ijt}}\left[\log W_{ijt} \mid \tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it}\right] = p_{jt} + \boldsymbol{\lambda}_j' \tilde{\boldsymbol{\theta}}_{it} + h_j(\boldsymbol{\tau}_{it}).$$

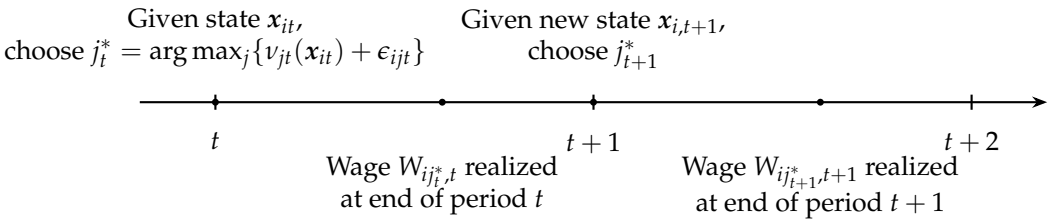


Figure 5: Timing of choices and wage realizations.

Without the worker index, and with its dependence on the price path made explicit,

[11] The taste shocks ϵ_{ijt} serve two purposes. First, they add idiosyncratic noise to the occupational choice in every period. When entering the labor market, all workers in a given education group enter with the same prior beliefs and no experience, so without the shocks all workers in a group would make the same choice. The shocks instead disperse entrants across occupations and, in later periods, help rationalize occupational choices and moves that the systematic part of the model cannot account for. Appendix E.1 illustrates the entry choice with and without the shocks. Second, they smooth the integrated value function and the conditional choice probabilities, which keeps the Bellman fixed point and the likelihood well-behaved for estimation.

the worker's dynamic program is

$$
\begin{aligned}
V_t\big(g, \tilde{\boldsymbol{\theta}}_t, \tilde{\boldsymbol{\Sigma}}_t, \boldsymbol{\tau}_t, d_{t-1}; \{p_{js}\}_{s\geq t}\big) = \max_{j\in\mathcal{J}} \Big\{ & p_{jt} + \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}}_t + h_j(\boldsymbol{\tau}_t) + \eta_{jg} - \mathcal{C}\big(j, d_{t-1}\big) + \epsilon_{jt} \\
& + \beta\, \mathbb{E}_{\boldsymbol{\theta}, s_{jt}, \boldsymbol{\epsilon}_{t+1}}\big[V_{t+1}\big(g, \tilde{\boldsymbol{\theta}}_{t+1}, \tilde{\boldsymbol{\Sigma}}_{t+1}, \boldsymbol{\tau}_{t+1}, j; \{p_{js}\}_{s\geq t+1}\big)\big] \Big\}, \\
\text{subject to:} \quad \boldsymbol{\tau}_{t+1} &= \boldsymbol{\tau}_t + \boldsymbol{\omega}_j, \\
s_{jt} &= \boldsymbol{\lambda}_j'\boldsymbol{\theta} + \varepsilon_{jt}, \\
\tilde{\boldsymbol{\theta}}_{t+1} &= \tilde{\boldsymbol{\theta}}_t + \boldsymbol{K}_{jt}\big(s_{jt} - \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}}_t\big), \\
\tilde{\boldsymbol{\Sigma}}_{t+1} &= \big(\boldsymbol{I}_K - \boldsymbol{K}_{jt}\boldsymbol{\lambda}_j'\big)\tilde{\boldsymbol{\Sigma}}_t, \\
\text{given} \quad \big(\tilde{\boldsymbol{\theta}}_0, \tilde{\boldsymbol{\Sigma}}_0, \boldsymbol{\tau}_0, d_{-1}\big) &= \big(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g, \mathbf{0}, \varnothing\big) \text{ for an entrant of group } g.
\end{aligned}
\tag{26}
$$

The flow payoff is the expected log wage, in which the posterior mean $\tilde{\boldsymbol{\theta}}_t$ replaces the latent skills $\boldsymbol{\theta}_i$ and the zero-mean wage shock ε_{ijt} drops out under log utility; the continuation expectation integrates over the predictive distribution of the wage signal s_{jt}, which drives the Bayesian update of beliefs, and the next-period taste shock vector $\boldsymbol{\epsilon}_{i,t+1}$. The next-period covariance is deterministic given the chosen occupation, so the only stochastic state evolution is the posterior mean $\tilde{\boldsymbol{\theta}}_{t+1}$. Let $v_{jt}(\cdot)$ denote the choice-specific value of occupation j, the term in braces in (26) excluding the taste shock ϵ_{jt}. Integrating out ϵ yields the multinomial logit conditional choice probabilities:

$$
\mathbb{P}_t\big(d_t = j \mid g, \tilde{\boldsymbol{\theta}}_t, \tilde{\boldsymbol{\Sigma}}_t, \boldsymbol{\tau}_t, d_{t-1}; \{p_{js}\}_{s\geq t}\big) = \frac{\exp\Big(v_{jt}\big(g, \tilde{\boldsymbol{\theta}}_t, \tilde{\boldsymbol{\Sigma}}_t, \boldsymbol{\tau}_t, d_{t-1}; \{p_{js}\}_{s\geq t}\big)\,/\varrho\Big)}{\sum_{m\in\mathcal{J}} \exp\Big(v_{mt}\big(g, \tilde{\boldsymbol{\theta}}_t, \tilde{\boldsymbol{\Sigma}}_t, \boldsymbol{\tau}_t, d_{t-1}; \{p_{js}\}_{s\geq t}\big)\,/\varrho\Big)}.
\tag{27}
$$

With a continuum of workers, these conditional choice probabilities represent the mass of workers in each state choosing each occupation. The education group g is a permanent characteristic of each worker and it shifts the payoff of every occupation through the occupation-specific amenities η_{jg}. In the empirical implementation, rather than treating past occupation and education group as discrete states, I solve for a separate value function for each occupation–education pair and compute the choice probabilities by education group. I approximate the integrated value function using the Smolyak method with Chebyshev polynomials (Malin et al. 2011; Judd et al. 2014), which amounts to solving for the coefficients that satisfy the fixed-point condition for the ex-ante integrated value function $EV(\cdot)$ at a set of collocation points:

$$
EV(\boldsymbol{x}) = \varrho \log \sum_{j\in\mathcal{J}} \exp\left(\frac{p_j + \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}} + h_j(\boldsymbol{\tau}) + \eta_{jg} - \mathcal{C}(j, d_{-1}) + \beta\,\mathbb{E}[\,EV(\boldsymbol{x}') \mid \boldsymbol{x}, d = j]}{\varrho}\right),
\tag{28}
$$

where the expectation is taken over next-period state $\boldsymbol{x}'$ implied by the laws of motion under choice j. More details are provided in Appendix D.

3.5 Closing the Model

I close the model with a representative final-goods producer that aggregates occupation outputs $\{Y_{jt}\}_{j\in\mathcal{J}}$ into final output Y_t using a CES technology with elasticity of substitution $\sigma > 0$:

$$Y_t = \left(\sum_{j\in\mathcal{J}} \alpha_j Y_{jt}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \qquad \alpha_j > 0. \tag{29}$$

The $\{\alpha_j\}_{j\in\mathcal{J}}$ weights in the CES aggregator govern the relative importance of each occupation's output in final production; their overall scale serves as the price-level normalization (Section 3.6). Let P_t denote the price of the final good and P_{jt} the price of occupation j output. Profit maximization by the final goods producer yields the inverse demand function for occupation j output:

$$P_{jt} = P_t \, \alpha_j \left(\frac{Y_t}{Y_{jt}} \right)^{\frac{1}{\sigma}}. \tag{30}$$

This expression determines occupation prices as a function of the final-good price and relative outputs. Taking the final good as the numeraire ($P_t = 1$, Section 3.6) gives occupation prices directly.

3.6 Stationary Competitive Equilibrium

I define a stationary competitive equilibrium in which occupation prices are constant over time. With a continuum of workers, let $\pi(\boldsymbol{x})$ denote the mass of workers in state $\boldsymbol{x} = (g, \boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\Sigma}}, \boldsymbol{\tau}, d_{-1}) \in \mathcal{X}$ (excluding prices), where $\mathcal{X}$ is the worker state space and $g(\boldsymbol{x})$ denotes the education group of state $\boldsymbol{x}$. This state augments the decision state (23) with the worker's true latent skill $\boldsymbol{\theta}$[12]. Entrants are characterized by group masses π_g (with $\sum_{g\in\mathcal{G}} \pi_g = 1$) and initial distributions $\pi_g^0(\cdot)$ that draw true skill from the group prior, $\boldsymbol{\theta} \mid g \sim \mathcal{N}(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$, have rational initial beliefs $(\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\Sigma}}) = (\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$, and start without experience $\boldsymbol{\tau} = \mathbf{0}$. A stationary competitive equilibrium[13] consists of constant occupation prices $\{P_j\}_{j\in\mathcal{J}}$, a stationary mass distribution $\pi^*(\cdot)$ over worker states, aggregate output Y, occupation outputs $\{Y_j\}_{j\in\mathcal{J}}$, a value function $V(\cdot)$, and conditional choice probabilities $\mathbb{P}(d = \cdot \mid \cdot)$ satisfying the following conditions:

1. **Labor supply.** Taking the constant prices $\{P_j\}_{j\in\mathcal{J}}$ as given, workers solve the Bellman equation (26) with constant discount factor β, generating the value function $V(\cdot)$ and conditional choice probabilities (27). Experience accumulates according to (10), and beliefs update via Bayes' rule (18). In every period, workers

[12] The worker's problem depends on the state variables and the worker group $(g, \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\Sigma}}, \boldsymbol{\tau}, d_{-1})$: the value function and conditional choice probabilities condition on g through the amenity η_{jg} and on beliefs through expected payoffs, and they do not directly depend on the worker's true skill $\boldsymbol{\theta}$. However, true skill determines realized output $Y_j(\boldsymbol{x})$ and, through the realized wage signal $s_j = \boldsymbol{\lambda}_j'\boldsymbol{\theta} + \varepsilon_j$, the Bayesian belief update.

[13] I provide the algorithm used to compute the stationary competitive equilibrium in Appendix B.2.

face a constant and exogenous probability χ of leaving the labor market (this acts as a death shock). This exit risk is not internalized by workers and serves only to generate a well-defined stationary distribution: each worker who exits is replaced by an entrant from the same education group g, with skills drawn from the group skill distribution $\boldsymbol{\theta} \mid g \sim \mathcal{N}(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$.

2. **Labor demand and wages.** For each occupation $j \in \mathcal{J}$, aggregate output equals the integral of worker output contributions over the mass of workers:

$$Y_j = \int \left(\int \exp(\varepsilon)\, Y_j(\boldsymbol{x})\, dF_j(\varepsilon) \right) \mathbb{P}(d = j \mid \boldsymbol{x})\, \pi^*(\boldsymbol{x})\, d\boldsymbol{x}, \tag{31}$$

where $Y_j(\boldsymbol{x}) = \exp(\boldsymbol{\lambda}_j'\boldsymbol{\theta})\, H_j(\boldsymbol{\tau})$ is the task technology (5) net of the idiosyncratic shock, evaluated at the worker's true skill $\boldsymbol{\theta}$ and task experience $\boldsymbol{\tau}$, F_j is the distribution of the wage shock ε, and $\mathbb{P}(d = j \mid \boldsymbol{x})\pi^*(\boldsymbol{x})$ gives the mass of workers in state $\boldsymbol{x}$ choosing occupation j. Wages satisfy the piece-rate adjusted revenue condition (14) in each occupation.

3. **Price determination.** Occupation prices satisfy the inverse demand system (30) given aggregate output Y and occupation outputs $\{Y_j\}_{j \in \mathcal{J}}$. Because CES inverse demand pins down only relative prices, I take the final good as the numeraire and normalize its price to one. The price of the final good equals the dual CES unit-cost index of the occupation outputs, so the numeraire is

$$P = \left(\sum_{j \in \mathcal{J}} \alpha_j^{\sigma}\, P_j^{1-\sigma} \right)^{\frac{1}{1-\sigma}} = 1, \tag{32}$$

the cost function dual to the aggregator (29).[14] Rather than rescaling the weights to the simplex, I choose the scale of $\{\alpha_j\}$ so that (32) holds exactly at the Stage-1 estimated prices (Section 4), which makes the estimated price vector itself the stationary equilibrium under the $P = 1$ numeraire.

4. **Stationary distribution.** The mass distribution π^* is stationary (invariant), satisfying

$$\pi^*(\boldsymbol{x}') = \sum_{j \in \mathcal{J}} \int \left[(1 - \chi)\, Q(\boldsymbol{x}' \mid \boldsymbol{x}, j) + \chi\, \pi^0_{g(\boldsymbol{x})}(\boldsymbol{x}') \right] \mathbb{P}(d = j \mid \boldsymbol{x})\, \pi^*(\boldsymbol{x})\, d\boldsymbol{x}, \tag{33}$$

for all states $\boldsymbol{x}' \in \mathcal{X}$, where $Q(\cdot \mid \boldsymbol{x}, j)$ is the transition density over worker states induced by choosing occupation j: true skill $\boldsymbol{\theta}$ and group g are carried forward unchanged, task experience accumulates via (10), and the belief moments update by Bayes' rule (18) from the realized wage signal $s_j = \boldsymbol{\lambda}_j'\boldsymbol{\theta} + \varepsilon_j$, whose distribution is governed by the worker's true skill; and $\pi^0_{g(\boldsymbol{x})}(\cdot)$ is the initial mass distribution

[14] The dual index carries α_j^{σ}, not α_j; this is the unique scale-one unit cost consistent with (29) and its first-order conditions (30).

for entrants from the group $g(\boldsymbol{x})$ of the exiting worker, so that each exiting worker is replaced by an entrant from the same education group and the group masses π_g are preserved[15].

In practice, I compute the equilibrium as a damped fixed point in the vector of occupation log prices: given a price guess I solve the worker's dynamic program, simulate a stationary panel of workers (with exogenous exit and entrant replacement) to obtain occupation outputs $\{Y_j\}$, update prices from the CES inverse demand (30) subject to the numeraire normalization (32), and iterate to convergence. Appendix B.2 details the algorithm.

4 Identification and Estimation

To keep the state space tractable, I estimate the model with two task-specific skills, cognitive and manual, and aggregate the 103 three-digit ISCO occupations into $|\mathcal{J}| = 8$ clusters. I choose the clusters to minimize the within-cluster dispersion of the cognitive and manual task content across occupations within a given cluster, with a penalty for distorting the empirical occupational mobility graph; Appendix A.2 describes the algorithm and the resulting clusters. While this aggregation mechanically lowers measured mobility levels compared to the 3-digit ISCO analysis, the empirical facts documented in Section 2 remain qualitatively consistent within this clustered state space.

4.1 Identification of the Wage Equation under Learning

I first show that the wage equation is identified even though multidimensional skills are unobserved by both workers and the econometrician. The argument follows closely Arcidiacono et al. (2025) and is based on the formal nonparametric identification result of Bunting et al. (2024), adapted to the multidimensional-skill setting of this paper. The key identifying idea is that the model is a pure-learning environment, in the sense of Bunting et al. (2024): workers sort on beliefs about latent skills, and those beliefs are formed from the same wage and occupational histories observed by the econometrician. Therefore, conditional on the worker's history, occupational choice selects on no additional unobserved component, so that controlling for the worker history is enough to solve the selection problem.

Let $d_{it} \in \mathcal{J}$ denote worker i's occupation in period t, let $g(i) \in \mathcal{G}$ denote the worker's observed group, and let $\boldsymbol{\tau}_{it}$ be predetermined at the time of the occupational choice. The potential log wage in occupation j is

$$w_{it}(j) = p_j + h_j(\boldsymbol{\tau}_{it}) + \boldsymbol{\lambda}_j'\boldsymbol{\theta}_i + \varepsilon_{ijt}, \tag{34}$$

[15]Due to the learning structure, the dynamic problem of workers is Markovian. Hence under standard regularity conditions, the death-and-rebirth mechanism makes the worker-state dynamics regenerative and causes initial conditions to wash out over time. The process therefore admits a unique invariant distribution π^*, and simulation averages converge to expectations under π^*. Appendix B.2 describes the simulation procedure used to approximate this distribution in practice.

where $\boldsymbol{\theta}_i \in \mathbb{R}^K$ is a time-invariant latent skill vector and the observed log wage is $w_{it} = w_{it}(d_{it})$. The wage-side parameters are

$$\boldsymbol{\Gamma} \equiv \left\{ (\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)_{g\in\mathcal{G}}, (\boldsymbol{\lambda}_j)_{j\in\mathcal{J}}, (\boldsymbol{\delta}_j)_{j\in\mathcal{J}}, (\varsigma^2_{\varepsilon j})_{j\in\mathcal{J}}, (p_j)_{j\in\mathcal{J}} \right\},$$

where $\boldsymbol{\delta}_j$ collects the tenure steps of h_j and $\varsigma^2_{\varepsilon j}$ is the variance of ε_{ijt}. The piece-rate assumption implies that, after subtracting the occupation price and the human-capital component, both worker and econometrician observe a noisy linear measurement of $\boldsymbol{\lambda}_j'\boldsymbol{\theta}_i$.

Formally, the pure-learning structure implies

$$d_{it} \perp\!\!\!\perp \boldsymbol{\theta}_i \mid \mathcal{H}_{it},$$

where $\mathcal{H}_{it} \equiv (g(i), d_i^{t-1}, w_i^{t-1}, \boldsymbol{\tau}_i^t)$ is the observed history conditional on the observed education level $g(i)$. Conditional on the history, occupational choices carry no additional information about latent skills, so the endogenous-occupational-choice problem reduces to one of selection on observables. Combined with standard rank and support conditions and the anchor and weighted-zero-mean normalizations, this restriction delivers point identification of the wage-equation parameters $\boldsymbol{\Gamma}$ from the joint distribution of observed wages, occupational choices, groups, and predetermined state variables (Theorem 1 in the appendix).

Appendix C.1 gives the formal argument, which uses the inverse-probability weighting of Bunting et al. (2024). Intuitively, the "history-specific" CCPs serve as inverse-probability weights that recover the potential wage path under any supported occupational choice sequence. Under the linear-Gaussian wage equation this recovered path is multivariate normal: its means identify occupation prices plus an education-specific mean, and its within-worker covariances identify the latent skill covariance and the occupation-specific loadings. Repeated observations in the same occupation separate the permanent component from transitory noise and identify the occupation-specific task-experience returns. Occupation switchers identify how different occupations load on the common skill vector. As in standard factor models, loadings and factors are not separately identified without normalizations. Because I leave the variance-covariance matrix of group-specific skills unrestricted, I need to impose two restrictions on the loadings. For $K = 2$, I use an anchor occupation to fix the cognitive coordinate and one that fixes the manual coordinate; covariances of all other occupations with these two anchors recover their cognitive and manual loading vectors.

This identification argument can equivalently be viewed as a control-function argument: the observed history $\mathcal{H}_{it}$ is a sufficient control for endogenous occupational choice, so that the wage regression in expectation becomes

$$\mathbb{E}\left[w_{it}(j) \mid \mathcal{H}_{it},\, g(i),\, d_{it} = j\right] \;=\; p_j + h_j(\boldsymbol{\tau}_{it}) + \boldsymbol{\lambda}_j' \mathbb{E}\left[\boldsymbol{\theta}_i \mid \mathcal{H}_{it},\, g(i)\right],$$

where the posterior mean $\mathbb{E}[\boldsymbol{\theta}_i \mid \mathcal{H}_{it}, g(i)]$ enters as a control regressor and absorbs

selection based on beliefs.

4.2 Identification of the Dynamic Choice Parameters

Stage 1 identifies the wage-side parameters $\boldsymbol{\Gamma}$, so for every supported state $\boldsymbol{x}_{it}$ and every occupation j the econometrician can compute the expected log wage $\tilde{w}_j(\boldsymbol{x}_{it})$ and the distribution of the wage signals, hence the law of motion of beliefs. Workers' beliefs upon entry into the labor market coincide with the true distribution of skills identified in the first step (Section 3.3). This assumption, together with the identified wage signal and the law of motion of beliefs, allows the econometrician to compute the exact sequence of beliefs for every worker in the sample. Therefore, CCPs are identified directly from the joint distribution of choices and states. Because the data are generated by a stationary equilibrium with constant occupation prices, the choice-specific value index and the conditional choice probabilities are time invariant, $\nu_j(\boldsymbol{x})$ and $\mathbb{P}_j(\boldsymbol{x})$. Because the state $\boldsymbol{x}$ contains the education group g, ν_j, $\mathbb{P}_j$ and the ex-ante value EV are group-specific. With β fixed externally and ϱ calibrated, the only unknown choice parameters left are the amenities η_{jg} and the single switching cost κ_0 of (25), which I collect in

$$\boldsymbol{\Delta} \equiv \Big\{ (\eta_{jg})_{j\in\mathcal{J},\, g\in\mathcal{G}},\ \kappa_0 \Big\}.$$

I can identify these parameters using the level and composition of mobility across destinations observed in the data. Under the Type-I extreme value assumption, the observed log-odds recover differences in the choice-specific value function,

$$\varrho \, \log \frac{\mathbb{P}_j(\boldsymbol{x})}{\mathbb{P}_m(\boldsymbol{x})} = \nu_j(\boldsymbol{x}) - \nu_m(\boldsymbol{x}),$$

and the same logit structure that produces (27) can also be inverted, following Hotz and Miller (1993), to express the ex-ante value function EV in terms of any single choice using a probability correction term, that is $EV(\boldsymbol{x}) = \nu_j(\boldsymbol{x}) - \varrho \log \mathbb{P}_j(\boldsymbol{x})$ for any j. The correction $-\varrho \log \mathbb{P}_j(\boldsymbol{x})$ is the gap between the ex-ante value and the value of committing to j in a particular state $\boldsymbol{x}$: it is small when j is chosen often, and hence close to optimal, and large when j is chosen rarely.

Applying this inversion recursively along the path on which the worker keeps choosing the occupation she has just chosen yields the representation of Theorem 1 in Arcidiacono and Miller (2011), which writes the conditional value function as a discounted sequence of expected wages, each corrected by the choice-probability term $-\varrho \log \mathbb{P}_j$. Intuitively, I can express continuation values using CCPs that are directly identified from the data rather than through the unknown structural parameters. Collecting terms, the choice-specific value function is linear in the two unknowns,

$$\nu_j(\boldsymbol{x}) = \underbrace{\tilde{w}_j(\boldsymbol{x}) + \beta\, \Psi_j(\boldsymbol{x})}_{\text{known from Stage 1 and the CCPs}} + \frac{\eta_{jg}}{1-\beta} - \kappa_0\, \mathbf{1}_{\{j\neq d\}}, \qquad d \equiv d_{t-1} \text{ in } \boldsymbol{x},$$

where $\Psi_j(\boldsymbol{x})$ is the discounted stream of expected wages net of the choice-probability correction accumulated along that path, known from Stage 1 and the observed CCPs (Lemma 1 in Appendix C.2 gives its expression).

Two comparisons then separate the two kinds of parameter. Comparing two different occupation destinations $j \neq d$ and $m \neq d$ reached from a common origin d, a move costs κ_0 regardless of where the worker goes, so the switching cost cancels and the log-odds isolate the difference in amenities:

$$\eta_{jg} - \eta_{mg} = (1-\beta)\left[\varrho \log \frac{\mathbb{P}_j(\boldsymbol{x})}{\mathbb{P}_m(\boldsymbol{x})} - \big(\tilde{w}_j(\boldsymbol{x}) - \tilde{w}_m(\boldsymbol{x})\big) - \beta\big(\Psi_j(\boldsymbol{x}) - \Psi_m(\boldsymbol{x})\big)\right].$$

It is thus the relative frequency with which workers leaving a common origin go to each different destination that identifies amenities, and the normalization $\eta_{0g} = 0$ within each education group turns these differences into levels. Comparing instead a move to j with staying in d, the amenities are already known and the switching cost is the only remaining unknown:

$$\kappa_0 = \big(\tilde{w}_j(\boldsymbol{x}) - \tilde{w}_d(\boldsymbol{x})\big) + \beta\big(\Psi_j(\boldsymbol{x}) - \Psi_d(\boldsymbol{x})\big) + \frac{\eta_{jg} - \eta_{dg}}{1-\beta} - \varrho \log \frac{\mathbb{P}_j(\boldsymbol{x})}{\mathbb{P}_d(\boldsymbol{x})}.$$

Every term on the right is known, so κ_0 is identified.

The scale ϱ is conceptually separate from amenities and switching costs. Because expected log wages are measured in utility units, ϱ could in principle be identified from variation in expected wage differences. However, in this learning model, the same state variables that shift current expected wages also shift future beliefs and continuation values, so ϱ is weakly identified without variation in payoff that moves current utility but not the continuation value. I therefore calibrate ϱ to match the average occupational mobility rate in the large sample rather than estimating it jointly by maximum likelihood. Given this calibration, the normalization $\eta_{0g} = 0$ for every g, and support conditions requiring that the occupations used are chosen with positive probability in the states used, that each occupation is observed both as an origin and as a destination, and that both stays and moves are observed, the dynamic choice parameters $\big(\{\eta_{jg}\}_{j\in\mathcal{J},g\in\mathcal{G}}, \kappa_0\big)$ are identified. Theorem 2 in Appendix C.2 states this formally.

4.3 Two-Stage Estimation

I estimate the model in two stages. In the first stage I estimate the wage equation parameters and the group-specific distribution of latent skills using an Expectation–Maximization (EM) algorithm. A key object in the wage equation is the occupation price p_j. I assume that wages are generated by a unique stationary equilibrium (the one observed in the data), so that occupation prices are constant over time. In the empirical implementation I estimate p_j using occupation fixed effects, so that the wage

equation becomes

$$w_{ijt} = p_j + \boldsymbol{\lambda}_j'\boldsymbol{\theta}_i + h_j(\boldsymbol{\tau}_{it}) + \varepsilon_{ijt}, \qquad \varepsilon_{ijt} \sim \mathcal{N}(0, \varsigma_{\varepsilon j}^2), \tag{35}$$

where I treat the occupation price p_j as an estimable nuisance parameter. Latent skills satisfy $\boldsymbol{\theta}_i \mid g(i) \sim \mathcal{N}(\boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)})$. The first stage yields the wage-side parameters $\boldsymbol{\Gamma}$ defined in Section 4.1. In the second stage I estimate the choice-side parameters $\boldsymbol{\Delta}$ of Section 4.2 using the Rust (1987) Nested Fixed Point (NFXP) algorithm, taking $\widehat{\boldsymbol{\Gamma}}$ as given and setting the taste-shock scale $\varrho = 0.271$ to match the average occupational mobility rate in the large sample (see Section 4.2 for discussion). The observed-data likelihood for worker i factors into a wage component and a choice component:

$$L_i(w_i, d_i; \boldsymbol{\Gamma}, \boldsymbol{\Delta}) = L_i^{w}(w_i; \boldsymbol{\Gamma}) \cdot L_i^{c}(d_i; \boldsymbol{\Gamma}, \boldsymbol{\Delta}), \tag{36}$$

where $w_i = \{w_{it_0(i)}, \ldots, w_{iT(i)}\}$ and $d_i = \{d_{it_0(i)}, \ldots, d_{iT(i)}\}$ denote observed wage and choice histories[16].

The wage likelihood integrates out latent skills,

$$L_i^{w}(w_i; \boldsymbol{\Gamma}) = \int \left[\prod_{t=t_0(i)}^{T(i)} \prod_{j \in \mathcal{J}} f_{\varepsilon j}\Big(w_{ijt} - p_j - \boldsymbol{\lambda}_j'\boldsymbol{\theta} - h_j(\boldsymbol{\tau}_{it}) \Big)^{d_{ijt}} \right] \phi\Big(\boldsymbol{\theta}; \boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)}\Big)\, d\boldsymbol{\theta}, \tag{37}$$

where $f_{\varepsilon j}$ is the normal density with variance $\varsigma_{\varepsilon j}^2$ and $\phi(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the multivariate normal density. With $d_{ijt} \equiv \mathbf{1}\{d_{it} = j\}$ the indicator that worker i chooses occupation j in year t, the choice likelihood separates the first observed choice (which uses the group

[16] Starting from $L_i(w_i, d_i; \boldsymbol{\Gamma}, \boldsymbol{\Delta}) = \int f(w_i, d_i, \boldsymbol{\theta} \mid g(i); \boldsymbol{\Gamma}, \boldsymbol{\Delta})\, d\boldsymbol{\theta}$, the chain rule gives

$$f(w_i, d_i, \boldsymbol{\theta} \mid g(i)) = \phi(\boldsymbol{\theta}; \boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)}) \prod_t f(w_{it} \mid d_{it}, \boldsymbol{\theta}, \boldsymbol{\tau}_{it}; \boldsymbol{\Gamma}) \prod_t f(d_{it} \mid \mathcal{H}_{it}; \boldsymbol{\Gamma}, \boldsymbol{\Delta}).$$

By timing/independence, conditional on $(d_{it}, \boldsymbol{\theta}, \boldsymbol{\tau}_{it})$ wages depend only on $\boldsymbol{\Gamma}$, while choices depend on history only through $(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it}, \boldsymbol{\tau}_{it}, d_{it-1})$, hence $\prod_t f(d_{it} \mid \mathcal{H}_{it}; \boldsymbol{\Gamma}, \boldsymbol{\Delta})$ does not depend on $\boldsymbol{\theta}$. Therefore,

$$\begin{aligned} L_i(w_i, d_i; \boldsymbol{\Gamma}, \boldsymbol{\Delta}) &= \int \phi(\boldsymbol{\theta}; \boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)}) \prod_t f(w_{it} \mid d_{it}, \boldsymbol{\theta}, \boldsymbol{\tau}_{it}; \boldsymbol{\Gamma}) \prod_t f(d_{it} \mid \mathcal{H}_{it}; \boldsymbol{\Gamma}, \boldsymbol{\Delta})\, d\boldsymbol{\theta} \\ &= \left[\prod_t f(d_{it} \mid \mathcal{H}_{it}; \boldsymbol{\Gamma}, \boldsymbol{\Delta}) \right] \int \phi(\boldsymbol{\theta}; \boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)}) \prod_t f(w_{it} \mid d_{it}, \boldsymbol{\theta}, \boldsymbol{\tau}_{it}; \boldsymbol{\Gamma})\, d\boldsymbol{\theta} \\ &\equiv L_i^{c}(d_i; \boldsymbol{\Gamma}, \boldsymbol{\Delta}) \cdot L_i^{w}(w_i; \boldsymbol{\Gamma}). \end{aligned}$$

prior) from subsequent choices:

$$L_i^c(d_i;\boldsymbol{\Gamma},\boldsymbol{\Delta}) = \underbrace{\prod_{j\in\mathcal{J}}\Big[\mathbb{P}\Big(d_{it_0(i)}=j \mid \boldsymbol{\mu}_{g(i)},\boldsymbol{\Sigma}_{g(i)},\boldsymbol{\tau}_{it_0(i)},d_{i,t_0(i)-1};\boldsymbol{\Gamma},\boldsymbol{\Delta}\Big)\Big]^{d_{ij,t_0(i)}}}_{\text{first observed choice}} \times \underbrace{\prod_{t=t_0(i)+1}^{T(i)}\prod_{j\in\mathcal{J}}\Big[\mathbb{P}\big(d_{it}=j \mid \tilde{\boldsymbol{\theta}}_{it},\tilde{\boldsymbol{\Sigma}}_{it},\boldsymbol{\tau}_{it},d_{it-1};\boldsymbol{\Gamma},\boldsymbol{\Delta}\big)\Big]^{d_{ijt}}}_{\text{subsequent choices}}, \tag{38}$$

where CCPs are given by (27) and $(\tilde{\boldsymbol{\theta}}_{it},\tilde{\boldsymbol{\Sigma}}_{it})$ are the Bayesian posterior moments implied by $\boldsymbol{\Gamma}$.

Because $L_i = L_i^w \cdot L_i^c$, the log-likelihood is additively separable:

$$\log L_i(w_i,d_i;\boldsymbol{\Gamma},\boldsymbol{\Delta}) = \log L_i^w(w_i;\boldsymbol{\Gamma}) + \log L_i^c(d_i;\boldsymbol{\Gamma},\boldsymbol{\Delta}),$$

which permits sequential estimation: first obtain $\widehat{\boldsymbol{\Gamma}} = \arg\max_{\boldsymbol{\Gamma}} \sum_i \log L_i^w(w_i;\boldsymbol{\Gamma})$, then conditional on $\widehat{\boldsymbol{\Gamma}}$ compute $\widehat{\boldsymbol{\Delta}} = \arg\max_{\boldsymbol{\Delta}} \sum_i \log L_i^c(d_i;\widehat{\boldsymbol{\Gamma}},\boldsymbol{\Delta})$.

4.3.1 Estimation Stage 1: Expectation–Maximization Algorithm

Starting from an initial guess $\boldsymbol{\Gamma}^{(0)}$, the EM algorithm iterates:

- **E-step.** For current $\boldsymbol{\Gamma}^{(q)}$ and observed histories (w_i,d_i), compute the posterior distribution of $\boldsymbol{\theta}_i$,
$$f(\boldsymbol{\theta}_i \mid w_i,d_i;\boldsymbol{\Gamma}^{(q)}),$$
and form the expected complete-data log-likelihood.
- **M-step.** Update $\boldsymbol{\Gamma}^{(q+1)}$ by maximizing the expected complete-data log-likelihood. In practice this step updates $(\boldsymbol{\mu}_g,\boldsymbol{\Sigma}_g)$, the skill loadings $\boldsymbol{\lambda}_j$, the tenure steps $\boldsymbol{\delta}_j$, the shock variances $(\varsigma_{\varepsilon j}^2)$, and the occupation prices (p_j). The M-step explicitly operationalizes the control-function identification argument of Section 4.1: by regressing log wages on the E-step posterior mean of skills, $\hat{\boldsymbol{\zeta}}_i^{(q)} \equiv \mathbb{E}[\boldsymbol{\theta}_i \mid \boldsymbol{w}_i,\boldsymbol{d}_i;\boldsymbol{\Gamma}^{(q)}]$, and on the regressors entering $h_j(\boldsymbol{\tau}_{it})$, it absorbs selection on beliefs through the observed history.

Iterations continue until $\|\boldsymbol{\Gamma}^{(q+1)} - \boldsymbol{\Gamma}^{(q)}\|_2$ is small. Because EM guarantees only local convergence, I use multiple starting values and select the solution with the highest likelihood.

4.3.2 Estimation Stage 2: Nested Fixed Point Algorithm

In the second stage I estimate $\boldsymbol{\Delta}$ by maximizing the choice part of the log-likelihood given $\widehat{\boldsymbol{\Gamma}}$:

$$\widehat{\boldsymbol{\Delta}} = \arg\max_{\boldsymbol{\Delta}} \log L^c(\widehat{\boldsymbol{\Gamma}},\boldsymbol{\Delta}), \tag{39}$$

where

$$\log L^c(\widehat{\mathbf{\Gamma}}, \mathbf{\Delta}) = \sum_{i=1}^{N} \Bigg\{ \sum_{j \in \mathcal{J}} d_{ij,t_0(i)} \log \mathbb{P}\Big(d_{it_0(i)} = j \mid \boldsymbol{\mu}_{g(i)}, \mathbf{\Sigma}_{g(i)}, \boldsymbol{\tau}_{it_0(i)}, d_{i,t_0(i)-1}; \widehat{\mathbf{\Gamma}}, \mathbf{\Delta}\Big) + \sum_{t=t_0(i)+1}^{T(i)} \sum_{j \in \mathcal{J}} d_{ijt} \log \mathbb{P}\Big(d_{it} = j \mid \tilde{\boldsymbol{\theta}}_{it}, \tilde{\mathbf{\Sigma}}_{it}, \boldsymbol{\tau}_{it}, d_{it-1}; \widehat{\mathbf{\Gamma}}, \mathbf{\Delta}\Big) \Bigg\}.$$

The NFXP algorithm nests the solution of the worker dynamic problem inside the maximization over $\mathbf{\Delta}$. Fix a candidate $\mathbf{\Delta}$ and define the state $\boldsymbol{x} \equiv (g, \tilde{\boldsymbol{\theta}}, \tilde{\mathbf{\Sigma}}, \boldsymbol{\tau}, d_{-1})$, on which the value function is solved group by group. The inner step solves for the integrated ex-ante value function $EV(\cdot; \widehat{\mathbf{\Gamma}}, \mathbf{\Delta})$ as the fixed point (28), evaluated at $(\widehat{\mathbf{\Gamma}}, \mathbf{\Delta})$. Given the fixed point $EV(\cdot; \widehat{\mathbf{\Gamma}}, \mathbf{\Delta})$, the CCPs used in the likelihood are computed from (27).

The outer step updates $\mathbf{\Delta}$ to increase (39) using a numerical optimizer. The procedure iterates the two steps (i) solving the fixed point in (28) for EV at the current $\mathbf{\Delta}$ and (ii) updating $\mathbf{\Delta}$ given the resulting CCPs, until convergence.

4.3.3 Externally Set and Calibrated Parameters

Several parameters are set externally or calibrated to match aggregate statistics, as summarized in Table 3 in the Appendix. The discount factor is set to $\beta = 0.9$; education group shares π_g are set to the large-sample composition; and task importance weights ω_j are taken from Mihaylov and Tijdens (2019). The exit hazard $\chi = 0.08$ is set externally and implies an expected career length of 12.5 years, close to the average career span in the large sample[17].

The CES elasticity of substitution is set to $\sigma = 1.57$. The aggregator weights $\{\alpha_j\}$ are calibrated so that, evaluated at prices estimated as fixed effects from Stage-1, the CES inverse demand (30) holds exactly while the final-good price index (32) equals one. Inverting the CES demand (30) leaves the CES weights proportional to $P_j Y_j^{1/\sigma}$ up to a single scalar; I fix that scalar by normalizing the aggregate price level $P(\hat{P}) = 1$ and I obtain the following closed-form calibration equation

$$\alpha_j = \frac{P_j \, Y_j^{1/\sigma}}{\left(\sum_{m \in \mathcal{J}} P_m \, Y_m\right)^{1/\sigma}}, \qquad \text{equivalently} \qquad \alpha_j^{\sigma} = s_j \, P_j^{\sigma - 1}, \tag{40}$$

where $P_j = \exp(\hat{p}_j)$ is occupation j's output price estimated in Stage 1, Y_j its output (the observed occupation wage bill deflated by P_j, because under piece-rate pricing each occupation's revenue equals its wage bill), and $s_j = P_j Y_j / \sum_m P_m Y_m$ its wage-bill share. Because $\sum_j \alpha_j^{\sigma} P_j^{1-\sigma} = \sum_j s_j = 1$, this delivers $P(\hat{P}) = 1$ exactly.

[17] As explained in Section 3.6, it is exogenous and only serves the purpose of generating the stationary distribution. Workers do not internalize this exit rate when choosing occupation and behave as if they were infinitely lived; if they internalized it, they would excessively discount the future in the first years of their career.

4.4 Estimation Results

4.4.1 Stage 1 Estimation Results: Wage Equation Parameters

The EM algorithm jointly estimates occupation-specific skill and experience elasticities $(\boldsymbol{\lambda}_j, \boldsymbol{\delta}_j)$, idiosyncratic wage variances, occupation prices, and education-specific skill distributions $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$. Full estimation results are reported in Table 4 in the Appendix.

The wage-equation estimates deliver three findings. First, occupations differ substantially in the productivity of cognitive and manual skills, both in their skill loadings λ_{jk} and in their returns to experience[18]. Second, the estimated group skill distributions $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ (Figure 6) determine the initial comparative advantage of each education group, which shapes entrants' initial beliefs. The average level of skills rises with education in both dimensions, and the average comparative advantage in cognitive tasks widens: low- and medium-educated workers, on average, have a small comparative advantage in manual tasks, while high-educated workers have a relative comparative advantage in cognitive tasks. Cross-skill correlations indicate whether the data support a model of unidimensional or multidimensional skill. The estimated correlations are small in all three education groups: alongside workers who are high or low ability in both dimensions, many workers have a strong comparative advantage in one skill dimension. These estimated distributions imply that more highly educated workers have (and perceive) higher skills in cognitive tasks, making them more likely to start their career in more cognitive occupations, consistent with Deming (2025). Third, occupations have heterogeneous returns to both manual and cognitive task experience, and returns to cognitive experience are generally steeper than returns to manual experience (Figure 7).

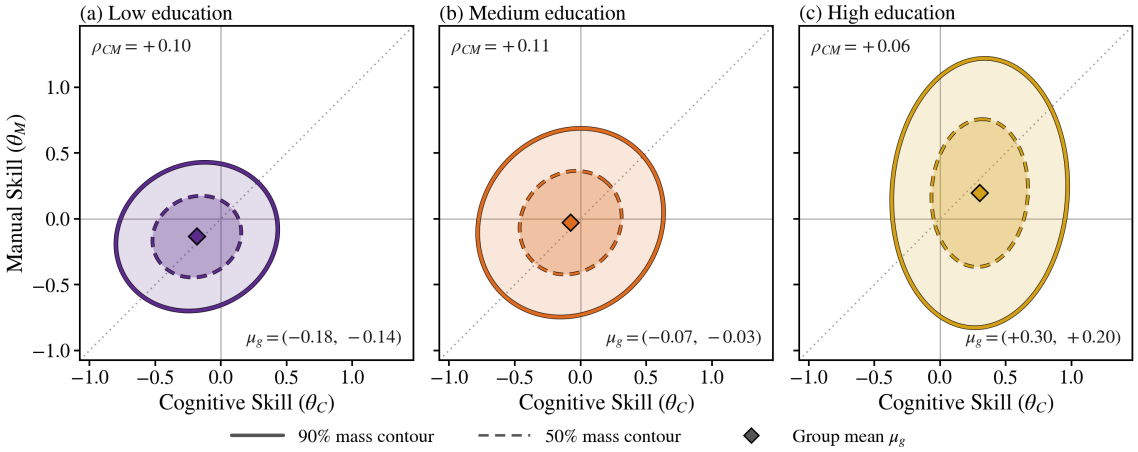


Figure 6: Estimated skill distributions by education group. Each panel plots the prior distribution of skills implied by the estimated group mean and covariance $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ in the space of cognitive skill θ_C (horizontal axis) and manual skill θ_M (vertical axis). The dashed and solid contours enclose 50% and 90% of the probability mass, the diamond is the group mean $\boldsymbol{\mu}_g$, and the dotted line is the $\theta_C = \theta_M$ line; each panel reports the within-group correlation between cognitive and manual skills, ρ_{CM}.

[18] Cluster 0 is the cognitive anchor and Cluster 1 the manual anchor.

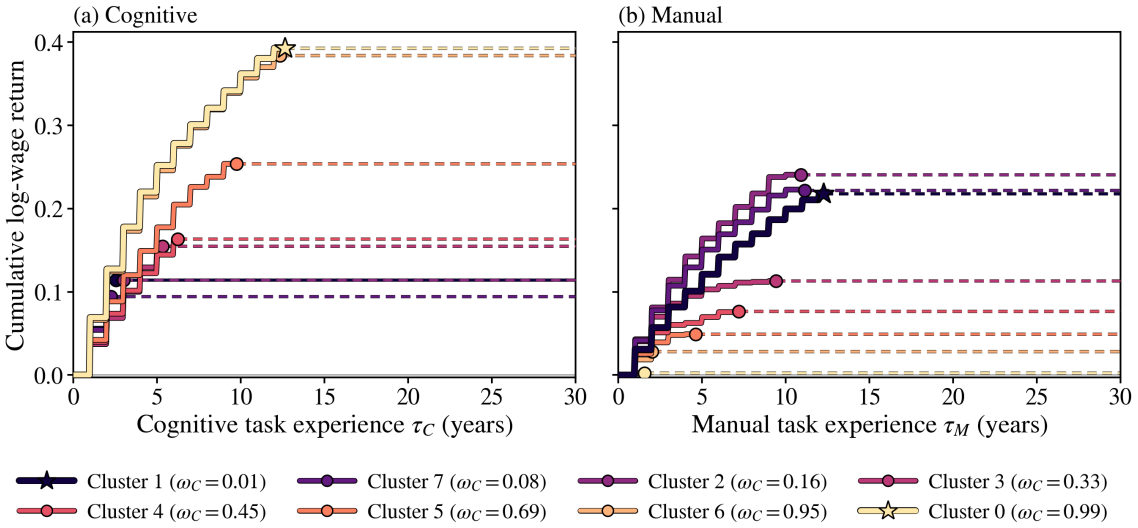


Figure 7: Estimated returns to task-specific experience by occupation. Each line plots the log-wage return to task experience, $\sum_b \delta_{jk}^{(b)} \mathbf{1}\{\tau \in b\}$, as task tenure τ accumulates from 0 to 30 years. Returns are estimated as a step function of tenure with annual bins, with the last bin being defined by the 95th percentile of observed task-specific tenure within that occupation. *Panel (a)*: returns to cognitive experience τ_C. *Panel (b)*: returns to manual experience τ_M. The marker on each curve is the occupation's 95th percentile of observed task tenure. Clusters are labeled by their cognitive task intensity ω_{jC} and shaded from navy (most manual) to cream (most cognitive); stars mark the anchor clusters (Cluster 0, cognitive; Cluster 1, manual).

4.4.2 Stage 2 Estimation Results: Dynamic Choice Parameters

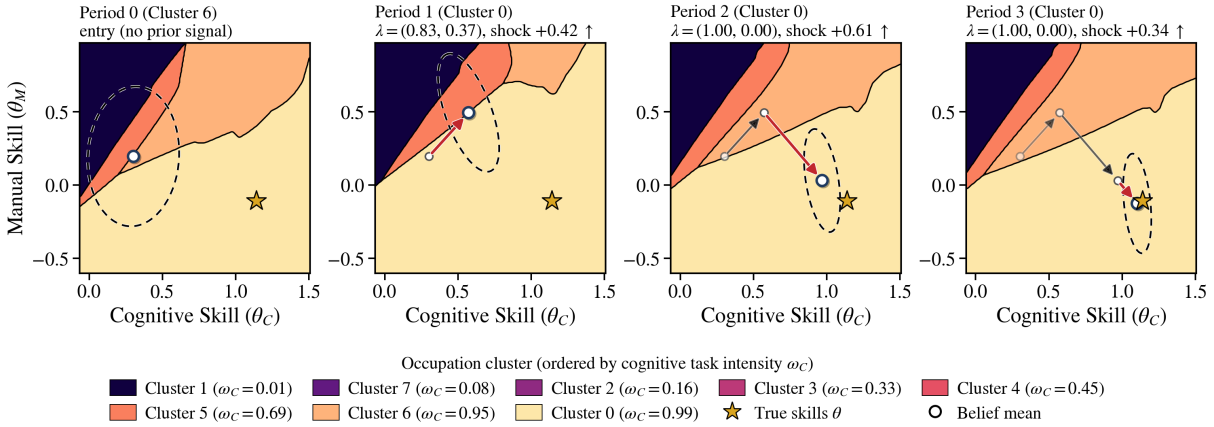
The NFXP estimates are reported in Table 5 in the Appendix. As discussed in Section 4.2, I set $\varrho = 0.271$ to match the average occupational mobility rate in the large sample, which still allows within-group heterogeneity in initial choices. The utility switching cost is moderate—a single constant of about 1.1 in log-wage units ($\kappa_0 = 1.103$)—suggesting that the life-cycle decline in occupational mobility is driven primarily by task-specific human capital accumulation and learning rather than by prohibitive switching costs. Occupation- and education-specific amenities η_{jg} display considerable heterogeneity. In the model, amenities act as compensating differentials that rationalize persistent occupational sorting patterns that cannot be explained by wage differentials alone. These parameters may also capture frictions the model does not include. For example, if access to some occupations requires a specific educational degree, η_{jg} would capture the fact that certain groups of workers do not move to these occupations despite having attractive wages.

5 Occupational Mobility and the Role of Learning

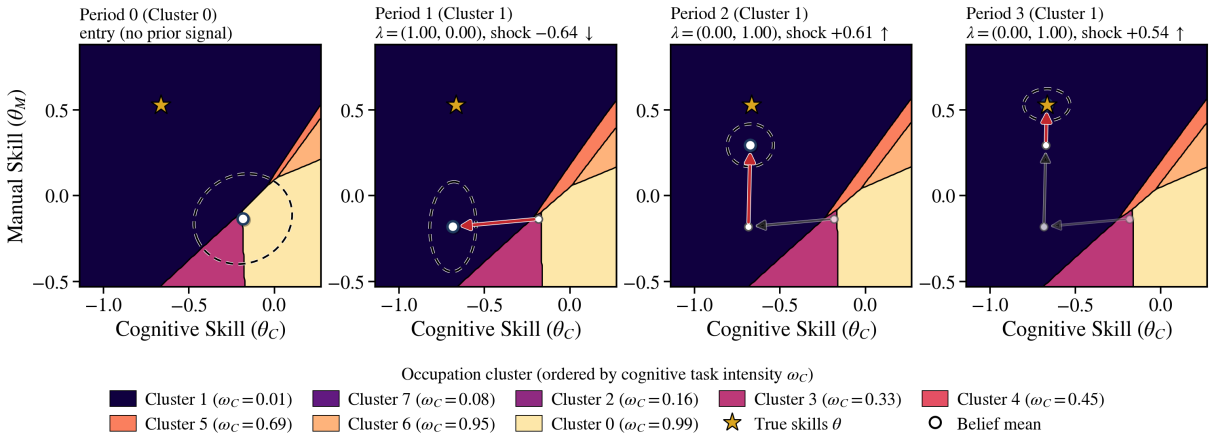
5.1 Learning Generates Directionality in the Task Space

Before assessing the model's fit, I simulate a simplified version of the model in which only learning operates, with no amenities, switching costs, human capital accumulation, or taste shocks. This simplified model isolates the role of learning in shaping

occupational mobility; I then add the other components back to assess their relative importance. Figure 8 plots, in the space of latent skills (θ_C, θ_M), each worker's belief about her own comparative advantage as it evolves during her first few periods in the labor market.



(a) Worker entering Cluster 6 and moving to the cognitive anchor (Cluster 0)



(b) Worker entering the cognitive anchor (Cluster 0) and moving to the manual anchor (Cluster 1)

Figure 8: Belief dynamics in the space of latent skills for two simulated workers over their first four periods; each row is one worker. Horizontal axis: cognitive skill θ_C; vertical axis: manual skill θ_M. The shaded regions give the occupation chosen as a function of the belief mean with amenities set to zero, ordered by cognitive task intensity ω_C (legend). The star is the worker's true skills $\boldsymbol{\theta}$, the circle her posterior belief mean, and the dashed ellipse her posterior covariance. The red arrow is the belief-mean update from the period's wage signal and the gray arrows trace past updates; panel headers report the period, the cluster held, the loadings $\boldsymbol{\lambda} = (\lambda_C, \lambda_M)$ of the cluster whose wage generated the signal, and the realized signal.

Panel (a) of Figure 8 shows the sequence of choices of a worker who starts in an occupation whose wages load on both skills (Cluster 6) and whose true skills lie in the cognitive-intensive part of the map. After period 0, this worker receives a positive signal in this occupation, so she updates her beliefs on both task dimensions and moves diagonally toward the upper-right quadrant of the skill space. She then receives a positive signal in the cognitive-intensive occupation (Cluster 0), whose wages load only on the cognitive task. Accordingly, she revises her mean cognitive belief upward

and her mean manual belief downward, because the correlation between the two skills turns negative after the first signal. This sequence of signals pushes the worker toward the cognitive-intensive occupations.

Panel (b) shows a similar pattern for a worker who starts in the cognitive-intensive occupation and ends in the manual-intensive occupation, where her comparative advantage lies. Unlike the first worker, she sorts into highly specialized occupations, so her belief moves are purely horizontal or vertical: each signal updates only one skill dimension. For example, after period 0 she receives a negative signal in the cognitive-intensive occupation that lowers only her cognitive belief and moves her to the manual-intensive occupation, where positive signals confirm her manual comparative advantage, and she settles there for the rest of her career.

5.2 The Drivers of Occupational Mobility

I now use the estimated model to decompose the mobility patterns documented in Section 2 into the contributions of learning about comparative advantage and of task-specific human capital accumulation.[19] To do so, I compare the estimated model with two counterfactual choice rules, *learning off* and *experience off*.[20]

Figure 9 shows that the model reproduces both patterns of occupational mobility. Panel (a) shows the hazard of switching by experience: mobility is high at entry, around 16% in the data and 15% in the model, and falls to about 5% after two decades. The model matches both the level and the decay. The profile of the hazard is not targeted in estimation (only the average mobility rate is, through ϱ); it is generated by learning and the accumulation of task-specific experience. Learning shapes this profile in two distinct ways. Early in the career, when uncertainty about comparative advantage is largest, learning generates mobility: without learning, mobility at entry would be about 11% instead of 15%. Later in the career, learning reduces mobility: once workers have learned their skills they have little reason to change occupation, whereas without learning mobility would rise again to about 10% after thirty years. Task-specific human capital instead lowers mobility throughout the career: without it, workers would keep switching at about 15% during their first five years, and mobility would decline only slowly.

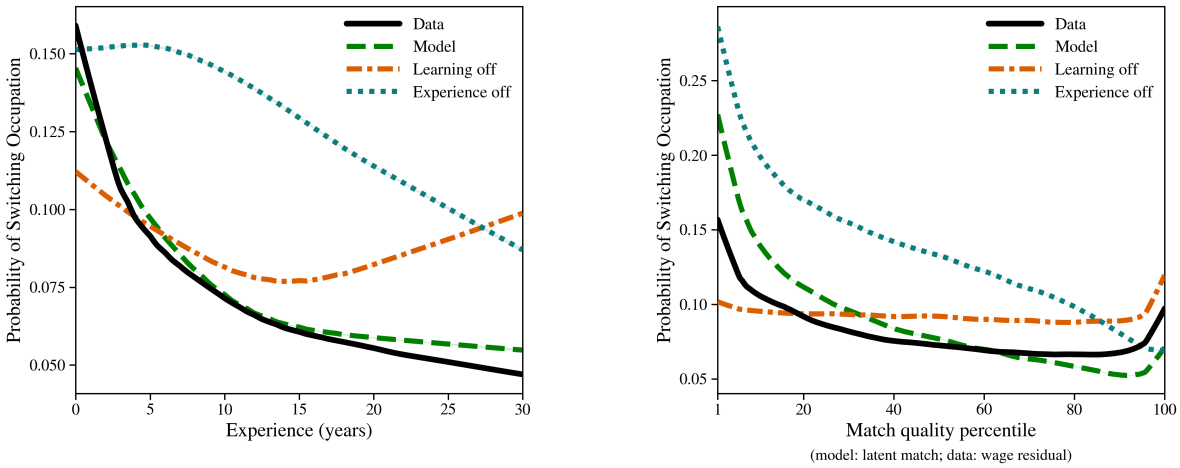
Panel (b) shows the U-shape of occupational mobility (Groes et al. 2015): the prob-

[19] Throughout this section a worker's position in the wage distribution is measured by a worker–occupation fixed effect $\widehat{m}_{ij}$, estimated from

$$w_{ijt} = m_{ij} + \iota_t + f_j(\text{tenure}_{ijt}) + v_{ijt},$$

where ι_t are year effects and f_j is an occupation-specific cubic in occupational tenure. It is the empirical counterpart of the model's match quality $\boldsymbol{\lambda}_j'\boldsymbol{\theta}_i$: it nets out the price of the occupation and the transitory wage shock, leaving the component of the wage specific to the worker–occupation pair, and workers are ranked on it within (occupation, year) cells.

[20] Both counterfactuals change only the rule that workers use to choose their occupation, evaluated at the realized states of the simulated baseline economy, and the model and counterfactual lines plot conditional choice probabilities of switching rather than realized choices. Appendix D.6 describes the two counterfactuals in detail.



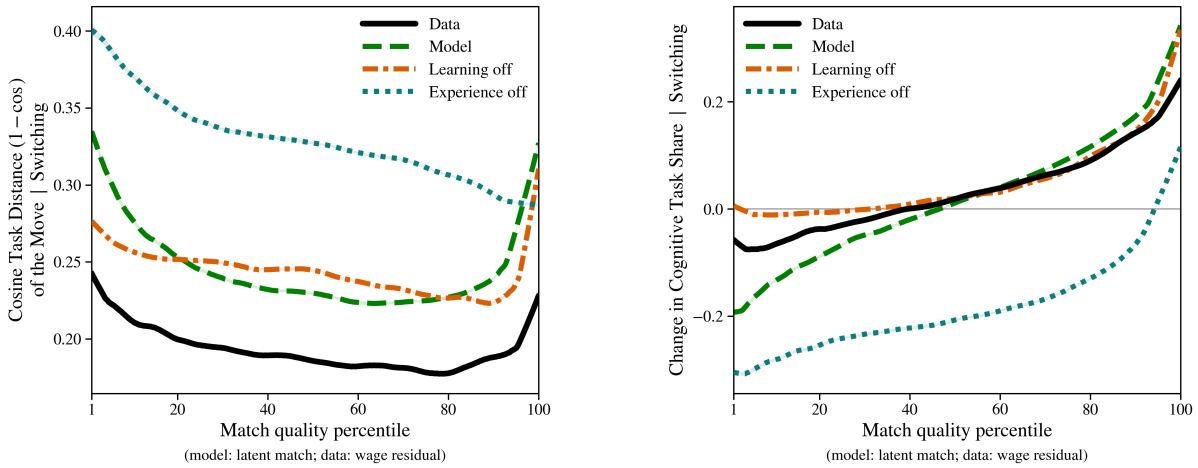
(a) Hazard of switching by experience

(b) Probability of switching by match quality

Figure 9: Occupational mobility, model versus data. Data (black), estimated model (green), *learning off* (orange), and *experience off* (teal). *Panel (a)*: probability of switching occupation by years of potential experience. *Panel (b)*: probability of switching by within-occupation match-quality percentile (model: latent match quality; data: worker–occupation fixed effect).

ability of switching occupation in the next period is higher for workers with low and high match quality within their occupation. The model reproduces the U-shape, although it overpredicts mobility at the very bottom of the distribution. The two counterfactuals show that the two arms of the U-shape have different drivers. Shutting down learning removes the left arm: mobility becomes flat across the match-quality distribution, so the high mobility of poorly matched workers is generated by learning about their comparative advantage. Shutting down experience accumulation instead removes the right arm: mobility falls monotonically with match quality, so the higher mobility of the best-matched workers reflects moves induced by experience accumulation.

Figure 10 turns to the task content of the moves. Panel (a) shows that workers at both ends of the match-quality distribution make larger moves in task space than workers in the middle, and Panel (b) shows the direction of the move: poorly matched workers move toward more manual-intensive occupations ($\Delta\omega_C < 0$), well-matched workers toward more cognitive-intensive ones ($\Delta\omega_C > 0$). The model reproduces both patterns, although it overpredicts the overall level of task moves. Learning drives the moves at the bottom: without learning, poorly matched switchers move less far in task space and no longer reallocate toward manual work. In the model, low match quality implies negative wage signals that make workers update their beliefs about their skills with a specific directionality, as shown in Figure 8. Experience accumulation drives the moves at the top: moving to more cognitive occupations has a higher continuation value, because cognitive experience becomes more important later in the career even for those workers with a manual comparative advantage. Without the experience accumulation channel, switchers would move toward more manual occupations throughout the distribution and the reallocation of well-matched workers toward cognitive occupations would almost disappear. Learning thus generates the downward,



(a) Task-content distance of the switch

(b) Change in cognitive task content

Figure 10: Task content of occupational switches by within-occupation match-quality percentile (model: latent match quality; data: worker–occupation fixed effect), in the data (black), estimated model (green), *learning off* (orange), and *experience off* (teal). *Panel (a)*: task-content distance of the switch, $1 - \cos(\boldsymbol{\omega}_{j_t}, \boldsymbol{\omega}_{j_{t+1}})$, computed on the task weights of the occupation clusters for both the data and the model. *Panel (b)*: change in the cognitive task share of the occupation held, $\Delta\omega_C$; negative values are moves toward more manual-intensive occupations.

task-reallocating moves of poorly matched workers, while experience accumulation generates the upward mobility of well-matched workers toward more cognitive occupations.

Appendix E.1 reports further untargeted moments: the distribution of wages at labor-market entry by education group and occupation of entry, the distribution of wages within and across occupations, the wage–tenure profile of occupational stayers, and the decline of mobility with occupational tenure. One of these moments deserves emphasis. In both the data and the model, the wage distributions of different occupations overlap substantially and share most of their support (Figure 29). The model generates the U-shape and the directionality of occupational mobility together with this overlap precisely because skills are multidimensional: occupations that pay similar wages on average can pay both low and high wages, but to workers with different comparative advantage. Occupational sorting operates on multidimensional skills and tasks, on the dynamic returns to task-specific experience, and on non-wage factors, rather than on a single ordering of occupations from low- to high-paying. This is an empirical improvement over the theory of Groes et al. (2015), in which workers sort on one-dimensional ability and the wage distributions of different occupations have little or no overlap.

The model also allows a finer decomposition of learning itself. Passive learning is the belief updating that follows any wage signal; active learning, or experimentation, operates through the continuation value, when workers choose an occupation in order to learn. In Appendix E.2 (Figures 33 and 34), I show that in the estimated model most of the effect of learning operates through passive learning, while active experimentation plays a smaller but quantitatively relevant role.

Appendix F.1 uses the same counterfactual choice rules to decompose life-cycle wage growth into the contributions of learning about comparative advantage and of the accumulation of task-specific human capital.

6 The Consequences of Skill Uncertainty

I now use the estimated model to quantify the wage cost of imperfect information about skills.[21] In the model, mismatch arises because the initial prior $\mathcal{N}(\boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)})$ is diffuse relative to each worker's realized $\boldsymbol{\theta}_i$: entrants sort on group means rather than on individual comparative advantage. Beliefs are correct on average, because entrants' priors equal the within-group distribution of $\boldsymbol{\theta}_i$, but each worker's skill vector is one draw from that distribution and can lie far from her prior. Throughout this section and the next I report the effects of each counterfactual on output and wages; Appendix F.2 reports the corresponding welfare effects, measured as consumption-equivalent variations. For all the counterfactuals that change workers' information, welfare moves in the same direction as output, with welfare gains between roughly a quarter and a half of the output gains. Welfare and output diverge when the switching cost itself is changed, as I discuss in Section 6.3, and in the forced-exploration exercise of Appendix F.4.

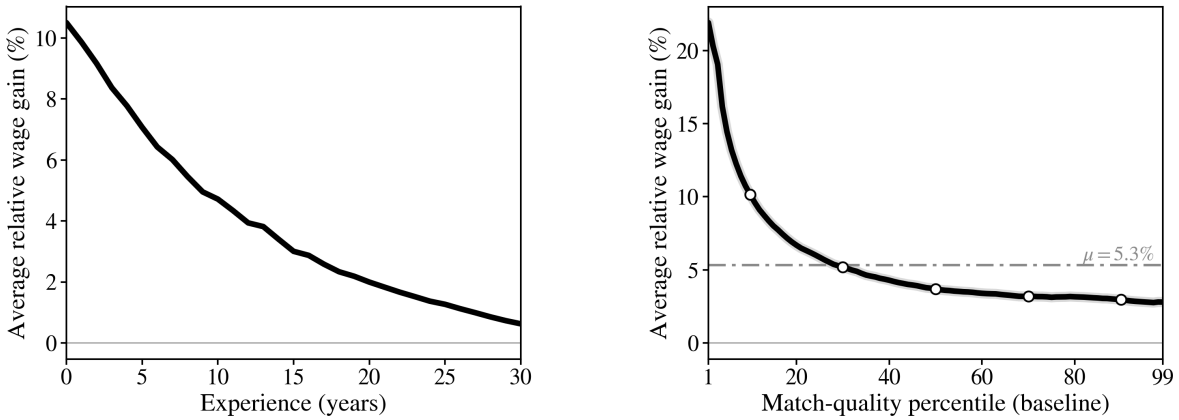
I construct a counterfactual general equilibrium in which workers have perfect information from the start of their careers: each worker's initial beliefs equal her true skills, and she faces no uncertainty.[22] In both the baseline and counterfactual economies, workers face identical switching costs, idiosyncratic taste shocks, and wage noise. The only difference is the distribution of beliefs across simulated workers at labor-market entry. In the stationary equilibrium, giving workers perfect knowledge of their comparative advantage at entry raises aggregate output by 5.1%.[23] Figure 11 then traces the individual wage gains from removing these frictions, over the life cycle and across the match-quality distribution within occupation.

This output gain reflects the misallocation of workers to occupations that do not match their true comparative advantage under imperfect information. The wage gains are concentrated at the start of the career and flatten over the life cycle. At labor-market entry, the average worker would earn about 10.5% more if she knew her true skills. This gap narrows as workers learn and change occupation accordingly, falling

[21] All stationary counterfactual economies in this and the following sections are computed by simulating $N = 200{,}000$ workers for 400 periods and discarding the first 200 periods as burn-in, so that statistics are computed on the stationary distribution.

[22] In practice, $\tilde{\boldsymbol{\theta}}_i = \boldsymbol{\theta}_i$ and $\tilde{\sigma}^2_{ik} \approx 0$ for all workers i and skill dimensions k.

[23] Aggregate output in the stationary equilibrium is $Y = \left(\sum_{j \in \mathcal{J}} \alpha_j Y_j^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$, where $Y_j = \int \exp(\varepsilon)\, Y_j(x)\, \mathbf{1}_{\{d=j\}}\, \pi^*(x)\, dx\, dF_j(\varepsilon)$ is occupation j's aggregate output, $Y_j(x)$ is individual output from the task technology (5) net of the idiosyncratic shock, and π^* is the stationary distribution of worker states. The mismatch cost is computed as $(Y^{PI} - Y^{II})/Y^{II}$, where Y^{PI} and Y^{II} are aggregate output in the perfect and imperfect information stationary equilibria, respectively, both solved at the same model parameters.



(a) Wage gains over the life cycle

(b) Wage gains by match-quality percentile

Figure 11: Wage gains from removing information frictions. *Panel (a)*: average relative wage gain over the life cycle (horizontal axis: years of potential experience; vertical axis: percent wage gain). *Panel (b)*: average relative wage gain by the worker's within-occupation match-quality percentile in the baseline economy (horizontal axis); the dashed horizontal line marks the unconditional mean gain, 5.3%. Shaded bands are bootstrap confidence intervals.

to about 2% after twenty years of experience and fading toward zero (about 0.6%) after 30 years. Mismatch driven by imperfect information is largely an early-career phenomenon: the cost of imperfect information is borne mostly in the first decade, when there is more skill uncertainty and workers are still discovering their comparative advantage, and it is progressively resolved as learning and re-sorting bring imperfect-information workers closer to their perfect-information counterparts. Because learning, imperfectly transferable experience, and switching costs create strong path dependence, some imperfect-information workers never fully converge to their optimal career, although on average the residual gain closes gradually.

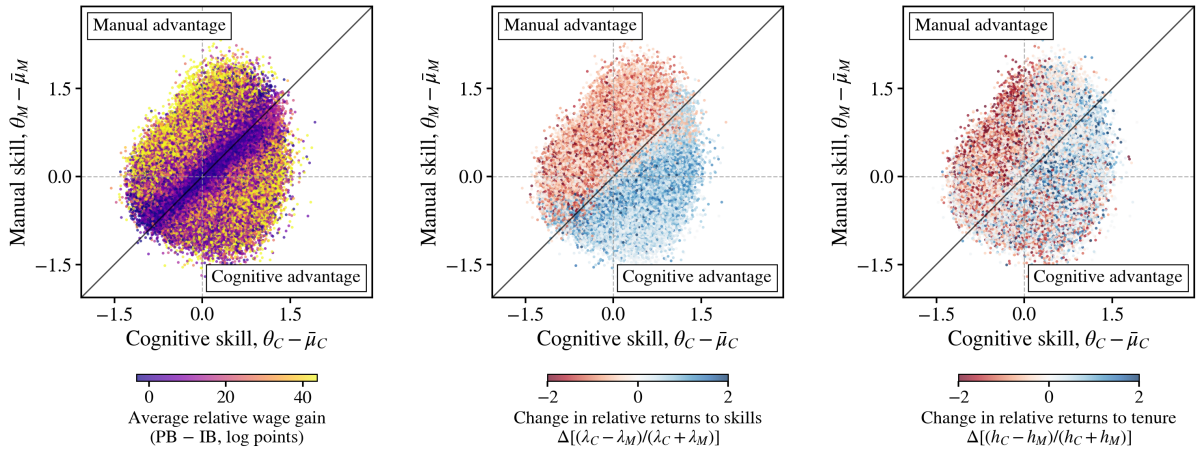
Decomposing the average life-cycle wage gain between the two economies, I find that it comes almost entirely from better matching on comparative advantage rather than from experience accumulated along the right dimension: the total gain and its matching component decay together, from about 10.5% at entry toward zero late in the career.[24] The full decomposition is reported in Figure 35 in the Appendix.

Gains are also largest for the workers who are worst matched in the baseline (Panel (b)). The gain falls steeply in match quality: workers in the bottom percentile of their occupation's match-quality distribution would earn over 20% more under perfect information, against roughly 3% at the top, with an unconditional mean of 5.3%. Perfect information therefore does not raise all wages proportionately: it repairs the bottom of the match-quality distribution, where imperfect beliefs have led workers into occupations that fit them poorly.

[24]The residual can change because, although I hold luck fixed, the variance of the idiosyncratic shocks changes, as it is also driven by the occupational choice.

6.1 Who Benefits Most, and Why

Not all workers reallocate under perfect information. The model includes other important determinants of occupational choice—idiosyncratic taste shocks, compensating differentials, and switching costs—that have been shown to matter for labor mobility. Under perfect information, a sizeable share of workers would choose a different initial occupation, and many more would switch occupations at some point during their career. Among those who do reallocate, the wage gains are substantial, and Figure 12 shows both who they are and how they move.



(a) Average wage gain

(b) Change in relative returns to skills

(c) Change in relative returns to tenure

Figure 12: Wage gains and reallocation in the space of true skills when information frictions are removed. In each panel a point is a worker who chooses a different occupation under perfect than under imperfect information, positioned by her true log cognitive skill θ_C (horizontal axis) and true log manual skill θ_M (vertical axis), both measured relative to the population mean $\bar{\mu}_k = \sum_g \pi_g \mu_{gk}$; the solid line is the 45° line $\theta_C - \bar{\mu}_C = \theta_M - \bar{\mu}_M$. *Panel (a)*: color is the worker's average relative wage gain over the life cycle, in log points. *Panel (b)*: color is the change, from the imperfect- to the perfect-information economy, in the relative returns to cognitive versus manual skills of the occupation held. *Panel (c)*: the change in the relative returns to cognitive versus manual tenure, evaluated at the worker's task tenures in each economy (periods in which both task tenures are below one year are excluded). In Panels (b) and (c) blue denotes a move toward a more cognitive occupation and red a move toward a more manual one.

Panel (a) plots the life-cycle-average wage gain of each reallocated worker in the space of true skills. Specialists, workers with a strong comparative advantage in one dimension, have the largest gains from perfect information, while generalists, workers near the 45° line with similar cognitive and manual skills, gain relatively little. Specialists have high returns to sorting correctly on skills: a worker with strong cognitive comparative advantage loses if she mistakenly enters a manual-intensive occupation, or even a more balanced occupation. Generalists, in contrast, are less likely to be mismatched on comparative advantage, so the cost of sorting "incorrectly" is smaller than for specialists. Information treatments about skills, and career guidance more generally, may therefore deliver the largest wage gains to workers with a strong comparative

advantage in one skill dimension.

Panels (b) and (c) show the mechanism: coloring the same workers by the direction of their reallocation reveals a sharp separation along the 45° line. Workers with $\theta_C - \bar{\mu}_C > \theta_M - \bar{\mu}_M$ (cognitive specialists, below the 45° line) systematically move to more cognitive occupations when given perfect information, while workers with $\theta_M - \bar{\mu}_M > \theta_C - \bar{\mu}_C$ move to more manual ones. Panel (b) shows this for the relative returns to skills $\boldsymbol{\lambda}_j$; Panel (c) shows an analogous but weaker and noisier pattern for the relative returns to tenure, driven mainly by workers with manual comparative advantage, who move toward occupations with higher returns to manual experience. This suggests that while the contribution of experience to the aggregate wage gain is small, task-specific experience still plays a role in generating the wage costs that keep workers from switching to the right occupation as they discover their true comparative advantage.

6.2 The Effects of Information Interventions

I now use the model to assess how much of the output and wage losses can be recovered by implementing realistic information interventions, and how their effectiveness changes when they are provided at different stages of the career. I take the field experiment of Kiss et al. (2023) as the empirical anchor. The paper studies two related information interventions; I calibrate the model intervention to the effects estimated in the first experiment, in which young jobseekers receive a short computerized assessment covering numeracy, communication, and concept formation skills. The information treatment consists of individualized feedback about where the treated worker stands in the skill distribution of treated individuals. This information treatment is therefore informative about workers' comparative advantage in skills. Treated workers update their beliefs about their own skills and re-direct their search toward occupations that better match their assessed comparative advantage. A desirable feature of this class of intervention is that it is relatively cheap to deploy.[25] The estimated treatment effect is a 13.7 percentage point increase in the share of workers whose stated career aspiration is aligned with their measured comparative advantage.

I embed this intervention in the model as follows. At entry ($t = 0$), before making her first occupational choice, each worker i receives K independent noisy Gaussian signals, one for each latent skill dimension (in the estimated model, $K = 2$, corresponding to the cognitive and manual dimensions):

$$s^0_{i,k} = \theta_{i,k} + \zeta_{i,k}, \qquad \zeta_{i,k} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_\zeta), \qquad k = 1, \ldots, K. \tag{41}$$

[25] The cost of the computerized assessment and information treatment reported by Kiss et al. (2023) is $45.74 USD PPP per worker. I convert it to 2020 EUR via the OECD USD–EUR PPP rate ($1 = € 0.92), yielding € 42.08; deflating to 1995 real EUR via the Portuguese CPI (2020–to–1995 deflator of 0.65) gives € 27.35 per worker, which is the cost of the information treatment per worker, expressed in the model and data wage units (real hourly EUR 1995) used in this paper.

Stacking the signals into $\boldsymbol{s}_i^0 = (s_{i,1}^0, \ldots, s_{i,K}^0)'$, I obtain $\boldsymbol{s}_i^0 = \boldsymbol{\theta}_i + \boldsymbol{\zeta}_i$ with $\boldsymbol{\zeta}_i \sim \mathcal{N}(\mathbf{0}, \sigma_\zeta^2 \boldsymbol{I}_K)$. Because the prior $\boldsymbol{\theta}_i \mid g \sim \mathcal{N}(\boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)})$ and the signal vector are jointly Gaussian, the post-assessment belief is the standard closed-form Gaussian update

$$\tilde{\boldsymbol{\theta}}_{i0} = \boldsymbol{\mu}_{g(i)} + \boldsymbol{\Sigma}_{g(i)} \left(\boldsymbol{\Sigma}_{g(i)} + \sigma_\zeta^2 \boldsymbol{I}_K \right)^{-1} \left(\boldsymbol{s}_i^0 - \boldsymbol{\mu}_{g(i)} \right), \tag{42}$$

$$\tilde{\boldsymbol{\Sigma}}_{i0} = \boldsymbol{\Sigma}_{g(i)} - \boldsymbol{\Sigma}_{g(i)} \left(\boldsymbol{\Sigma}_{g(i)} + \sigma_\zeta^2 \boldsymbol{I}_K \right)^{-1} \boldsymbol{\Sigma}_{g(i)}. \tag{43}$$

The worker is myopic about the arrival of this signal and uses the same value function $EV(\cdot)$ as in the baseline economy. The signal variance σ_ζ^2 is common across all skill dimensions and is the only free parameter of the experiment. I calibrate the noise of the signal so that a signal delivered before labor-market entry raises the share of workers whose beliefs about their comparative advantage match their true comparative advantage by 13.7 percentage points, the average treatment effect in Kiss et al. (2023).[26] The target is the model counterpart: the difference between post- and pre-assessment alignment rates.[27] The calibration yields $\sigma_\zeta \approx 0.606$. I then feed this value into the estimated model and re-simulate the economy, holding all other parameters at their estimated values.

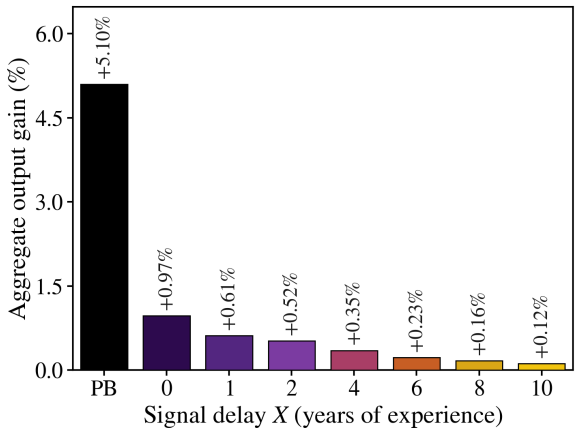


Figure 13: Aggregate output gain (percent) relative to the baseline economy when the calibrated skill assessment is delivered $X \in \{0, 1, 2, 4, 6, 8, 10\}$ years after labor-market entry, holding the calibrated signal precision σ_ζ^2 fixed at its at-entry value. Each bar is a separately solved stationary equilibrium. The leftmost bar (perfect information, PB in the figure) is the output gain from giving workers perfect information about their skills at entry, the upper bound the assessment could in principle recover.

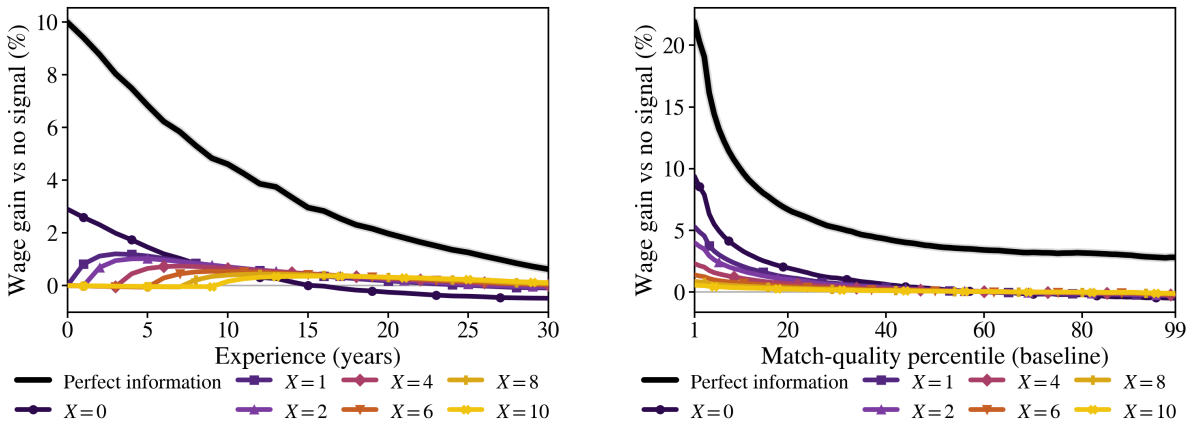
Relative to the baseline economy, providing entrants with the calibrated skill signal

[26] I follow their Section 2.6 definition. A large reference pool of latent types $\boldsymbol{\theta}$ is drawn from the mixture distribution of education groups and used to compute the marginal quintile cut-points of each skill dimension. A worker is then assigned to one of five quintile bins on each skill according to her true type, her prior mean, and her post-assessment posterior mean. Her belief is aligned if the skill in which her believed quintile is higher coincides with the skill in which her true quintile is higher; workers whose true type falls in the same bin on both skills (no unique comparative advantage) are dropped, as in Kiss et al. (2023).

[27] The function mapping signal noise to the average treatment effect is monotone, as the posterior collapses to the truth as $\sigma_\zeta^2 \to 0$ and to the prior as $\sigma_\zeta^2 \to \infty$, so this moment identifies σ_ζ^2.

raises aggregate output by 0.97%, that is, around 19% of the total misallocation generated by information frictions about skills. The wage gains arising from the information treatment are strongly front-loaded in the career: average wages are about 2.9% above the baseline at entry, then decline as better-matched entrants accumulate experience and baseline workers catch up after updating their beliefs and sorting accordingly, crossing zero after roughly fifteen years and settling mildly negative (-0.5%) thereafter (Figure 14, Panel (a)). These gains come almost entirely from better matching on skills at the first occupational choice; the late-career reduction in the gap between the two economies reflects cross-task experience accumulation (Appendix F.1), as the policy induces early specialization and so reduces some of the exploratory experience that baseline workers accumulate under imperfect information. Imperfect information thus generates mismatch, but for workers who eventually reach their ideal career path, the cross-returns to experience in other tasks can be beneficial.

The treatment does not help everyone equally, and Panel (b) in Figure 14 shows how the gains are distributed. The wage gains are concentrated at the bottom of the match quality distribution within occupation: workers in the lowest percentiles in the baseline economy, who are more likely to be mismatched, gain about 9%; the gain fades to roughly zero by the median. A single noisy assessment provided early in the career recovers about two fifths of the perfect-information gain for the worst-matched workers (9% against 22%), and essentially nothing for the well-matched.



(a) Wage gains over the life cycle

(b) Wage gains by match-quality percentile

Figure 14: Wage gains from the skill assessment, delivered $X \in \{0, 1, 2, 4, 6, 8, 10\}$ years after labor-market entry, holding the calibrated signal precision σ_ζ^2 fixed. *Panel (a)*: life-cycle profile of average wage gains (percent) relative to the no-signal baseline, by delay, with the perfect-information counterfactual (solid black) as a benchmark; horizontal axis is years of experience. *Panel (b)*: average wage gain relative to the no-signal baseline by the worker's within-occupation match-quality percentile in the baseline economy, for each delay, with the perfect-information counterfactual (solid black) as a benchmark.

To see how the value of the policy depends on its timing, I repeat the experiment delivering the signal in (41) at $t \in \{0, 1, 2, 4, 6, 8, 10\}$ periods after entry, holding its precision σ_ζ^2 fixed and keeping workers myopic about when it will arrive. The later the assessment comes, the lower are the realized output and wage gains: the output

gain is largest at entry (Figure 13), drops by roughly half within the first two years, and is nearly gone by the end of the first decade. A delayed assessment generates only a small hump once it arrives, mainly because imperfectly transferable task-specific experience creates lock-in: even when a worker's skills would be more productive in another occupation, switching costs her part of her task-specific experience. This implicit switching cost rises with labor-market experience and directly erodes the effect of a delayed signal.

In Appendix F.4 I study an experiment that directs each labor-market entrant, for one period only, into the occupation that is most informative about her comparative advantage, after which she chooses freely. The forced exploration reduces wages early in the career but improves learning and sorting from the second period onward. The large initial reduction is driven not only by forcing workers into occupations for which they are poorly matched, but also by the downward-sloping demand curve for occupation-specific goods. Forcing everyone to move into their most informative occupation pushes prices down in the informative occupations while raising prices elsewhere. Overall, aggregate output rises by 1.11%.

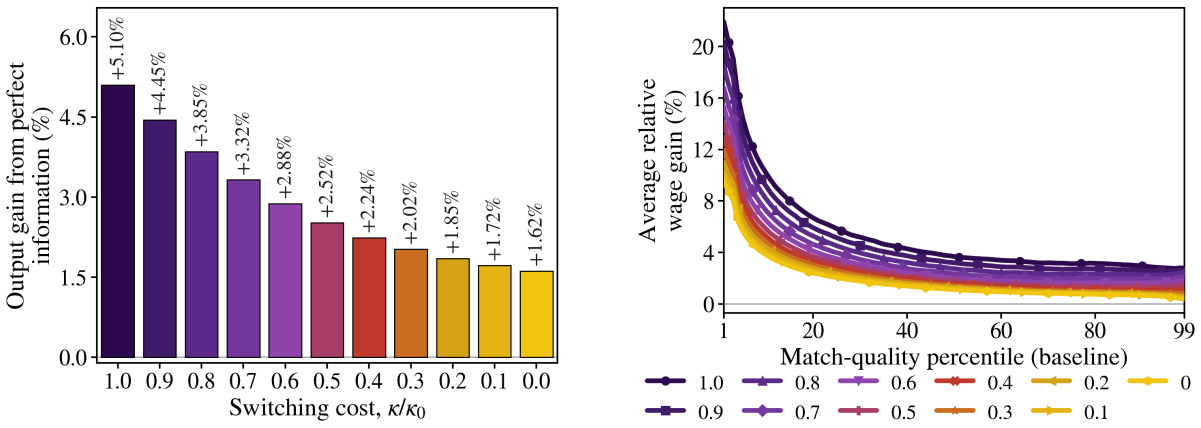
6.3 Switching Costs and Information Frictions

The cost of imperfect information depends on how easily workers can switch once they learn where their best fit is. Absent any switching cost, a worker who discovers she is mismatched could simply switch at any point in her career. Switching costs and information frictions are thus complementary sources of misallocation, and I now study how they interact. I reduce the switching cost κ from its estimated value κ_0 to zero in steps of ten percent of κ_0, re-solving the stationary equilibrium at each level, and I run two comparisons.

The first holds the switching cost fixed and compares the imperfect-information economy to its perfect-information counterpart; in other words, I measure the cost of information frictions at each level of mobility friction. This cost falls monotonically as mobility becomes cheaper (Figure 15): the output gap between the perfect- and imperfect-information economies declines from about 5.1% at the estimated switching cost to under 2% when moving is free (Panel (a)), and the gain from perfect information shrinks at every point of the match-quality distribution as the switching cost falls (Panel (b)). Lower moving costs let workers switch once they learn their types, so information frictions generate less misallocation.

The second comparison is across the imperfect-information economies themselves, and asks how aggregate output responds to the switching cost that workers actually face. Here output is not monotonically decreasing in the switching cost (Figure 16): it first rises as mobility becomes cheaper, with a peak when switching costs are 70% of the estimated value, and then falls, turning negative and reaching a loss of about 2.4% when switching is completely free[28].

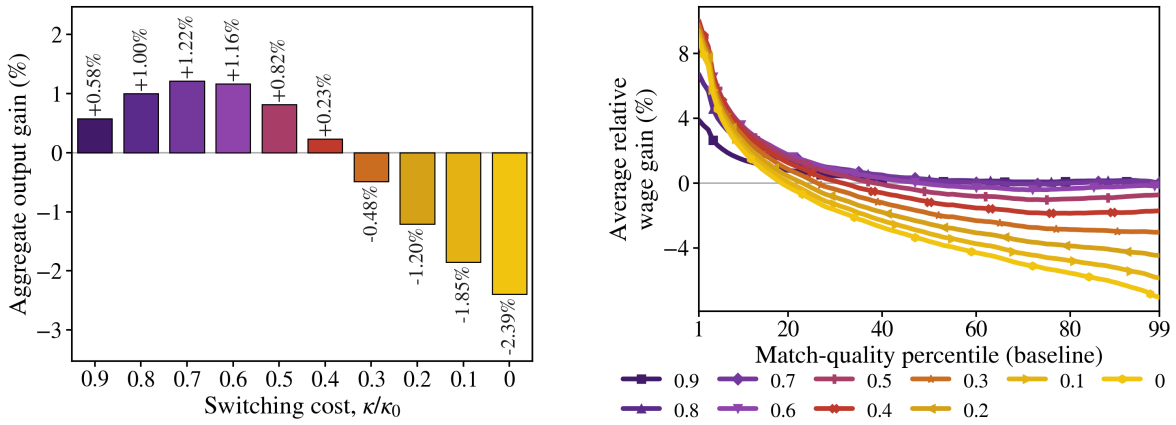
[28] In these counterfactuals, moving still carries the implicit cost of forgone task-specific human capital.



(a) Output gain from perfect information

(b) Wage gain from perfect information by match quality

Figure 15: The cost of information frictions at different levels of the switching cost. At each value of κ/κ_0 both the imperfect- and the perfect-information stationary equilibria are re-solved, and the figure reports the gap between them. *Panel (a)*: aggregate output gain from moving to perfect information, $(Y^{PI} - Y^{II})/Y^{II}$, in percent, by κ/κ_0. *Panel (b)*: average relative wage gain from perfect information by the worker's within-occupation match-quality percentile, by κ/κ_0. The corresponding life-cycle wage profiles are in Appendix F.3.



(a) Aggregate output

(b) Wage gains by match quality

Figure 16: Aggregate output and wage gains across imperfect-information economies with different switching costs. Each curve or bar is a separately solved stationary equilibrium, indexed by the switching cost as a fraction of its estimated value ($\kappa/\kappa_0 = 1$ is the baseline, 0 is costless mobility). *Panel (a)*: percent change in CES aggregate output relative to the baseline economy. *Panel (b)*: average relative wage gain over the baseline by the worker's within-occupation match-quality percentile, by κ/κ_0. The corresponding life-cycle wage profiles are in Appendix F.3.

A small reduction in the switching cost raises output, because workers can freely reallocate toward occupations in which they have a comparative advantage as they learn. They often benefit from the returns to their accumulated experience along the way: an occupation may expose a worker to tasks she would not have performed elsewhere, building experience that remains valuable in the occupation she eventually moves to. A large reduction, however, makes workers excessively responsive to idiosyncratic taste shocks. This excessive churning pushes them into occupations where their skills and human capital are not particularly productive. Panel (b) of Figure 16 shows who bears the resulting losses: wages fall for the well-matched workers, at the top of the within-occupation match-quality distribution. The poorly matched workers gain from cheaper mobility at almost every switching-cost level; but a worker who was already well matched has nothing to gain from moving. The aggregate output loss from excessive churning is therefore driven by the top of the match-quality distribution. The two comparisons offer different insights into the nature and causes of misallocation: a lower cost of moving always reduces the misallocation generated by information frictions, but some friction is useful because it leads workers to accumulate task-specific human capital.

Output is not welfare, and the two diverge in this experiment. Appendix F.2 reports, for each counterfactual, the consumption-equivalent welfare gain: the uniform and permanent wage increase that would make the average worker in the baseline economy as well off as in the counterfactual. This measure prices everything workers value in the model, that is, expected wages, amenities, the switching cost itself, and the option value of the taste shocks, whereas output counts only production. Lowering the switching cost raises welfare monotonically, by 4.2% at $\kappa/\kappa_0 = 0.7$ and by 61% when mobility is free, even where output falls (Figure 39). The switching cost is a utility cost paid by every mover, and its estimated value of 1.10 utils is equivalent to a wage change of about 110 log points, so removing it is a direct gain to movers; in addition, workers re-sort on amenities and taste shocks that output does not count. The lower output at low switching costs therefore does not indicate that the additional mobility is inefficient from the workers' point of view: workers internalize the task-specific experience they forgo and still choose to move, because the switching-cost savings and the non-wage components of their choice are worth more to them. The welfare gain from perfect information, in contrast, falls with the switching cost like the output gain, from 2.0% at the estimated switching cost to 0.5% when moving is free.

I close this subsection by providing two practical examples of the frictions the switching cost captures. First, if occupations are concentrated in different local labor markets then workers have to change location in order to switch occupation, so part of κ_0 may reflect the monetary and psychic cost of geographic mobility (Kennan and Walker 2011). Reducing switching costs in this case can be thought of as providing moving vouchers to mismatched workers. Second, reallocating across occupations often requires passing through an unemployment spell, as in the island and sectoral-reallocation tradition in which workers move by first becoming unemployed (Lucas

and Prescott 1974; Chodorow-Reich and Wieland 2020), what Pilossoph (2012) calls "move unemployment" and Alvarez and Shimer (2011) call "search unemployment". In this case, κ_0 represents the cost of being unemployed, and switching-cost reductions can be read as changes to unemployment benefits, such as their duration or level.

7 Artificial Intelligence and Skill Uncertainty

The estimated model is well suited to studying the interaction of technological change and skill uncertainty. I focus on the job transformation (Freund and Mann 2025) induced by Large Language Model (LLM) adoption, and I model it as a task-biased shock that lowers the importance of cognitive tasks across occupations. This affects the time workers spend on cognitive tasks in proportion to each occupation's exposure to large language models. I simulate counterfactual scenarios with and without imperfect information to show how output and the cost of mismatch driven by information frictions change not only in the long run, but also on impact and along the transition path.

7.1 AI Shock Design

I measure each occupation's exposure to AI following the framework of Eloundou et al. (2024), which treats a task as exposed if access to a large language model would cut the time needed to complete it by at least half. I apply the rubric to the 296 Level-3 skills of the ESCO classification, assigning each skill to one of four exposure categories with weights $e_{E0} = 0$, $e_{E1} = 1$, and $e_{E2} = e_{E3} = 0.5$. Manual skills are unexposed ($e = 0$), since language models cannot perform physical tasks. Averaging skill exposure by ESCO importance within each occupation and aggregating to the $|\mathcal{J}| = 8$ clusters defines a measure of cognitive exposure $x_{jC} \in [0, 1]$ for each cluster, with $x_{jM} = 0$ implying that manual tasks are not exposed to LLMs.

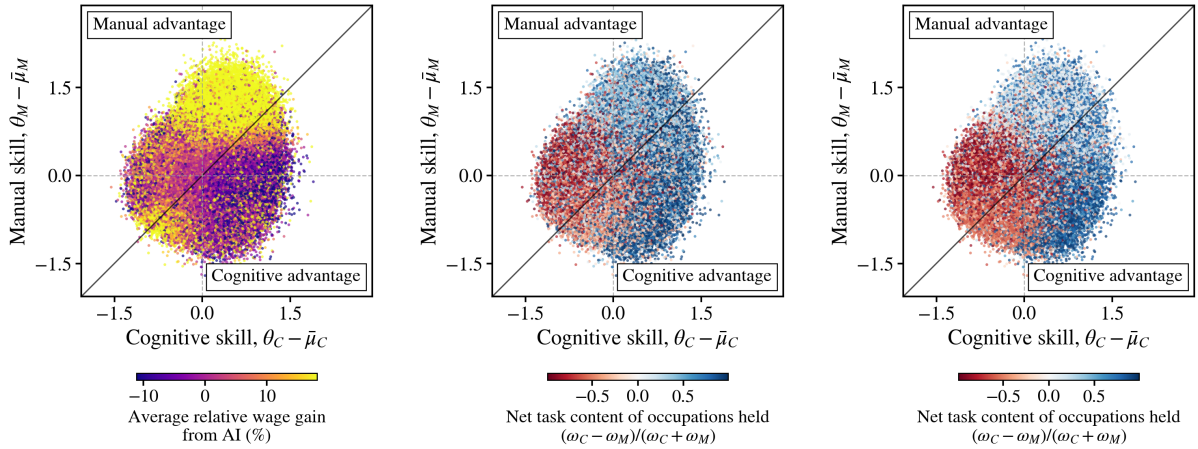
AI lowers the time spent on the cognitive task in proportion to exposure. Through the model's optimal time allocation, the post-shock task weights are

$$\omega_{jC}^{\text{AI}} = \omega_{jC}\,(1 - x_{jC}), \qquad \omega_{jM}^{\text{AI}} = \omega_{jM}, \tag{44}$$

renormalized so that $\omega_{jC}^{\text{AI}} + \omega_{jM}^{\text{AI}} = 1$, so the manual weight rises mechanically. Because the effective wage parameters scale with the task weights ($\lambda_{jk} = \omega_{jk} a_{jk}$ and $h_j(\boldsymbol{\tau}) = \sum_k \omega_{jk} h_{jk}(\tau_k)$), the shock scales the skill loadings and the returns to experience by the common ratio $\omega_{jk}^{\text{AI}}/\omega_{jk}$. Lower cognitive weight therefore reduces the return to cognitive skills and experience. Additionally, because task-specific experience accumulates in proportion to the task weights (equation (10), now with $\boldsymbol{\omega}_j^{\text{AI}}$), a decrease in cognitive task importance implies less time spent on cognitive tasks and therefore slower accumulation of human capital along that dimension. Manual-intensive clusters are left largely untouched.

7.2 The Direct Effects of AI

I compare the pre- and post-shock stationary equilibria under imperfect information. AI raises aggregate productivity, but its wage effects are heterogeneous and polarizing. Figure 17 maps these effects onto the space of true skills and reveals the mechanism.



(a) Average wage gain from AI

(b) Net task content of occupations held, pre-AI

(c) Net task content of occupations held, post-AI

Figure 17: The direct effects of AI in the space of true skills. In each panel a point is a simulated worker placed at her true log cognitive skill θ_C (horizontal axis) and true log manual skill θ_M (vertical axis), both measured relative to the population mean $\bar{\mu}_k = \sum_g \pi_g \mu_{gk}$; the solid line is the 45° line $\theta_C - \bar{\mu}_C = \theta_M - \bar{\mu}_M$, which separates cognitive specialists (below) from manual specialists (above). *Panel (a)*: color is the worker's average relative wage gain from AI over the life cycle, in percent (yellow = gains, dark = losses). *Panels (b) and (c)*: color is the average net task comparative advantage $(\omega_C - \omega_M)/(\omega_C + \omega_M)$ of the occupations the worker holds over her career—blue = cognitive-intensive, red = manual-intensive—in the pre-AI (b) and post-AI (c) stationary equilibrium.

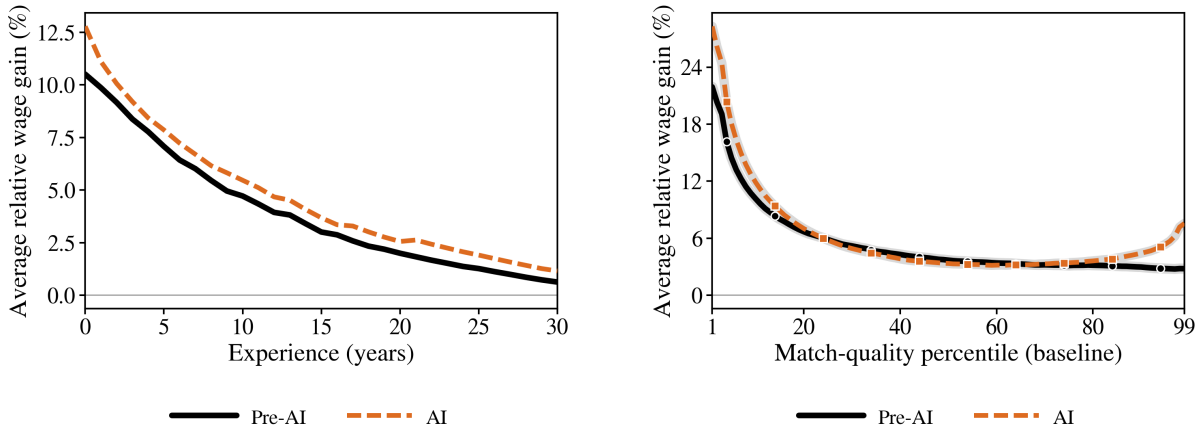
Panel (a) plots the average life-cycle wage gain from AI for each worker at her true skills. Manual specialists, high-ability manual workers above the 45° line where $\theta_M - \bar{\mu}_M > \theta_C - \bar{\mu}_C$, gain the most, because AI raises the relative return to their manual skills. Cognitive specialists in the lower-right region lose as the tasks they are best at become less important. Generalists near the 45° line are roughly unaffected.

Panels (b) and (c) trace these wage effects to occupational reallocation, showing the net task content of the occupations a worker holds over the life cycle before and after the shock. Workers, especially those with a manual comparative advantage, sort into more manual-intensive occupations after AI (the red region expands). Cognitive specialists instead remain in cognitive-intensive occupations but earn lower returns there. High-ability generalists, by contrast, benefit by moving toward more balanced occupations. These effects in the skill space aggregate into the familiar polarization pattern of the routinization literature (Autor et al. 2003): across the baseline wage distribution the wage gains are U-shaped—workers in the middle lose while those at the bottom and top gain—and employment shifts out of cognitive-intensive clusters and into manual-intensive ones, whose occupation prices rise in general equilibrium (Figure 38 in the

Appendix).

7.3 AI and the Implications of Skill Uncertainty

The analysis so far compares stationary equilibria and therefore captures the long-run incidence of AI in a world with information frictions. I now ask how AI changes the implications of skill uncertainty itself, that is, how it reshapes the wage cost of imperfect information studied in Section 6. Figure 18 recomputes the wage gain from giving workers perfect information about their skills, once in the baseline economy and once in the post-AI economy, along the same two dimensions as Figure 11: over the life cycle and across the match-quality distribution. In the aggregate, the output loss generated by imperfect information about skills increases from 5.1% in the baseline to 5.8% under AI.



(a) Wage gains over the life cycle: pre-AI vs AI

(b) Wage gains by match-quality percentile: pre-AI vs AI

Figure 18: The wage cost of skill uncertainty before and after AI. Both panels report the average relative wage gain from moving to perfect information, computed separately in the pre-AI (black, solid) and post-AI (orange, dashed) stationary equilibria. *Panel (a)*: by years of potential experience. *Panel (b)*: by the worker's within-occupation match-quality percentile in the corresponding baseline economy. Shaded bands are bootstrap confidence intervals.

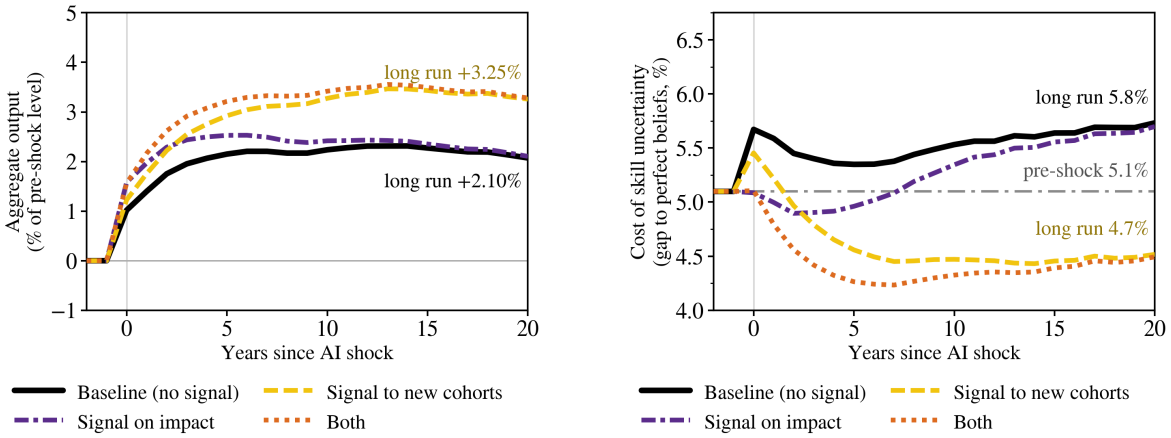
AI raises the cost of skill uncertainty over the whole career path. Panel (a) shows that the wage gain from perfect information is higher in the economy affected by the AI shock at every level of experience: at labor-market entry it rises from about 10.5% to about 12.8%, and the wedge between the two curves persists as experience accumulates. By raising the dispersion of returns across occupations, AI makes correct sorting more valuable, so the cost of holding the wrong occupation under imperfect beliefs is higher. Panel (b) shows who accounts for this increase. The extra cost of skill uncertainty under AI is concentrated at the two ends of the match-quality distribution. For the worst-matched workers within occupation, the gain from perfect information rises from about 22% to about 28%. At the top of the distribution AI also increases the gains from removing information frictions: well-matched workers within occupation now

gain 7.6%, compared with 3% in the pre-AI scenario.

Appendix F.1 repeats the decomposition of life-cycle wage growth into learning and human capital accumulation for the post-AI economy.

7.4 Output and Misallocation along the Transition Path

I now compute the transition path from the pre-shock to the post-shock equilibrium[29], which shows the consequences of information frictions on impact and during the adjustment. I also assess the effectiveness of the same calibrated information treatment in reducing misallocation over the transition path. I model the AI shock as an MIT shock: it is an unanticipated shock and workers have perfect foresight over the entire subsequent path of prices and technology. The sequence of equilibrium prices along the transition path is obtained via a backward–forward shooting algorithm, keeping the CES weights fixed at their calibrated values and the aggregate price level as a numeraire in each period. Appendix B.3 describes the shooting algorithm in detail.



(a) Aggregate output

(b) Cost of skill uncertainty

Figure 19: The AI transition with and without skill signals, by years since the AI shock (horizontal axis). Baseline imperfect-information economy with no signal (black, solid); skill signal delivered to all workers on impact (purple, dash-dotted); skill signal delivered to every new cohort at entry (yellow, dashed); both (orange, dotted). *Panel (a)*: aggregate output, percent deviation from the pre-shock steady state. *Panel (b)*: cost of skill uncertainty, the output gap to the perfect-information economy (percent); the dash-dotted gray line marks its pre-shock level. Labels report the pre-shock and long-run values.

Two results stand out. First, the AI shock raises aggregate output on impact and along the entire transition (Figure 19, Panel (a)). Aggregate output jumps to about +1.0% above its pre-shock level on impact, keeps rising as workers reallocate across occupations, peaks at roughly +2.3% after about fourteen years, and settles at its new long-run level of about +2.1%. Output does not jump to its new long-run level on impact because experienced workers are locked into career paths chosen under the

[29]The transition path is simulated with the same population of $N = 200{,}000$ workers used for the stationary counterfactuals, and its pre- and post-shock endpoints correspond to the pre-AI and post-AI stationary equilibria of Section 7.3. Output along the path is measured relative to the pre-AI stationary output level.

old task content of occupations: part of the gain materializes only as new cohorts, who sort on the new task content from entry, replace them. Second, AI increases the misallocation generated by information frictions about skills (Figure 19, Panel (b)). The output gain from moving to perfect information jumps from about 5.1% before the shock to about 5.7% on impact, as the shock makes many existing matches worse, falls back to about 5.35% within five years as incumbents learn and reallocate, and then rises again, as cohorts enter the new economy, to roughly 5.8% in the new long run.

I then ask how effective the information treatment calibrated in Section 6.2 is along the transition, comparing two ways of delivering it. A one-shot skill signal delivered to all workers on impact raises output on impact from about $+1.0\%$ to about $+1.6\%$ and temporarily pushes the cost of skill uncertainty below its pre-shock level, but its effect is transitory: as treated cohorts retire, both output and the cost of skill uncertainty converge back to the untreated path. A skill signal delivered to every new cohort at entry instead has a small effect on impact, when only the first cohort is treated, but its effect builds up as treated cohorts replace untreated ones. In the long run it raises output to about $+3.25\%$ above the pre-shock level, 1.1 percentage points more than the untreated economy, and it lowers the cost of skill uncertainty from about 5.8% to about 4.7%, closing about a fifth of the gap. Delivering both signals combines the two profiles: the economy follows the one-shot path on impact and the entry-signal path in the long run. From a policy perspective, these results suggest that information provision is more valuable in the economy transformed by AI than in the pre-AI economy, since the long-run output gain of the entry signal is larger than the 0.97% it delivers before the shock. The timing of the intervention matters: a one-shot assessment of incumbent workers smooths the reallocation on impact, while a permanent assessment of new entrants delivers most of the long-run gains.

8 Conclusion

In this paper, I develop and estimate a general equilibrium dynamic Roy model in which workers learn about their multidimensional skills and accumulate task-specific human capital. Using Portuguese administrative data, I extend the occupational mobility facts of Groes et al. (2015) to the task content of moves and to the wage change upon switching, by the worker's wage position in the origin occupation. I find that changes in task content upon switching are also U-shaped in workers' positions within their origin occupations and directional in task space. Moreover, switching raises the wages of poorly matched workers the most. Through the lens of the model, the upward mobility of well-matched workers reflects experience accumulation, whereas the downward, task-reallocating moves of poorly matched workers reflect learning about comparative advantage.

These findings suggest that mismatched workers could benefit from having more precise beliefs about their skills earlier in their career. Removing information frictions would raise aggregate output by 5.1%. Individual wage gains are largest at entry, fade

to roughly zero over the career, and are concentrated among the worst-matched workers within occupations and among specialists. Policy can recover part of this loss. An information treatment calibrated to Kiss et al. (2023) and delivered at entry raises aggregate output by 0.97%, but its effect decays quickly when it comes later in the career, because of switching costs and imperfectly transferable human capital. Information frictions also interact with switching costs: modest reductions in switching costs help workers reallocate after learning about their comparative advantage, while eliminating switching costs induces excessive churning, destroys task-specific experience, and lowers output.

Finally, a task-biased technology shock induced by Large Language Model adoption raises aggregate output but increases the misallocation generated by information frictions, widening the output gap relative to the perfect-information economy. Information provision therefore becomes more valuable: delivered to every new cohort throughout the transition, the skill signal raises long-run output by about 1.1%, compared with 0.97% before the shock.

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A Appendix: Data and Additional Figures

Samples and Selection Criteria. The empirical analysis and structural estimation rely on two related samples constructed from *Quadros de Pessoal*.

The *large sample* follows the standard selection criteria of Rute Cardoso and Portela (2009), Card et al. (2016), and Fonseca et al. (2018). I use it to document the main patterns of occupational mobility at the three-digit occupation level, and its moments are reliable enough to serve as external validation targets for the structural model.

The *estimation sample* is designed specifically for structural estimation using the life-cycle model. Starting from the large sample, I apply the following additional selection criteria: (i) I keep only workers first observed after 1995, with age at entry below 30 and zero initial experience—ensuring I observe workers from the beginning of their careers; (ii) I retain each worker's first consecutive employment spell (allowing for the 2001 data gap), so that experience can be tracked precisely from entry; (iii) I require at least 5 distinct years of observation per worker; and (iv) I trim extreme hourly wages at the 1st and 99th percentiles.

Because the estimation sample focuses on young entrants, it has a different age and experience composition than the large sample. Table 1 compares the two samples side by side. The empirical patterns of occupational mobility are qualitatively the same in the two samples. The descriptive empirical analysis and the structural model comparison use related but not identical wage objects. In the empirical section, I use residualized wages in the large sample to isolate the component of within-occupation wage variation driven by unobserved fit or comparative advantage. I estimate the structural model instead on a more restricted sample and a reduced occupational state space, so I compare its mobility moments with the data at the level of the occupation clusters, ranking workers within occupation and year on the worker–occupation fixed effect of their wages in the large sample.

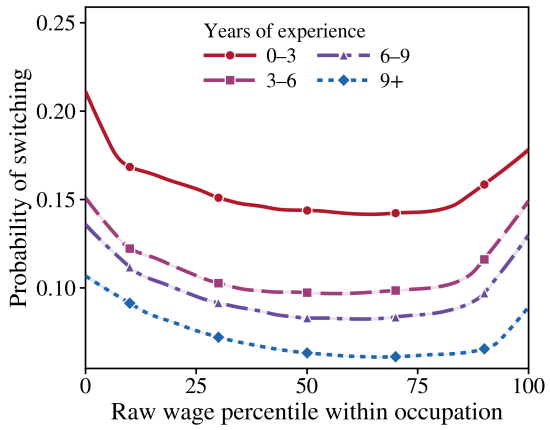
Table 1: Comparison of the Large Sample and the Estimation Sample

Large Sample		Estimation Sample	
Statistic	**Value**	**Statistic**	**Value**
Total observations	29,428,668	Total observations	2,434,141
Number of unique workers	4,425,614	Number of unique workers	283,765
Number of unique firms	618,009	Number of unique firms	131,626
Number of unique occupations	103	Number of unique occupations	101
Years covered	1995–2019	Years covered	1996–2019
Average real hourly wage (€)	6.32	Average real hourly wage (€)	5.92
Average occupational tenure (years)	4.22	Average occupational tenure (years)	6.11
Average total experience (years)	11.77	Average total experience (years)	7.29
Gender & Education (% share)		**Gender & Education (% share)**	
Men	57.79	Men	55.91
Women	42.21	Women	44.09
Low education (1–2)	67.90	Low education (1–2)	33.43
Medium education (3)	20.07	Medium education (3)	33.30
High education (4–5)	12.03	High education (4–5)	33.26

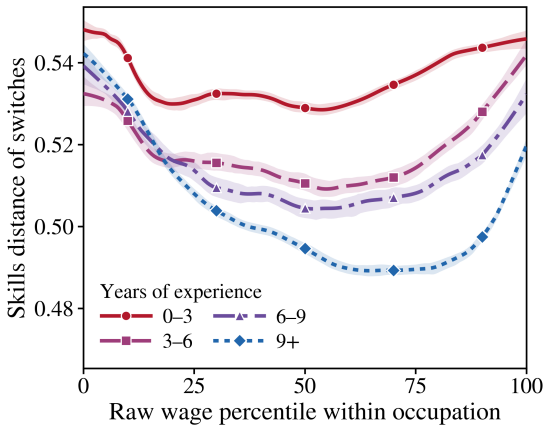
Notes: The large sample is designed to maximize representativeness and the number of observations for descriptive facts. The estimation sample is constructed from the large sample by applying the following additional restrictions: (i) entry age below 30 with zero initial experience; (ii) first consecutive employment spell only (the 2001 data gap is allowed); (iii) at least 5 distinct years observed per worker; (iv) hourly wages trimmed at the 1st and 99th percentiles. Low education = Ensino Básico; Medium = Ensino Secundário; High = Ensino Superior (Bachelor or more). Occupational tenure (end-of-spell) and total experience are computed as the maximum per worker for both samples. In the large sample, gender and education shares are computed over unique workers at their first observation; these are the education shares used in the quantitative counterfactuals.

Task and Skill Classifications. The task classification developed by Mihaylov and Tijdens (2019) classifies occupations into non-routine abstract, non-routine interactive, non-routine manual, routine cognitive, and routine manual tasks. The occupational code I use is the Portuguese CNP1994, which embeds ISCO-88; from 2010, *Quadros de Pessoal* adopts CPP2010, which is based on ISCO-08, the classification used by Mihaylov and Tijdens (2019). To ensure consistency across the classification change, I develop a mapping algorithm that combines the official crosswalk between CNP1994 and CPP2010 with empirical mappings based on workers who remained at the same firm during the 2009–2010 transition. This hybrid approach minimizes misclassification of occupational switches around the code change. I abstract from the routine dimension and focus on two broad categories: cognitive and manual skills. Cognitive skills sum non-routine abstract, routine cognitive, and non-routine interactive tasks; manual skills sum non-routine manual and routine manual tasks. I group tasks this way because these two dimensions are the most relevant for the direction of occupational mobility, and because the quantitative model has two skill dimensions for tractability, so the same grouping in the data gives a cleaner mapping between data and model.

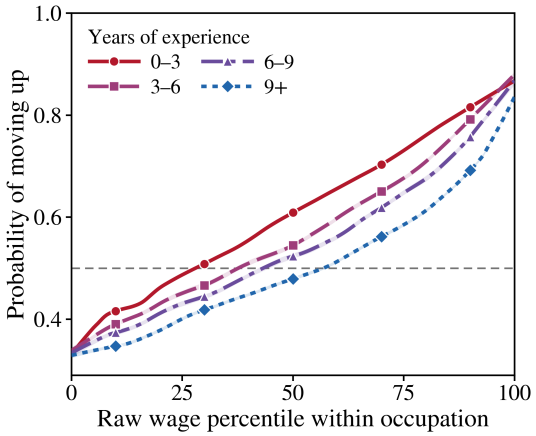
Robustness: Raw Hourly Wages. The empirical analysis in Section 2 uses residualized wages as a proxy for unobserved worker ability or match quality. Figure 20 replicates the main empirical facts from that section using raw real hourly wages to construct the within-occupation percentiles. The patterns—the pseudo U-shape in switching, the directionality in both wages and skills, and the task share changes—remain qualitatively robust when using raw wages instead of residualized wages.



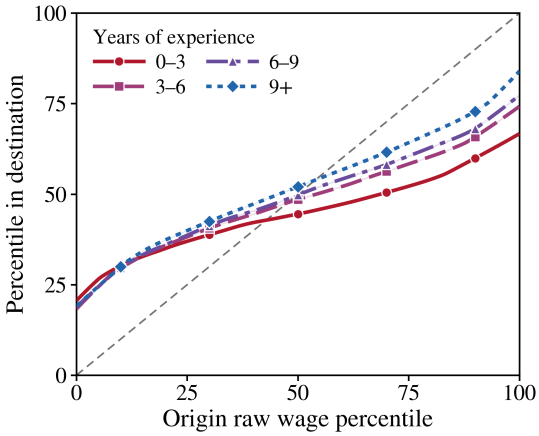
(a) Probability of switching occupations



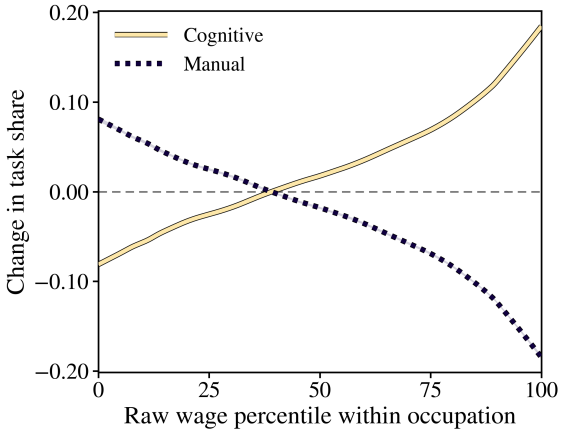
(b) Skills distance conditional on switching



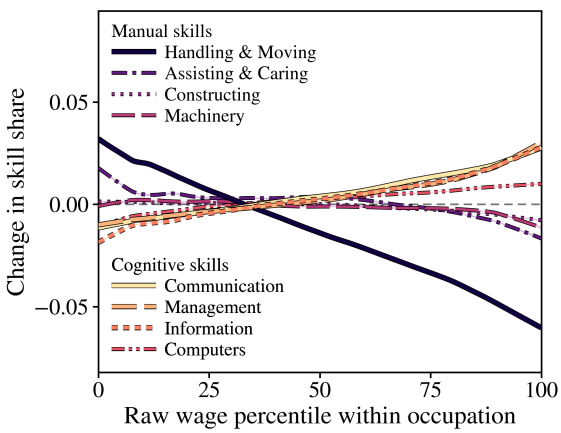
(c) Probability of moving to a higher-paying occupation



(d) Rank in the destination occupation



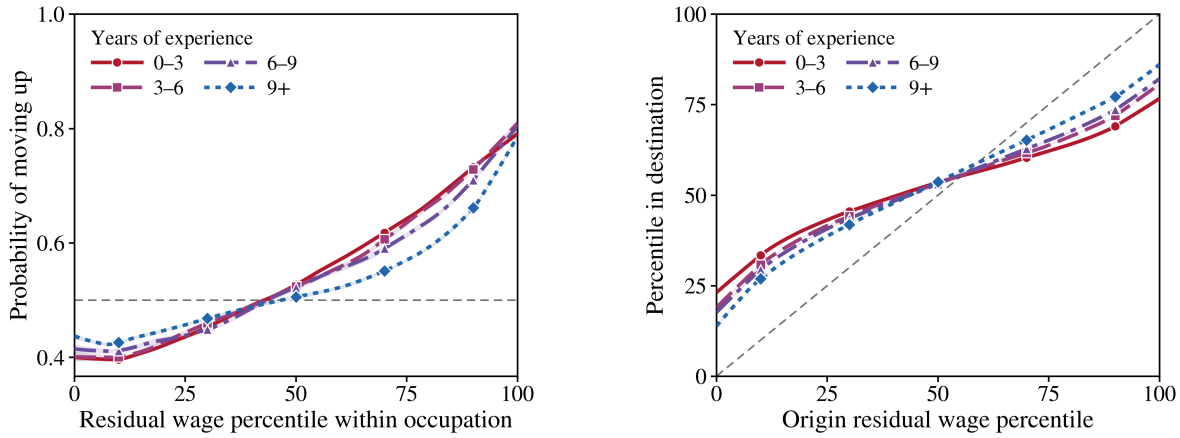
(e) Change in Cognitive–Manual task shares



(f) Change in ESCO skill shares

Figure 20: Robustness: Raw Wages. The figure replicates the main empirical facts using within-occupation percentiles of raw rather than residualized hourly wages. Panels (a) and (b): probability of switching occupations and skills distance conditional on switching, by experience group. Panels (c) and (d): probability of moving to an occupation with a higher average wage and percentile in the destination occupation, by experience group. Panels (e) and (f): change in Cognitive–Manual task shares and ESCO skill shares conditional on switching.

Directionality and Rank-to-Rank Mobility by Experience. Figure 21 reports the directionality of occupational mobility documented in Section 2. Panel 21(a) shows that the probability of moving to a higher-wage occupation conditional on switching rises monotonically with the worker's residualized within-occupation wage percentile, with the slope flattening as experience accumulates. Panel 21(b) shows the corresponding rank-to-rank pattern: low-ranked workers in the origin occupation tend to land in higher relative positions at destination, while high-ranked workers fall toward the middle of the destination distribution. Both patterns weaken with experience, consistent with workers learning their comparative advantage and progressively sorting into better-matched occupations.



(a) Directionality by Experience

(b) Rank-to-Rank by Experience

Figure 21: *Panel (a)*: Probability of moving to a higher-wage occupation conditional on switching (vertical axis) as a function of residualized within-occupation wage percentile (horizontal axis), for different experience groups. *Panel (b)*: Percentile position in destination occupation (vertical axis) against percentile position in origin occupation (horizontal axis) for occupational switchers, by experience group.

Skills Distance: Summary Statistics. Table 2 reports summary statistics for the skills distance measure defined in equation (2). Panel A shows the distribution of pairwise distances across all 115 three-digit ISCO-88 occupations mapped to the ESCO skill classification (6,555 unique pairs), a superset of the 103 occupations observed in the large sample. The mean pairwise distance is 0.71 (std. 0.19), indicating that the average pair of occupations has substantially different skill requirements. Panel B restricts to the 2.13 million actual occupation switches observed in the data. The mean distance for switchers is 0.52 (std. 0.22, median 0.58), well below the unconditional mean: workers tend to switch to occupations with similar skill profiles. The 10th percentile of switcher distances is 0.21, while the 90th percentile is 0.80.

Among all occupation pairs, whether or not a switch between them is observed, the most distant pair is glass and ceramics kiln operators and shoe cleaning and street services workers ($d = 0.99$), and the most similar pair (excluding identical occupations) is stall and market salespersons and street food vendors ($d = 0.04$). Women switch to slightly more distant occupations on average (mean $d = 0.53$) than men (mean

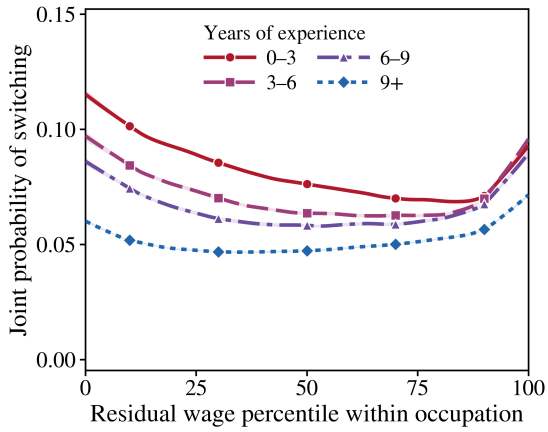
$d = 0.51$). Panel C reports the most common occupational switch by education group, by gender, and by gender $\times$ education. Two moves dominate: between mining and quarrying laborers and building frame workers for low-educated workers (education levels 1–2) and for men ($d = 0.21$; 23,000 transitions among the low-educated), and between accounting clerks and bookkeepers for medium- and high-educated workers (levels 3 and 4–5) and for women ($d = 0.36$; 11,000 and 9,000 transitions). High-educated men are the exception: their most frequent switch is between physical science technicians and architects and engineers ($d = 0.43$).

Table 2: Skills Distance Summary Statistics. Cosine distance $d(j, j') = 1 - \text{Sim}(j, j')$ computed using the 296-dimension ESCO skill classification. ISCO-88 three-digit occupation codes.

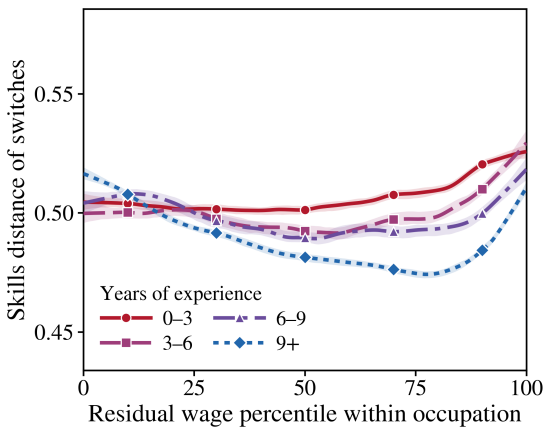
	Mean	Std	Median	p10	p25	p75	p90
Panel A: All occupation pairs (6,555 pairs)							
All pairs	0.71	0.19	0.76	0.44	0.63	0.85	0.91
Panel B: Actual switchers (2,127,097 transitions)							
All	0.52	0.22	0.58	0.21	0.33	0.69	0.80
Men	0.51	0.22	0.57	0.21	0.33	0.69	0.79
Women	0.53	0.22	0.58	0.22	0.34	0.69	0.81

Panel C: Most common occupation switch by group			
Group	Occupation switch	d	N
Low educ. (1–2)	Laborers $\rightarrow$ Building frame workers	0.21	23,070
Medium educ. (3)	Accounting clerks $\rightarrow$ Bookkeepers	0.36	10,987
High educ. (4–5)	Accounting clerks $\rightarrow$ Bookkeepers	0.36	9,364
Men	Laborers $\rightarrow$ Building frame workers	0.21	24,311
Women	Accounting clerks $\rightarrow$ Bookkeepers	0.36	16,774
Men, Low	Laborers $\rightarrow$ Building frame workers	0.21	23,016
Men, Medium	Finance dealers $\rightarrow$ Shop salespersons	0.59	7,174
Men, High	Phys. science tech. $\rightarrow$ Engineers	0.43	5,224
Women, Low	Domestic helpers $\rightarrow$ Cooks	0.33	11,301
Women, Medium	Accounting clerks $\rightarrow$ Bookkeepers	0.36	7,570
Women, High	Accounting clerks $\rightarrow$ Bookkeepers	0.36	6,374

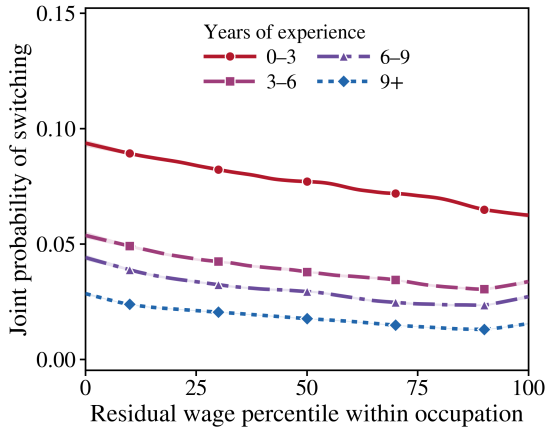
Large Sample Construction. I construct my measure of earnings by summing a worker's base salary and any regular earnings supplements and dividing the amount by her normal hours of work; this yields the same "straight-time" hourly wage used in Card et al. (2016). I obtain real hourly wages by deflating the "straight-time" wage using the CPI with 1995 as base year. From the initial dataset I eliminate entries if the worker ID is reported multiple times per year; for each worker I drop the entire history if sex changes over time, age is inconsistent, or the worker is not aged between 20 and 55. I also drop observations if the hiring date for any job is missing or inconsistent across observations. I drop the entire history of a worker observed at different firms in two



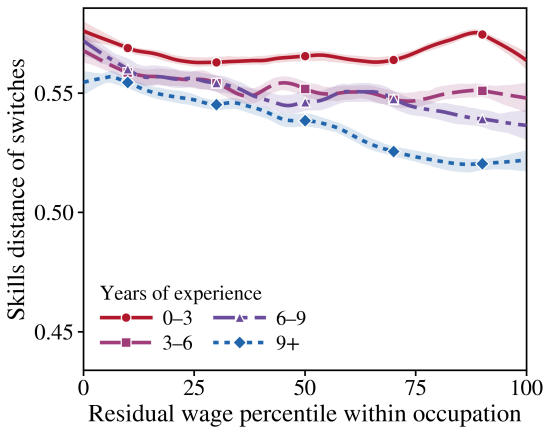
(a) Within firm: probability of switching occupation



(b) Within firm: skills distance of switches



(c) Across firms: probability of switching occupation



(d) Across firms: skills distance of switches

Figure 22: Occupational mobility within and across firms, by residualized within-occupation wage percentile (horizontal axis) and experience group. *Panels (a) and (c)*: the figures decompose the probability of switching occupation between t and $t+1$ in Figure 2(a) into the joint probability of switching occupation and staying in the same firm (Panel (a)) and the joint probability of switching occupation and changing firm (Panel (c)); the two panels sum to the total probability of switching and share a common vertical scale. *Panels (b) and (d)*: skills distance between origin and destination occupation, conditional on switching occupation within the same firm and across firms, on a common vertical scale.

consecutive years if the hiring date of the second job equals that of the first. I also drop workers with missing education or with education decreasing over time. The sample is further restricted to full-time workers (at least 30 hours per week or 130 hours per month) who earn at least the minimum wage. Following Fonseca et al. (2018), I drop workers in occupations and industries associated with the primary sector, given that the survey is optional for those. I also drop workers in public administration and defense[30] and drop a worker history if she has been in sectors with fewer than 100 workers reported or in occupations with fewer than 100 workers reported in a given

[30]These correspond to industries with NACE code A,B,C,L and Q before 2007, when the CAE rev2-rev2.1 was used in Portugal, and to the industries A,B,O and U after the CAE rev3 industry classification was introduced in Portugal, starting from 2007. Occupations dropped belong to the 2-digit groups 11, 61 and 92 for the CNP 1994 and 11, 61, 62, 63 and 92 for the CPP 2010.

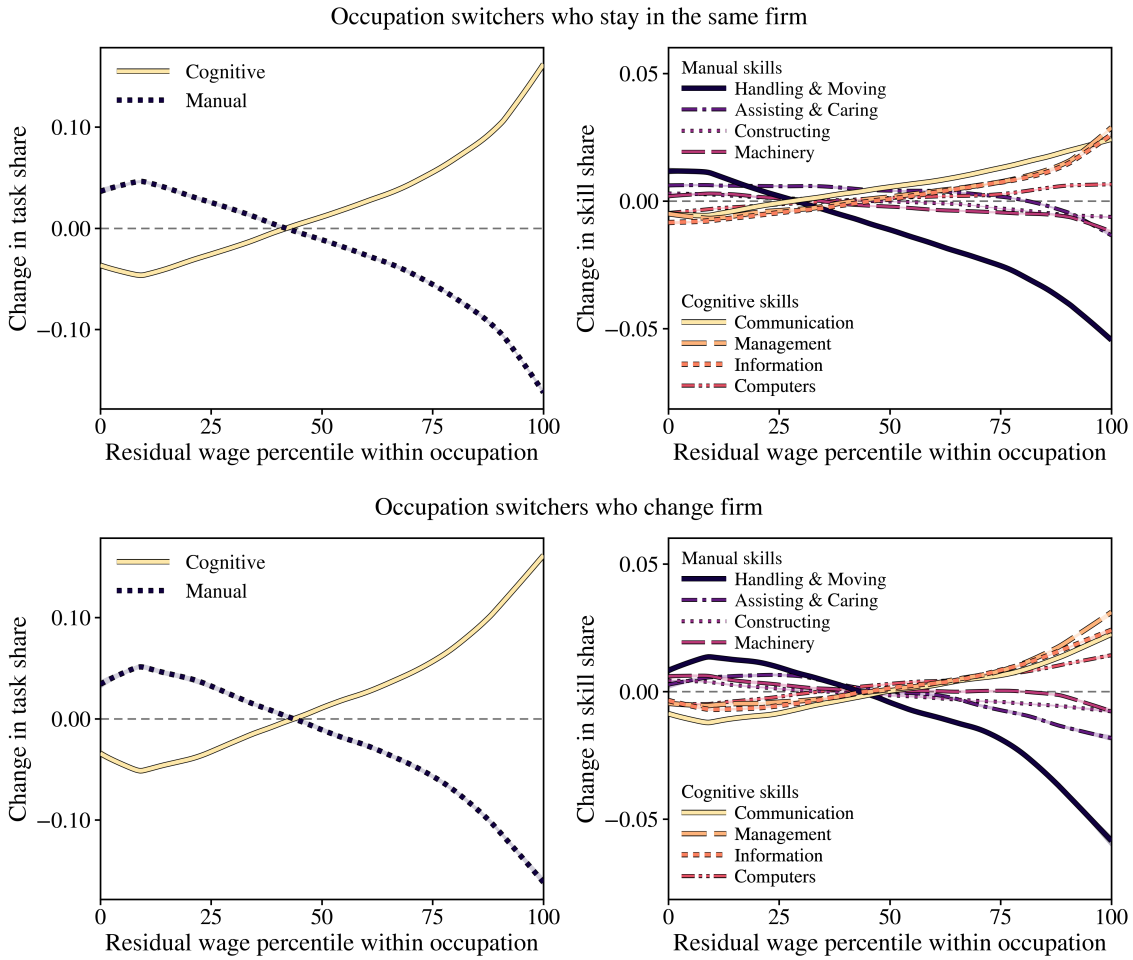
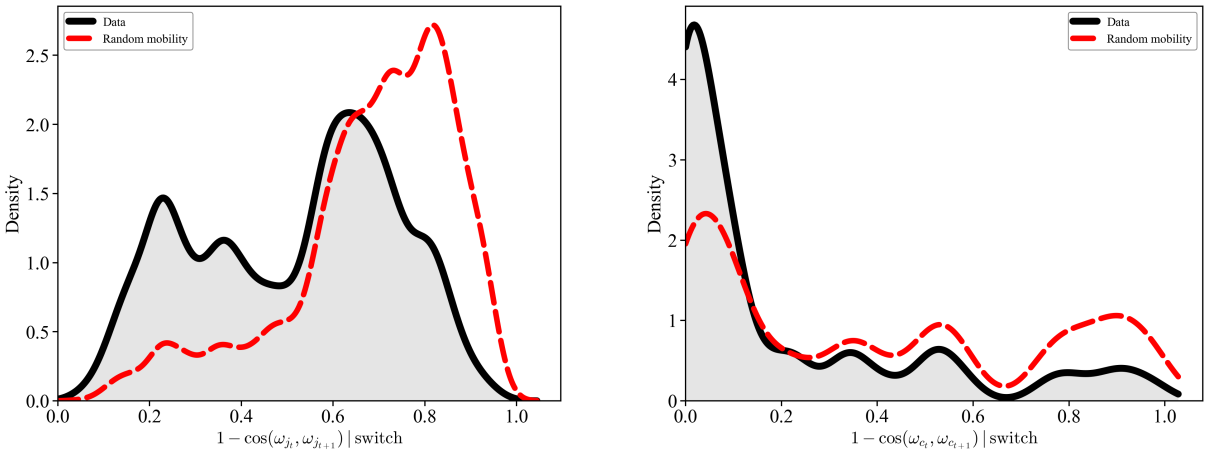


Figure 23: Change in task content for occupation switchers who stay in the same firm (top row) and who change firm (bottom row). Vertical axis: change in the task share (left, Cognitive–Manual classification) or skill share (right, ESCO skill dimensions) between destination and origin occupation; horizontal axis: residualized wage percentile within the origin occupation. Each column shares the same vertical scale across the two rows.

year. As I am also interested in occupational mobility within firms, I drop workers in firms with fewer than 3 employees. Industry codes follow CAE Rev. 2 (Classificação Portuguesa de Actividades Económicas, Revisão 2) from 1995, CAE Rev. 2.1 (Revisão 2.1) from 2003, and subsequently CAE Rev. 3 (Revisão 3)[31]. As for occupations, the National Classification of Occupations 1994 (CNP/94) was used from 1994 until 2009 and was then replaced by the Portuguese Classification of Occupations 2010 (CPP/2010) that is embedded in the ISCO-08 international classification. I develop an algorithm to minimize the number of occupational switch misclassifications in the years around the occupational code change.

[31]Between 2002 and 2003 the system of industry classification used for Quadros de Pessoal changed from CAE Rev 2 to CAE Rev 2.1. At the 2-digit level, only two industries were introduced: 96 Activ. de produção de bens pelas famílias p/ uso próprio; 97 Activ. de produção de serviços pelas famílias p/ uso próprio.



(a) ESCO skill distance (296 dimensions)

(b) Cognitive–Manual task distance (occupation clusters)

Figure 24: Task distance of occupational switches, observed (black) versus a random mobility counterfactual (red, dashed) in which each switcher's destination is drawn proportional to employment shares, following Gathmann and Schönberg (2010). Task distance is $1 - \cos(\boldsymbol{\omega}_{j_t}, \boldsymbol{\omega}_{j_{t+1}})$. *Panel (a)*: the 296-dimension ESCO skill space, between three-digit occupations. *Panel (b)*: the Cognitive–Manual task space of Mihaylov and Tijdens (2019), between the eight occupation clusters used in the structural model. In both, observed moves are closer in task space than random reallocation would predict.

A.1 Wage Changes upon Occupational Switches

I compare the wages of workers who switch occupation in the large sample with those of stayers who start from the same position in the same occupation. Let w_{it} denote the log real hourly wage of worker i in year t. As in the rest of the empirical analysis, a switcher is a worker observed in occupation j_t in year t and in a different three-digit occupation j_{t+1} in year $t+1$; stayers are workers observed in the same occupation in t and $t+1$. The initial position of a worker is the percentile of the residual wage $\widehat{r}_{i,t-1}$ of equation (1), as in Facts 1 and 2, measured in the year before the switch and ranked within the origin occupation and year. Ranking on year $t-1$ keeps w_{it}, which enters the wage change, out of the ranking, so that a transitory wage shock in t does not generate mean reversion mechanically. I therefore restrict the sample to workers observed in the origin occupation in both $t-1$ and t.

Wage changes by initial position within occupation. Panel (a) of Figure 4 reports, for each horizon $h \in \{0, \ldots, 4\}$, the coefficients β_{hq} of the regression

$$w_{i,t+1+h} - w_{it} = \sum_{q=1}^{10} \beta_{hq}\, S_{it}\, \mathbf{1}\{q_{it} = q\} + v^{h}_{c(i,t)} + \xi'_h\, X_{it} + e_{ith}, \tag{45}$$

where S_{it} indicates a switch between t and $t+1$, q_{it} is the decile of the initial position, v^h_c are fixed effects for origin occupation $\times$ year $\times$ decile cells, and X_{it} includes education fixed effects, experience and experience^2. Standard errors are clustered by worker. Horizon $h = 0$ is the first year in the new occupation. The coefficient β_{hq} is the cumula-

tive wage change of switchers relative to stayers who start from the same decile of the same occupation in the same year, so that it nets out the mean reversion in wages that is common to the two groups. The wage change upon switching declines monotonically with the initial position. Relative to stayers, switchers in the bottom quintile gain 6 log points on impact and 6 after four years, switchers around the median gain 2 and 3, and switchers in the top decile lose 3 on impact and close the gap after four years.

Wage changes by initial position and task content of the switch. Panel (b) of Figure 4 groups switches by the change in the cognitive task share of the occupation, $\Delta\omega_C = \omega_{C,j_{t+1}} - \omega_{C,j_t}$, measured with the Cognitive–Manual classification of Mihaylov and Tijdens (2019): ↑Cognitive if $\Delta\omega_C > 0.10$ and ↑Manual if $\Delta\omega_C < -0.10$. Switches with a smaller change in the cognitive task share are included in the regression as a third group and are not reported in the figure. Panel (b) reports the coefficients of equation (45) on impact ($h = 0$) and after four years ($h = 4$) when the switch indicator is interacted with both the decile of the initial position and the three groups, so that each group of switchers is compared with the stayers in the same origin occupation, year, and decile. Workers in the bottom quintile gain relative to stayers whatever the direction of the move. On impact they gain 7 log points when they move to more cognitive occupations and 5 when they move to more manual occupations; after four years the gains are 6 and 4. The return to moving toward more manual occupations falls with the initial position: around the median it is 0 log points on impact and 1 after four years, and in the top decile it is -6 on impact and -3 after four years. In the top decile, workers moving to more cognitive occupations lose 2 log points on impact and gain 2 after four years. In both groups the wage change is more dispersed across initial positions on impact than after four years. Workers at the bottom of the wage distribution of their occupation are also those who move more often toward manual occupations (34% of the switches in the bottom decile against 15% in the top decile).

Event studies. Figure 25 reports the dynamics around the switch. Let $\Delta w_{it} \equiv w_{it} - w_{it-1}$ and let $q \in \{\text{bottom quintile}, \text{middle}, \text{top quintile}\}$ index the initial position; in this paragraph q_{it} denotes the position group of worker i in year t rather than the decile. For each relative time $\tau \in \{-3, \ldots, 4\}$ I run a separate OLS regression

$$\Delta w_{i,t+1+\tau} = \sum_q \beta_{\tau q}\, S_{it}\, \mathbf{1}\{q_{it} = q\} + v^{\tau}_{c(i,t)} + \xi'_\tau\, X_{it} + e_{it\tau}, \tag{46}$$

where v^{τ}_c are fixed effects for origin occupation × year × position group cells. The coefficients $\beta_{\tau q}$ measure the difference in annual wage growth between switchers and stayers with the same initial position, and $\tau = 0$ is the wage change spanning the switch, $w_{i,t+1} - w_{it}$. To present wage levels around the event I cumulate the estimated

growth effects and normalize at $\tau = -1$, similar to Heise and Porzio (2022):

$$C_q(k) \;=\; \sum_{\tau=-3}^{k} \hat{\beta}_{\tau q} \;-\; \sum_{\tau=-3}^{-1} \hat{\beta}_{\tau q}, \qquad k \in \{-3, \ldots, 4\},$$

with 95% confidence bands computed from the joint worker-clustered covariance of the estimates across τ. In Panel (a), switchers in the bottom quintile gain 6 log points relative to stayers at the switch, and the gain is persistent: it is 6 after four years. Switchers in the middle of the distribution gain 2 on impact and 4 after four years, while switchers in the top quintile lose 2 on impact and recover the loss within four years.

Before the switch, wage growth of switchers and stayers with the same initial position is similar: the cumulative difference is below 1 log point in absolute value in all groups. Ranking workers on the wage in t instead selects switchers whose last wage in the origin occupation contains a transitory component. Under that ranking, the wage of switchers in the bottom quintile falls by 2 log points relative to stayers in the three years before the switch, and that of switchers in the top quintile rises by 3; the bottom-quintile gain would then be 7 log points on impact and 6 after four years, and the top-quintile loss 3 on impact and 1 after four years.

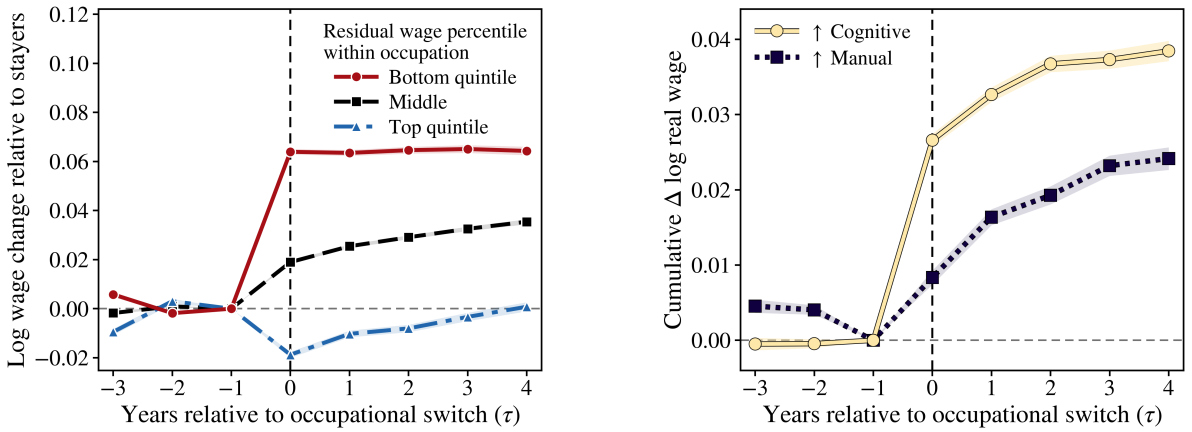
Panel (b) groups switchers by the task content of the switch, with the groups defined above. I estimate equation (46) with the switch indicator interacted with the groups and all stayers as the omitted category, and I demean the dependent variable and the controls by calendar year in place of the cell fixed effects. Workers moving to more cognitive occupations gain 3 log points on impact and 4 after four years, while workers moving toward more manual occupations gain 1 on impact and 2 after four years.

A.2 Occupation Clustering

Let $o = 1, \ldots, n$ index three-digit ISCO occupations and let $c \in \{0, \ldots, J-1\}^n$ denote a partition into J clusters. Each occupation has a standardized task-content vector $x_o \in \mathbb{R}^2$ that summarizes its cognitive and manual content, a mass weight s_o equal to its share of person-periods in the panel, and pairwise mobility flows $F_{oo'}$ between different occupations measured in the large sample. The partition minimizes

$$L(c) \;=\; \underbrace{\sum_{o} s_o \, \| x_o - \bar{x}_{c_o} \|^2}_{\text{task-content dispersion}} \;+\; \lambda_F \underbrace{\sum_{o,o'} F_{oo'} \, \mathbf{1}\{c_o = c_{o'}\}}_{\text{absorbed flow}},$$

where $\bar{x}_c$ is the mass-weighted centroid of cluster c. The first term ensures that occupations with similar task content are clustered together; the second term penalizes observed worker flows that are absorbed into a cluster, so that aggregated cluster-to-cluster transitions remain informative and representative with respect to the original occupational mobility graph. I set $\lambda_F = 10$, which trades off within-cluster disper-



(a) Event study of wages around the switch, by initial position in the origin occupation

(b) Event study of wages around the switch, by task content of the switch

Figure 25: Event studies of wages around occupational switches. Horizontal axis: event time relative to the switch, where $\tau = -1$ is the last year in the origin occupation (year t) and $\tau = 0$ is the first year in the new occupation (year $t+1$); vertical axis: cumulative change in the log real wage of switchers relative to stayers, normalized at $\tau = -1$. *Panel (a)*: workers in the bottom quintile (red, solid), the middle (black, dashed), and the top quintile (blue, dash-dot) of the residual wage distribution of the origin occupation in year $t-1$, relative to stayers with the same initial position. *Panel (b)*: switches toward more cognitive occupations (light yellow, solid) and more manual occupations (dark blue, dotted), relative to all stayers. Shaded areas are 95% confidence bands clustered by worker.

sion against the cross-cluster mobility share. To make sure that two clusters have a clear cognitive and manual interpretation, I fix two seed sets of occupations, each of which must be contained in a single cluster. These occupations are chosen based on their extreme task content, and are either fully manual or fully cognitive. I minimize $L(c)$ subject to two feasibility constraints: the cluster-level mobility graph must remain connected, and every non-seed cluster must retain at least 5% of total mass. The procedure is run from several starting points (mass-weighted k-means initializations) and the best partition is retained. Figure 26 plots the two terms of the loss function against J. I select $J = 8$ (vertical dashed line), which balances parsimony in the number of clusters against within-cluster task-content dispersion and the cross-cluster mobility share. For larger J the measure improves but the number of observations included in each cluster falls below the 5% threshold, which can weaken the identification of some cluster-specific components of the model. I run the clustering under the two classifications of the main text: an ESCO variant in which each granular ESCO skill is labeled as either cognitive or manual, and (C_o, M_o) is occupation o's share of cognitive vs. manual skills aggregated from those labels using its ESCO skill mix, and a task-based variant in which $(C_o^{\mathrm{MT}}, M_o^{\mathrm{MT}})$ is computed from the routine/non-routine task shares of Mihaylov and Tijdens (2019). Cluster labels are deterministic: the cluster containing the cognitive seed set is Cluster 0, the cluster containing the manual seed set is Cluster $J-1$, and the remaining clusters, Clusters 1 to $J-2$, are ordered by ascending mean cognitive share. The seed sets only serve the clustering; the anchor occupations used to identify the wage equation are chosen separately as the clusters with the highest cognitive and manual task shares, Clusters 0 and 1. The manual anchor is therefore the non-seed

cluster with the lowest cognitive share, not the cluster containing the manual seed set.

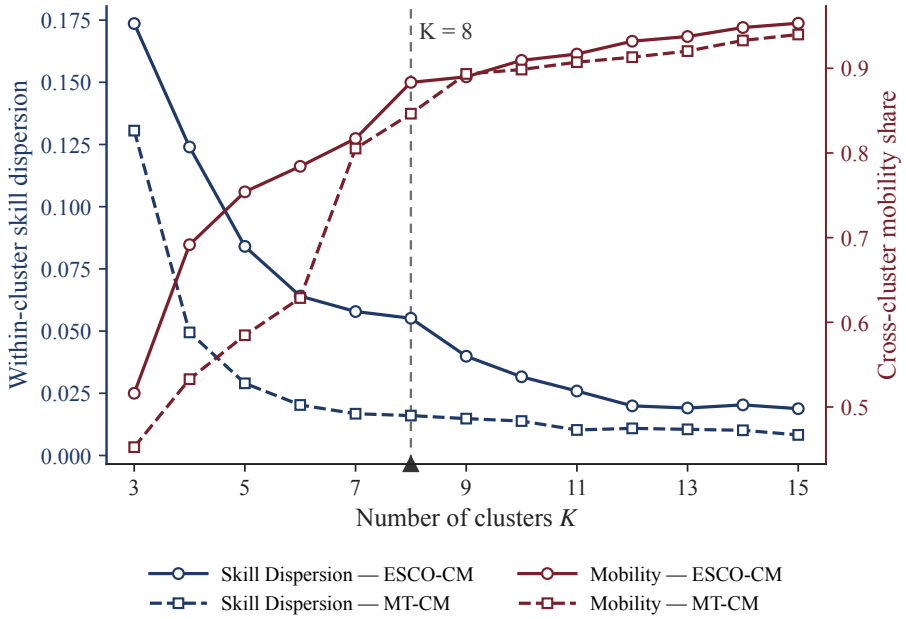


Figure 26: Choice of the number of clusters. The horizontal axis is the number of clusters J, from 3 to 15, with $\lambda_F = 10$ fixed. For each J, the curves report the two terms of the loss function at the partition that minimizes $L(c)$: the mass-weighted within-cluster dispersion of task content and the cross-cluster mobility share. Each term is shown for the ESCO-based cognitive and manual shares (C_o, M_o) and for the task-based shares $(C_o^{\text{MT}}, M_o^{\text{MT}})$ of Mihaylov and Tijdens (2019); the vertical dashed line marks the selected $J = 8$.

In a model of occupational mobility, the level of occupational aggregation has first-order consequences both for estimation and for the quantitative importance of the model's mechanisms. Grouping occupations helps identification by increasing the amount of within-occupation variation available to estimate task-specific parameters, especially returns to task tenure: if occupations are defined too granularly, there is too little variation in accumulated experience along each task dimension k within a given occupation j to identify the coefficients on task-specific human capital. When occupations are too granular, mobility patterns become sparse, the occupational network loses connectivity, and cells become too small to separately identify occupation-specific prices, skill loadings, experience returns, and realistic residual variances. Very fine occupations may also be observationally distinct but economically similar from the point of view of my model. The facts I document suggest that the drivers of occupational mobility are best understood in the task space, which the cognitive and manual dimensions summarize well. Clustering also has a cost. In a discrete choice framework it shrinks the worker's choice set, and fewer alternatives, with less dispersion of expected wages across them, mechanically reduce the mobility the model generates; clustering therefore affects the estimated switching costs and compensating differentials.

B Appendix: Model

B.1 Firm's Time Allocation Problem

Following Althoff and Reichardt (2026), the firm controls the worker's time allocation statically, choosing $\{\ell_{jk}\}_{k=1}^{K}$ to maximize expected log output conditional on the symmetric posterior $(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it})$ shared with the worker, where the expectation is taken under the learning block's posterior $\boldsymbol{\theta}_i \mid \mathcal{I}_{it} \sim \mathcal{N}(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it})$.

Task-level output of worker i in occupation j at time t is

$$y_{jk} = \ell_{jk} \exp\big(a_{jk}\,\theta_{ik}\big)\, H_{jk}(\tau_{ikt}), \qquad H_{jk}(\tau) \;\equiv\; \exp\big(h_{jk}(\tau)\big), \tag{47}$$

where $\ell_{jk} \in [0,1]$ is the fraction of unit time allocated to task k, subject to $\sum_{k=1}^{K} \ell_{jk} = 1$; a_{jk} is the primitive task-level semi-elasticity of output with respect to latent skill; and $H_{jk}(\cdot)$ is the multiplicative task-level human-capital function, with $h_{jk}(\cdot)$ its log. Tasks aggregate according to the normalized Cobb-Douglas technology with weights $\omega_{jk} > 0$ satisfying $\sum_{k=1}^{K} \omega_{jk} = 1$:

$$Y_{ijt} = \prod_{k=1}^{K} \left(\frac{y_{jk}}{\omega_{jk}} \right)^{\omega_{jk}}. \tag{48}$$

Taking logs,

$$\log Y_{ijt} = \sum_{k=1}^{K} \omega_{jk} \log \ell_{jk} + \sum_{k=1}^{K} \omega_{jk}\, a_{jk}\, \theta_{ik} + \sum_{k=1}^{K} \omega_{jk}\, h_{jk}(\tau_{ikt}) - \sum_{k=1}^{K} \omega_{jk} \log \omega_{jk}. \tag{49}$$

Log output is linear in $\boldsymbol{\theta}_i$, so taking the expectation under the symmetric posterior replaces θ_{ik} with $\tilde{\theta}_{ikt}$:

$$\mathbb{E}\big[\log Y_{ijt} \,\big|\, \tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it}\big] = \sum_{k=1}^{K} \omega_{jk} \log \ell_{jk} + \sum_{k=1}^{K} \omega_{jk}\, a_{jk}\, \tilde{\theta}_{ikt} + \sum_{k=1}^{K} \omega_{jk}\, h_{jk}(\tau_{ikt}) - \sum_{k=1}^{K} \omega_{jk} \log \omega_{jk}. \tag{50}$$

Only the first term depends on $\{\ell_{jk}\}$. The firm's problem therefore reduces to

$$\max_{\{\ell_{jk}\}} \sum_{k=1}^{K} \omega_{jk} \log \ell_{jk} \quad \text{s.t.} \quad \sum_{k=1}^{K} \ell_{jk} = 1. \tag{51}$$

Forming the Lagrangian $\mathcal{L} = \sum_k \omega_{jk} \log \ell_{jk} - \gamma\big(\sum_k \ell_{jk} - 1\big)$, the first-order conditions are

$$\frac{\omega_{jk}}{\ell_{jk}} = \gamma \quad \Rightarrow \quad \ell_{jk} = \frac{\omega_{jk}}{\gamma}. \tag{52}$$

Summing over k and using $\sum_k \ell_{jk} = \sum_k \omega_{jk} = 1$ yields $\gamma = 1$, so the optimal time allocation is

$$\ell^{*}_{jk} = \omega_{jk}. \tag{53}$$

The optimal time allocation thus depends only on the task weights ω_{jk}, not on the worker's skills, experience, or beliefs about her skills. The assumption of Cobb-Douglas aggregation of tasks, with a unitary elasticity of substitution, makes the beliefs in expected log output additively separable from $\{\ell_{jk}\}$, so beliefs play no role in the task assignment. Generalizing the production function to a CES aggregator in tasks, or a dynamic task assignment chosen by the firm, would break this separability and make the allocation belief-dependent, but is beyond the scope of this paper.

B.2 Stationary Equilibrium Algorithm

A stationary equilibrium is a constant vector of occupation log prices $\tilde{\boldsymbol{p}} = (\log P_j)_{j\in\mathcal{J}}$ at which every occupation's goods market clears. The CES aggregator weights $\{\alpha_j\}$ are held fixed at their pre-shock calibration and any candidate price vector is renormalized so that the aggregate price index equals one, $P(\tilde{\boldsymbol{p}}) = \left(\sum_j \alpha_j^{\sigma} P_j^{1-\sigma}\right)^{1/(1-\sigma)} = 1$ (Section 3.6). I compute the stationary equilibrium of the model by iterating on the time-independent price vector: at each iteration, I solve the worker's dynamic problem given the current prices, simulate a synthetic panel, and compute the update for the price vector using the labor demand equation. When simulating, workers exit at rate χ[32] and are replaced by entrants drawn from the group priors. Once I discard an initial burn-in, this death-and-rebirth process makes initial conditions irrelevant and the simulated panel approximates the ergodic cross-section $\pi^{\star}(\tilde{\boldsymbol{p}}^{\star})$ resulting from the equilibrium price vector $\tilde{\boldsymbol{p}}^{\star}$.

In the simulation, I aggregate occupation-specific output using choice probabilities as weights. Rather than assigning each worker to a single occupation, which would make output jump for small moves in the price vector, I let her contribute to every occupation in proportion to her conditional choice probability ($\mathbb{P}_{ij}(\tilde{\boldsymbol{p}})$), which smooths the response of output to prices and sharply reduces simulation noise. The probability is given by

$$\begin{aligned} \mathbb{P}_{ij}(\tilde{\boldsymbol{p}}) &= \frac{\exp\big(v_{ij}(\tilde{\boldsymbol{p}})/\varrho\big)}{\sum_{k\in\mathcal{J}} \exp\big(v_{ik}(\tilde{\boldsymbol{p}})/\varrho\big)}, \\ v_{ij}(\tilde{\boldsymbol{p}}) &= p_j + \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}}_i + h_j(\boldsymbol{\tau}_i) + \eta_{jg(i)} - \mathcal{C}\big(j, d_{i,-1}\big) + \beta\,\mathbb{E}\big[EV(\boldsymbol{x}') \mid \boldsymbol{x}_i, j\big]. \end{aligned} \tag{54}$$

Output, by contrast, is produced at the worker's true skills $\boldsymbol{\theta}_i$: her expected wage in occupation j, in levels and with the log-normal correction for noise $\mathbb{E}[e^{\varepsilon}] = e^{\varsigma_{\varepsilon j}^2/2}$, is $\overline{W}_{ij}(\tilde{\boldsymbol{p}}) = \exp\big(p_j + \boldsymbol{\lambda}_j'\boldsymbol{\theta}_i + h_j(\boldsymbol{\tau}_i) + \tfrac{1}{2}\varsigma_{\varepsilon j}^2\big)$. Occupation output is the probability-weighted sum of these wages deflated by the price in that occupation,

$$Y_j(\tilde{\boldsymbol{p}}) = \frac{1}{P_j}\sum_i \mathbb{P}_{ij}(\tilde{\boldsymbol{p}})\,\overline{W}_{ij}(\tilde{\boldsymbol{p}}) = \sum_i \mathbb{P}_{ij}(\tilde{\boldsymbol{p}})\,\exp\Big(\boldsymbol{\lambda}_j'\boldsymbol{\theta}_i + h_j(\boldsymbol{\tau}_i) + \tfrac{1}{2}\varsigma_{\varepsilon j}^2\Big), \tag{55}$$

[32] Workers do not internalize the exit rate when they choose, so their dynamic problem remains effectively infinite-horizon.

with $P_j = e^{p_j}$, so the price cancels and output is properly measured in efficiency units.

Given the occupation outputs and aggregate output, I obtain a new price vector from the CES inverse demand (30). I then update the price guess with a damped step and iterate until the price vector converges. The algorithm box below summarizes the procedure.

Algorithm: Stationary Equilibrium (price tâtonnement)

Inputs: the estimated wage and skill parameters $\widehat{\mathbf{\Gamma}} = \{(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)_{g\in\mathcal{G}}, (p_j, \boldsymbol{\lambda}_j, \delta_j, \varsigma^2_{\varepsilon j})_{j\in\mathcal{J}}\}$ and dynamic-choice parameters $\widehat{\boldsymbol{\Delta}} = \{(\eta_{jg}), \kappa_0\}$; CES weights $\{\alpha_j\}$ and elasticity σ; discount β, exit probability χ, taste scale ϱ; damping ζ, tolerance *tol*.

Output: equilibrium log prices $\tilde{\boldsymbol{p}}^\star$, value coefficients $\boldsymbol{\Phi}^\star$, outputs $\{Y_j\}$, ergodic cross-section $\pi^\star$.

Initialize $\tilde{\boldsymbol{p}}^{(0)}$ at the Stage-1 prices, renormalized to $P(\tilde{\boldsymbol{p}}^{(0)}) = 1$, where $P(\tilde{\boldsymbol{p}}) = \left(\sum_j \alpha_j^\sigma P_j^{1-\sigma}\right)^{1/(1-\sigma)}$ with $P_j = e^{p_j}$.

Repeat for $n = 0, 1, \ldots$ until $\max_{j\in\mathcal{J}} |\tilde{p}_j^{(n+1)} - \tilde{p}_j^{(n)}| < tol$:

- **Solve.** Compute $\boldsymbol{\Phi}^{(n)}$ as the fixed point of the integrated Bellman operator at prices $\tilde{\boldsymbol{p}}^{(n)}$ (discount β; χ not internalized).
- **Simulate.** Draw a worker panel with common random numbers; workers exit at rate χ and are replaced by entrants from the group priors; discard the burn-in to obtain the ergodic cross-section.
- **Aggregate.** For each worker i and occupation j, form the choice probability $\mathbb{P}_{ij}(\tilde{\boldsymbol{p}}^{(n)})$ of (54) and the output $Y_j = \sum_i \mathbb{P}_{ij} \exp\!\left(\boldsymbol{\lambda}_j'\boldsymbol{\theta}_i + h_j(\boldsymbol{\tau}_i) + \frac{1}{2}\varsigma^2_{\varepsilon j}\right)$ as in (55).
- **Prices.** Set $\hat{P}_j = \alpha_j \left(Y/Y_j\right)^{1/\sigma}$, with $Y = \left(\sum_j \alpha_j Y_j^{(\sigma-1)/\sigma}\right)^{\sigma/(\sigma-1)}$, and renormalize $\hat{\boldsymbol{P}} \leftarrow \hat{\boldsymbol{P}}/P(\hat{\boldsymbol{P}})$.
- **Update.** $\tilde{\boldsymbol{p}}^{(n+1)} = (1-\zeta)\,\tilde{\boldsymbol{p}}^{(n)} + \zeta \log \hat{\boldsymbol{P}}$, with $\zeta \in (0, 1)$.

Return $\tilde{\boldsymbol{p}}^\star$, $\boldsymbol{\Phi}^\star$, $\{Y_j\}$, $\pi^\star$.

B.3 Transition Path Algorithm

The perfect-foresight transition path after an unanticipated technology shock at $t = 0$ is the sequence of price vectors of the (non-stationary) competitive equilibria along the transition to the new stationary price vector. Each of these competitive equilibria is computed under the same fixed weights $\{\alpha_j\}$ and the same per-period numeraire $P(\tilde{\boldsymbol{p}}_t) = 1$. The computation mirrors the stationary algorithm (solve, simulate, aggregate, update), applied to the whole price path by backward–forward shooting.

The technology shock considered in the experiment changes the task weights $\boldsymbol{\omega}_j$, with AI lowering the cognitive-task weight in proportion to each occupation's exposure; the wage parameters inherit the change through the effective mapping, $\lambda_{jk}^{\mathrm{AI}} = \lambda_{jk}\,\omega_{jk}^{\mathrm{AI}}/\omega_{jk}$, and the log human-capital bundle h_j inherits the new task weights $\boldsymbol{\omega}_j^{\mathrm{AI}}$, while noise variances, amenities, and switching costs are unchanged. Let Λ collect the parameters of this technology. Adoption is instantaneous: the technology mapping skills and experience into output jumps from Λ_{pre} to Λ_{post} at $t = 0$.

I first compute the two fixed points that determine pre- and post-shock stationary competitive equilibria, each of which determines the equilibrium price vector and the dynamic program of the workers. The two endpoints deliver $(\tilde{\boldsymbol{p}}_{\text{pre}}, \boldsymbol{\Phi}_{\text{pre}})$ and $(\tilde{\boldsymbol{p}}_{\text{post}}, \boldsymbol{\Phi}_{\text{post}})$, where $\boldsymbol{\Phi}$ collects the Smolyak value-function coefficients that solve the Bellman equations at these two stationary points. The terminal value of the transition path for prices is pinned at $\tilde{\boldsymbol{p}}_{\text{post}}$. Given a guess for the price path, I solve for the optimal coefficients approximating the value functions by backward induction from the terminal condition $\boldsymbol{\Phi}_{T-1} = \boldsymbol{\Phi}_{\text{post}}$,

$$\boldsymbol{\Phi}_t = \mathcal{S}^{-1}\Big[\varrho \log \sum_{j \in \mathcal{J}} \exp\Big(\big\{r_j(\boldsymbol{x}; \tilde{\boldsymbol{p}}_t, \Lambda_t) + \beta\, \mathbb{E}\big[EV_{t+1}(\boldsymbol{x}') \mid \boldsymbol{x}, j\big]\big\}/\varrho\Big)\Big], \qquad t = T-2, \ldots, 0, \tag{56}$$

where $\mathcal{S}^{-1}$ is the Chebyshev/Smolyak inversion[33], $r_j(\boldsymbol{x}; \tilde{\boldsymbol{p}}_t, \Lambda_t)$ is the flow payoff of occupation j in state $\boldsymbol{x}$ (the expected log wage at the period-t prices and technology, plus the amenity η_{jg}, minus the switching cost $\mathcal{C}(j, d_{-1})$), and the expectation integrates the Gaussian belief update by Gauss–Hermite quadrature. I then forward simulate a single panel in which workers face the time-varying (non-stationary) value functions, prices, and technology, and I obtain occupation outputs Y_{jt} in each period with the aggregation (54)–(55). Finally, I use the per-period CES inverse demand to get the updating price sequence, which I renormalize to the numeraire in each period. I iterate on the path until the largest price change across iterations and occupations falls below a tolerance. The procedure is summarized in the algorithm box below.

[33] I solve the model by collocation: I compute the value function at the collocation nodes and obtain the Chebyshev coefficients by inverting the interpolation (basis) matrix at those nodes.

Algorithm: Transition Path (backward–forward shooting)

Inputs: the estimated parameters $(\widehat{\mathbf{\Gamma}}, \widehat{\mathbf{\Delta}})$; pre- and post-shock technologies Λ_{pre} and Λ_{post}; CES weights $\{\alpha_j\}$ and elasticity σ; discount β, exit probability χ, taste scale ϱ; horizon T, damping ζ, tolerance *tol*.

Output: equilibrium log-price path $\{\tilde{\boldsymbol{p}}_t^\star\}_{t=0}^{T-1}$, value coefficients $\{\mathbf{\Phi}_t\}$, outputs $\{Y_{jt}\}$.

Endpoints. Solve the pre- and post-shock stationary equilibria with the stationary algorithm to obtain $(\tilde{\boldsymbol{p}}_{\text{pre}}, \mathbf{\Phi}_{\text{pre}})$ and $(\tilde{\boldsymbol{p}}_{\text{post}}, \mathbf{\Phi}_{\text{post}})$.

Initialize the price path at $\tilde{\boldsymbol{p}}_t^{(0)} = \tilde{\boldsymbol{p}}_{\text{post}} + (\tilde{\boldsymbol{p}}_{\text{pre}} - \tilde{\boldsymbol{p}}_{\text{post}})\, e^{-t/t_d}$, where t_d is a decay constant, with $\tilde{\boldsymbol{p}}_{T-1}^{(0)} = \tilde{\boldsymbol{p}}_{\text{post}}$ and each period renormalized to $P(\tilde{\boldsymbol{p}}_t^{(0)}) = 1$.

Repeat for $n = 0, 1, \ldots$ until $\max_{t<T-1,\, j\in\mathcal{J}} |\tilde{p}_{jt}^{(n+1)} - \tilde{p}_{jt}^{(n)}| < tol$:

- **Solve backward.** Set $\mathbf{\Phi}_{T-1} = \mathbf{\Phi}_{\text{post}}$ and, for $t = T-2, \ldots, 0$, compute $\mathbf{\Phi}_t$ from the one-step Bellman equation (56) at prices $\tilde{\boldsymbol{p}}_t^{(n)}$ and technology Λ_t, given the next-period coefficients $\mathbf{\Phi}_{t+1}$.
- **Simulate forward.** Draw one worker panel with common random numbers; burn in under Λ_{pre}, then from $t = 0$ let workers choose under $\{\mathbf{\Phi}_t, \tilde{\boldsymbol{p}}_t^{(n)}, \Lambda_t\}$.
- **Aggregate.** In each period t, form the choice probabilities $\mathbb{P}_{ijt}$ of (54) and the outputs Y_{jt} of (55).
- **Prices.** In each period t, set $\hat{P}_{jt} = \alpha_j\, (Y_t / Y_{jt})^{1/\sigma}$, renormalize $\hat{\boldsymbol{P}}_t \leftarrow \hat{\boldsymbol{P}}_t / P(\hat{\boldsymbol{P}}_t)$, and keep the terminal period at $\tilde{\boldsymbol{p}}_{\text{post}}$.
- **Update.** For $t < T-1$, $\tilde{\boldsymbol{p}}_t^{(n+1)} = (1-\zeta)\, \tilde{\boldsymbol{p}}_t^{(n)} + \zeta \log \hat{\boldsymbol{P}}_t$, with $\zeta \in (0, 1)$.

Return $\{\tilde{\boldsymbol{p}}_t^\star\}$, $\{\mathbf{\Phi}_t\}$, $\{Y_{jt}\}$.

C Appendix: Proofs

C.1 Identification of the Wage Equation Parameters under Learning

This subsection proves identification of the wage equation in the presence of multidimensional skills that are unobserved both by the econometrician and by the workers. The argument follows the pure-learning identification logic of Bunting et al. (2024), adapted to the multidimensional skill environment of this paper. The key idea is that workers sort into occupations based on beliefs about their latent skills, but those beliefs are formed from the same wage and occupational histories observed by the econometrician. Controlling for the worker's history therefore controls for belief-based selection, because workers hold no private information beyond what the econometrician observes.

The wage equation is

$$w_{it}(j) = p_{jt} + h_j(\boldsymbol{\tau}_{it}) + \boldsymbol{\lambda}_j' \boldsymbol{\theta}_i + \varepsilon_{ijt}, \tag{57}$$

where p_{jt} is the occupation-time log price, $h_j(\boldsymbol{\tau}_{it})$ is the human-capital component, $\boldsymbol{\lambda}_j$ is the occupation loading on the latent skill vector, $\boldsymbol{\theta}_i$ is fixed over the worker's life,

and ε_{ijt} is the transitory wage shock.

I make the following assumptions.

Assumption 1 (Linear-Gaussian skills and wage shocks) *Latent skills are Gaussian within group, $\boldsymbol{\theta}_i \mid g(i) = g \sim N(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ with $\boldsymbol{\Sigma}_g \succ 0$, and are fixed over the worker's life. Wage shocks are $\varepsilon_{ijt} \sim N(0, \varsigma^2_{\varepsilon j})$ with $\varsigma^2_{\varepsilon j} > 0$, independent across workers and periods, independent of $\boldsymbol{\theta}_i$, and realized after the occupational choice is made.*

Assumption 2 (Pure-learning selection) *Let*

$$\mathcal{H}_{it} = \left(g(i),\, d_i^{t-1},\, w_i^{t-1},\, \boldsymbol{\tau}_i^t,\, \{p_{js}\}_{j\in\mathcal{J}, s\geq t}\right)$$

be the history observed by the econometrician when worker i chooses d_{it}. Workers observe no private signal about $\boldsymbol{\theta}_i$ beyond this history:

$$d_{it} \;\perp\!\!\!\perp\; \left(\boldsymbol{\theta}_i,\, \{\varepsilon_{ijs} : s \geq t,\, j \in \mathcal{J}\}\right) \;\mid\; \mathcal{H}_{it}.$$

Assumption 3 (Overlap) *Every occupation used in the identifying moments is chosen with positive probability in the histories used:* $\Pr(d_{it} = j \mid \mathcal{H}_{it}) > 0$.

Assumption 4 (Repeated observations and mobility) *Within each occupation j, task experience $\boldsymbol{\tau}_{it}$ varies across the workers observed in the same occupation and year. Across occupations, workers are mobile enough that, for each occupation j and each anchor $a_k \in \mathcal{J}_A = \{a_1, \ldots, a_K\}$, wages in j and in a_k are observed for the same worker at two distinct dates.*

Assumption 5 (Normalizations) *The K anchor occupations load on a single skill each, $\boldsymbol{\lambda}_{a_k} = \boldsymbol{e}_k$ for $k = 1, \ldots, K$; the location of the skill distribution is fixed by $\sum_{g\in\mathcal{G}} \pi_g \boldsymbol{\mu}_g = \mathbf{0}$, with π_g the population share of group g; and the returns to task experience are measured from a base level, $h_j(\boldsymbol{\tau}^0) = 0$ for every j.*

Assumption 2 is the central restriction. Workers sort into occupations on the basis of their beliefs, but those beliefs are formed from histories that the econometrician also observes, so conditional on $\mathcal{H}_{it}$ their choice carries no additional information about unobserved skills. Assumption 3 guarantees that inverse-probability weights used to control for selection are well identified. Assumption 4 ensures that workers are observed for enough periods and that there are enough movers across occupations to identify the various components of the wage equation: experience variation within an occupation–year separates the returns to task experience from the occupation price; mobility across occupations pins down the wage covariances that in turn separate the permanent multidimensional skill component and the stationary price vector from idiosyncratic noise. Assumption 5 is needed to separately identify each component of the wage equation. In particular it fixes the rotation, location and scale of the latent

skill space, which the wage equation leaves free: the anchors pin down the rotation and scale, the zero-mean condition on the population of workers the location, and the base experience level the split between the price of an occupation and the return to experience.

Theorem 1 (Identification of the wage equation parameters under learning) *Suppose Assumptions 1–5 hold. Then the wage-equation parameters*

$$\left\{ (p_{jt}, \boldsymbol{\lambda}_j, \boldsymbol{\delta}_j, \varsigma^2_{\varepsilon j})_{j \in \mathcal{J}, t}, \; (\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)_{g \in \mathcal{G}} \right\}$$

are identified from the joint distribution of observed wages, occupational choices, groups, and predetermined state variables.

Proof of Theorem 1.

I first show that choice probabilities conditional on the observed history serve as inverse-probability weights that control for belief-driven selection into occupations. Fix a worker in group g, and consider a realized occupational history

$$d^t = (d_1, \ldots, d_t).$$

After subtracting the occupation price and human-capital component, each past wage is a noisy signal of the worker's latent skill:

$$S_{is} \equiv w_{is}(d_s) - p_{d_s s} - h_{d_s}(\boldsymbol{\tau}_{is}) = \boldsymbol{\lambda}'_{d_s} \boldsymbol{\theta}_i + \varepsilon_{i d_s s}.$$

Stack the signals observed before period t into

$$\boldsymbol{S}_{i,t-1} = (S_{i1}, \ldots, S_{i,t-1})'.$$

Let $\boldsymbol{\Lambda}_{i,t-1}$ be the matrix whose sth row is $\boldsymbol{\lambda}'_{d_s}$, and let

$$\boldsymbol{\Omega}_{i,t-1} = \operatorname{diag}\left(\varsigma^2_{\varepsilon d_1}, \ldots, \varsigma^2_{\varepsilon d_{t-1}} \right).$$

Because the prior over $\boldsymbol{\theta}_i$ is Gaussian and the wage signals are linear and Gaussian, the posterior is Gaussian:

$$\boldsymbol{\theta}_i \mid \mathcal{H}_{it}, g(i) = g \sim N\left(\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it} \right),$$

where

$$\tilde{\boldsymbol{\Sigma}}_{it} = \left(\boldsymbol{\Sigma}_g^{-1} + \boldsymbol{\Lambda}'_{i,t-1} \boldsymbol{\Omega}^{-1}_{i,t-1} \boldsymbol{\Lambda}_{i,t-1} \right)^{-1},$$

and

$$\tilde{\boldsymbol{\theta}}_{it} = \tilde{\boldsymbol{\Sigma}}_{it} \left(\boldsymbol{\Sigma}_g^{-1} \boldsymbol{\mu}_g + \boldsymbol{\Lambda}'_{i,t-1} \boldsymbol{\Omega}^{-1}_{i,t-1} \boldsymbol{S}_{i,t-1} \right).$$

These are also the belief objects that enter the worker's state $\boldsymbol{x}_{it}$. Conditional on the

history, the potential log wage in occupation j is normally distributed according to

$$w_{it}(j) \mid \mathcal{H}_{it}, g(i) = g \sim N\left(p_{jt} + h_j(\boldsymbol{\tau}_{it}) + \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}}_{it},\ \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\Sigma}}_{it}\boldsymbol{\lambda}_j + \varsigma_{\varepsilon j}^2\right).$$

The history therefore contains everything that is known about the worker's latent skills. Her beliefs coincide with the econometrician's beliefs only if her prior is the true skill distribution, but I do not need this to identify the wage equation: I only need that she holds no private information beyond what I observe.

The observed wage is a selected outcome, because workers choose occupations on the basis of their beliefs about their skills. These beliefs are a function of the observed history, so conditional on $\mathcal{H}_{it}$ the choice carries no further information about $\boldsymbol{\theta}_i$ or about future wage shocks (Assumption 2). Let $q_t(j \mid \mathcal{H}_{it}) \equiv \Pr(d_{it} = j \mid \mathcal{H}_{it})$ be the conditional choice probability, which I identify from the joint distribution of histories and choices. The timing of the model shows why this probability works as a weight: the worker enters period t with history $\mathcal{H}_{it}$, chooses her occupation, and only then observes her wage, so the joint density factors as

$$f_{w^T, D^T \mid g}(w^T, d^T) = \prod_{t=1}^{T} \underbrace{q_t(d_t \mid \mathcal{H}_{it})}_{\text{observed choices}}\ \underbrace{f_{w_t \mid d_t, \mathcal{H}_{it}, g}(w_t)}_{\text{observed wages}} = \prod_{t=1}^{T} \underbrace{q_t(d_t \mid \mathcal{H}_{it})}_{\text{selection}}\ \underbrace{f_{w_t(d_t) \mid \mathcal{H}_{it}, g}(w_t)}_{\text{potential wages}}.$$

The second equality is where the potential wage appears. The wage I observe in j is the potential wage in j, $w_{it} = w_{it}(d_{it})$; and since $w_{it}(j) = p_{jt} + h_j(\boldsymbol{\tau}_{it}) + \boldsymbol{\lambda}_j'\boldsymbol{\theta}_i + \varepsilon_{ijt}$ depends only on $(\boldsymbol{\theta}_i, \varepsilon_{ijt})$, which Assumption 2 makes independent of the choice given the history, the conditioning on $d_t = j$ can be dropped:

$$f_{w_t \mid d_t = j, \mathcal{H}_{it}, g}(w) = f_{w_t(j) \mid d_t = j, \mathcal{H}_{it}, g}(w) = f_{w_t(j) \mid \mathcal{H}_{it}, g}(w).$$

Conditional on their history, the workers who choose j are not a pool selected on skills. All the selection sits in the choice probabilities, so I can use them as weights: dividing by them recovers the potential wage path along any supported sequence $d^T = (d_1, \ldots, d_T)$,

$$\underbrace{\prod_{t=1}^{T} f_{w_t(d_t) \mid \mathcal{H}_{it}, g}(w_t)}_{\text{potential wage path}} = \frac{\overbrace{f_{w^T, D^T \mid g}(w^T, d^T)}^{\text{observed}}}{\underbrace{\prod_{t=1}^{T} q_t(d_t \mid \mathcal{H}_{it})}_{\text{identified from choices}}}.$$

Everything on the right is in the data, so the distribution of potential log wages along every supported path is identified. The weights scale each worker by the inverse of the probability of her path; without them, workers who believe that they have high cognitive skills, and who are therefore more likely to self-select into occupations with a high cognitive task content, would be over-represented in those occupations. Every moment I use below is a moment of this reweighted distribution.

Under the wage equation, the potential wage path associated with a sequence d^T,

$$w_i(d^T) = \big(w_{i1}(d_1), \ldots, w_{iT}(d_T)\big)',$$

is jointly Gaussian, because it is a linear map of the Gaussian skill vector $\boldsymbol{\theta}_i$ and the Gaussian wage shocks. The mean and covariance therefore characterize the path of potential wages, and repeated measurement identifies them: for a worker observed multiple times, the observed wages share her permanent skill $\boldsymbol{\theta}_i$ while shocks are independent over time. After reweighting, each wage is the same latent skill vector, with loadings that change if the worker switches occupation, plus independent noise. I write the three moments below, after netting out the returns to task experience.

Identification of the returns to task experience δ_j. Because $h_j(\cdot)$ is a known function up to unknown parameters, let $h_j(\boldsymbol{\tau}) = \boldsymbol{\delta}_j' \boldsymbol{D}(\boldsymbol{\tau})$ be a linear-in-parameters representation, where

$$\boldsymbol{D}(\boldsymbol{\tau})$$

is a vector of indicators for the task-specific tenure bins, and $\boldsymbol{\delta}_j$ stacks the corresponding coefficients. Using the reweighted potential wage distribution, compare two wages for the same worker in the same occupation j at two dates t and s:

$$w_{it}(j) - w_{is}(j) = (p_{jt} - p_{js}) + \big[h_j(\boldsymbol{\tau}_{it}) - h_j(\boldsymbol{\tau}_{is})\big] + \varepsilon_{ijt} - \varepsilon_{ijs}.$$

Comparing a worker with herself, in the same occupation, differences out her permanent skill $\boldsymbol{\lambda}_j' \boldsymbol{\theta}_i$; the price is absorbed by occupation-time effects (and cancels out if prices are constant, i.e., if the model is generated by a stationary equilibrium). What is left is the return to the task experience she accumulated in between. The wage shocks are mean zero and independent of predetermined experience, so $\boldsymbol{\delta}_j$ is identified, provided experience varies across the workers observed in the same occupation and year—otherwise experience is indistinguishable from calendar time. With $h_j(\cdot)$ in hand, define the human-capital-adjusted log wage

$$Z_{it}(j) \equiv w_{it}(j) - h_j(\boldsymbol{\tau}_{it}).$$

Then

$$Z_{it}(j) = p_{jt} + \boldsymbol{\lambda}_j' \boldsymbol{\theta}_i + \varepsilon_{ijt}.$$

Everything that follows uses the means and covariances of $Z_{it}(j)$, which is the wage stripped of experience and therefore a direct measurement of skill.

The moments of the adjusted wage. Three moments of the reweighted distribution of adjusted wages are identified, and they are all I use below. Each of them looks at the workers observed in a given occupation at a given date, so it is enough to reweight each worker by the probability of the choice she made at that date, $q_t(j \mid \mathcal{H}_{it})$. The logic

is the same as above: conditional on her history, the choice of a worker reveals nothing further about her skills or her future wage shocks (Assumption 2), so the workers who choose j at t, weighted by the inverse of the probability of doing so, are representative of all the workers with the same history (Assumption 3 ensures that this probability is positive). For these workers, the observed adjusted wage Z_{it} is the potential one, $Z_{it}(j)$.

The first is the mean adjusted wage in occupation j, group g, period t,

$$m_{jgt} \equiv \mathbb{E}\left[\frac{\mathbf{1}\{d_{it} = j\}\, Z_{it}}{q_t(j \mid \mathcal{H}_{it})} \;\middle|\; g(i) = g\right] = \mathbb{E}[Z_{it}(j) \mid g(i) = g] = p_{jt} + \boldsymbol{\lambda}_j' \boldsymbol{\mu}_g.$$

Only the workers observed in j at t enter, each weighted by the inverse of the probability that, given her history, she chose j at t; after reweighting, their average adjusted wage is the average across all the workers of group g.

The second is the covariance between wages in occupations j and ℓ at two different dates, computed on the workers observed in j at t *and* in ℓ at s, each date reweighted by its own choice probability,

$$C_{j\ell,g} \equiv \mathbb{E}\left[\frac{\mathbf{1}\{d_{it} = j,\, d_{is} = \ell\}\, (Z_{it} - m_{jgt})(Z_{is} - m_{\ell gs})}{q_t(j \mid \mathcal{H}_{it})\, q_s(\ell \mid \mathcal{H}_{is})} \;\middle|\; g(i) = g\right] = \boldsymbol{\lambda}_j' \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_\ell, \qquad s < t,$$

which contains only permanent skill, since the transitory shocks are independent across periods. I reweight the two dates in sequence. The wage at the earlier date s already belongs to the history at t, so the weight for the choice at t leaves it unchanged; the weight for the choice at s then removes the selection into ℓ. The third is the same-period variance in occupation j,

$$V_{jg} \equiv \mathbb{E}\left[\frac{\mathbf{1}\{d_{it} = j\}\, (Z_{it} - m_{jgt})^2}{q_t(j \mid \mathcal{H}_{it})} \;\middle|\; g(i) = g\right] = \boldsymbol{\lambda}_j' \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j + \varsigma_{\varepsilon j}^2,$$

which contains skill and noise together. Comparing the weighted covariance $C_{j\ell,g}$ and the weighted variance V_{jg} allows me to identify the variance of skills of each group g and the occupation-specific noise separately. Without the weights each of these would be computed on a selected sample—the workers who chose that occupation, or that pair of occupations.

Identification of the skill distribution: the means μ_g and the anchor prices. Let the K anchor occupations be

$$\mathcal{J}_A = \{a_1, \ldots, a_K\},$$

with normalized loadings

$$\boldsymbol{\lambda}_{a_k} = \boldsymbol{e}_k.$$

Stack the mean adjusted wages of the anchor occupations:

$$\boldsymbol{m}_{Agt} = \left(m_{a_1gt}, \ldots, m_{a_Kgt}\right)'.$$

Also define the vector of anchor prices

$$\boldsymbol{p}_{At} = \left(p_{a_1t}, \ldots, p_{a_Kt}\right)'.$$

Because the anchor loading matrix is the identity, the mean adjusted wages identify the sum of the mean skill of group g in dimension k and the anchor price in the normalized occupation

$$\boldsymbol{m}_{Agt} = \boldsymbol{p}_{At} + \boldsymbol{\mu}_g.$$

The price is common across groups, the skill mean is not. Using the normalization

$$\sum_{g \in \mathcal{G}} \pi_g \boldsymbol{\mu}_g = \mathbf{0},$$

I obtain

$$\boldsymbol{p}_{At} = \sum_{g \in \mathcal{G}} \pi_g \boldsymbol{m}_{Agt}.$$

Then

$$\boldsymbol{\mu}_g = \boldsymbol{m}_{Agt} - \boldsymbol{p}_{At}.$$

The anchor prices and the group skill means are therefore identified: differences in mean anchor wages across groups reveal the average skills of each group, and the normalization fixes their level. Because skills are constant over the life cycle, the same $\boldsymbol{\mu}_g$ is recovered at every date, which is an overidentifying restriction.

Identification of the skill distribution: the covariance matrices $\boldsymbol{\Sigma}_g$. Let $\boldsymbol{C}_{AA,g}$ be the $K \times K$ covariance matrix whose (k, ℓ) element is

$$C_{a_k a_\ell, g} = \operatorname{Cov}\left(Z_{it}(a_k), Z_{is}(a_\ell) \mid g(i) = g\right), \qquad t \neq s.$$

Using the anchor normalization,

$$C_{a_k a_\ell, g} = \boldsymbol{e}_k' \boldsymbol{\Sigma}_g \boldsymbol{e}_\ell.$$

Therefore,

$$\boldsymbol{C}_{AA,g} = \boldsymbol{\Sigma}_g.$$

So the skill covariance matrix is read directly off wages in the anchor occupations: two wages of the same worker in the anchors in two different periods depend only on the worker's permanent skills, and their covariance is exactly $\boldsymbol{\Sigma}_g$.

Identification of the skill loadings $\boldsymbol{\lambda}_j$. For any non-anchor occupation j, define the vector of covariances between occupation j and the anchors:

$$\boldsymbol{C}_{jA,g} = \left(C_{ja_1,g}, \ldots, C_{ja_K,g}\right)'.$$

From the covariance formula,

$$\boldsymbol{C}'_{jA,g} = \boldsymbol{\lambda}'_j \boldsymbol{\Sigma}_g.$$

Since $\boldsymbol{\Sigma}_g$ has already been identified,

$$\boldsymbol{\lambda}'_j = \boldsymbol{C}'_{jA,g} \boldsymbol{\Sigma}_g^{-1}.$$

The loadings of every occupation are therefore identified by how strongly a worker's wage in occupation j co-moves with her wage in each anchor, which reveals how much occupation j pays for each skill. With more than one group, $\boldsymbol{\lambda}_j$ can be recovered from each group, and the equality of these estimates is, in principle, an overidentifying restriction.

Identification of the occupation prices p_{jt}. Once $\boldsymbol{\lambda}_j$ and $\boldsymbol{\mu}_g$ are known, the mean adjusted wage equation

$$m_{jgt} = p_{jt} + \boldsymbol{\lambda}'_j \boldsymbol{\mu}_g$$

identifies the occupation-time price:

$$p_{jt} = m_{jgt} - \boldsymbol{\lambda}'_j \boldsymbol{\mu}_g.$$

Prices are thus identified for every occupation and period: once I know what an occupation pays for skill, and how much skill each group has, whatever is left in the mean wage is the price. The same p_{jt} must emerge from every group, since all groups face the same prices.

Identification of the transitory wage variances $\varsigma^2_{\varepsilon j}$. Finally, the adjusted wage variance satisfies

$$V_{jg} = \boldsymbol{\lambda}'_j \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j + \varsigma^2_{\varepsilon j}.$$

The first term is now known, since $\boldsymbol{\lambda}_j$ and $\boldsymbol{\Sigma}_g$ have been identified. Subtracting it,

$$\varsigma^2_{\varepsilon j} = V_{jg} - \boldsymbol{\lambda}'_j \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j.$$

And what remains of the wage variance, once the part explained by skill is taken out, is the transitory noise.

Combining the arguments above identifies

$$\left\{ \left(p_{jt}, \boldsymbol{\lambda}_j, \boldsymbol{\delta}_j, \varsigma^2_{\varepsilon j}\right)_{j \in \mathcal{J}, t}, \left(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g\right)_{g \in \mathcal{G}} \right\}.$$

This proves Theorem 1. □

Specialization to two skills. In the empirical model there are two task-specific skills, cognitive and manual. Let a_C denote the cognitive anchor occupation and a_M the manual anchor occupation. The normalization is

$$\boldsymbol{\lambda}_{a_C} = (1,0)', \qquad \boldsymbol{\lambda}_{a_M} = (0,1)'.$$

The covariance matrix of latent skills for group g is identified from distinct-date covariances of adjusted wages in the two anchor occupations:

$$\boldsymbol{\Sigma}_g = \begin{pmatrix} C_{a_C a_C, g} & C_{a_C a_M, g} \\ C_{a_M a_C, g} & C_{a_M a_M, g} \end{pmatrix}.$$

For any other occupation j, its cognitive and manual loadings are recovered from its covariances with the two anchors:

$$\boldsymbol{\lambda}_j' = \left(C_{j a_C, g}, C_{j a_M, g}\right) \boldsymbol{\Sigma}_g^{-1}.$$

Thus, in the two-skill version of the model, the cognitive and manual task returns of every occupation are identified from how adjusted wages in that occupation covary with adjusted wages in the cognitive and manual anchor occupations.

C.2 Identification of the Dynamic Choice Parameters

With the wage equation identified, what remains to identify are the occupation-specific amenities $\{\eta_{jg}\}$ and the switching cost κ_0. Let $\boldsymbol{x}$ denote the state in equation (23), and let g and $d = d_{t-1}$ denote the group and origin occupation contained in $\boldsymbol{x}$. Write

$$\tilde{w}_j(\boldsymbol{x}) \equiv \mathbb{E}[\log W_{ijt} \mid \boldsymbol{x}].$$

For a function f, the assignment operator Q_j takes the expectation of f over next period's state when the worker is assigned to occupation j at t:

$$(Q_j f)(\boldsymbol{x}) \equiv \int f(\boldsymbol{x}')\, Q(d\boldsymbol{x}' \mid \boldsymbol{x}, j).$$

Q_j^s applies the same operator for s consecutive periods, so $Q_j^s f(\boldsymbol{x})$ propagates the state forward s periods under repeated choice of j. The choice is imposed exogenously: I set $d_{im} = j$ for every m from t to $t+s-1$. The conditional value function can be written using this operator as

$$v_j(\boldsymbol{x}) = \tilde{w}_j(\boldsymbol{x}) + \eta_{jg} - \kappa_0 \mathbf{1}_{\{j \neq d\}} + \beta (Q_j EV)(\boldsymbol{x}).$$

The conditional choice probabilities are

$$\mathbb{P}_j(\boldsymbol{x}) = \frac{\exp(\nu_j(\boldsymbol{x})/\varrho)}{\sum_{m\in\mathcal{J}}\exp(\nu_m(\boldsymbol{x})/\varrho)}.$$

With a Type-I extreme-value assumption and the re-centering of the shocks, the integrated value function is

$$EV(\boldsymbol{x}) = \varrho \log \sum_{m\in\mathcal{J}} \exp\left(\frac{\nu_m(\boldsymbol{x})}{\varrho}\right).$$

Define the choice-specific value correction

$$\psi_j(\boldsymbol{x}) \equiv EV(\boldsymbol{x}) - \nu_j(\boldsymbol{x}).$$

This term measures the value lost if the worker is required to commit to j in the current period, before observing the taste shocks, rather than being free to choose optimally, while future choices remain optimal. This is the probability correction term of Section 4.2. Below I use the inversion argument of Hotz and Miller (1993) to show that, under the multinomial logit assumption, these terms are recovered directly from the choice probabilities, $\psi_j(\boldsymbol{x}) = -\varrho \log \mathbb{P}_j(\boldsymbol{x})$, which can be read as a penalty for using j as the reference action. From the CCP representation this term has an intuitive revealed-preference interpretation: it is small when j is chosen often (close to optimal) and large when j is rarely chosen.

Assumption 6 (Stationarity and known primitives) *The data are generated by the stationary equilibrium of Section 3.6. The wage equation and wage-signal distribution are identified, so $\tilde{w}_j$ and Q_j are known for every occupation j. The discount factor $\beta \in (0,1)$ and taste-shock scale $\varrho > 0$ are fixed. The conditions of Lemma 2 hold, so the Bellman equation has a unique integrated value function EV.*

Assumption 7 (Support along the controlled path and limit conditions) *For every supported state $\boldsymbol{x}$ and every occupation j, the wage equation, CCPs, and transition law are identified at the states reached under $Q_j^s(\boldsymbol{x}, \cdot)$, for every $s \geq 1$, and $\mathbb{P}_j > 0$ on those states. Moreover,*

$$\sum_{s=1}^{\infty} \beta^{s-1} Q_j^s\left[|\tilde{w}_j| + \varrho|\log \mathbb{P}_j|\right](\boldsymbol{x}) < \infty, \qquad \lim_{S\to\infty} \beta^S Q_j^{S+1}|EV|(\boldsymbol{x}) = 0.$$

I write $\boldsymbol{x}_{dg} = (\,\cdot\,, d, g)$ for the state of a worker whose last-period occupation was d and who belongs to group g, with $\cdot$ collecting the remaining state variables (mean and variance-covariance of the belief distribution, and task-specific tenure).

Assumption 8 (Comparison support) *For every group g and occupation $j \neq 0$, there is a supported state $\boldsymbol{x}_{dg}$ with $d \notin \{j, 0\}$. In addition, there is a supported state $\boldsymbol{x}_{dg}$ whose occupation d admits a destination $j \neq d$. Thus observing CCPs from a common state permits a comparison between two moves from a common origin for each (j, g), and at least one comparison between moving to any occupation and staying in the current occupation.*

The identification argument has two steps. First, the Hotz–Miller inversion and the known transition law recover continuation values from future wages and observed choice probabilities. Second, comparing two moves from a common origin identifies amenities, after which comparing a move with a stay identifies the switching cost.

Lemma 1 (CCP inversion along the j-assignment path) *Under Assumptions 6–7, for every supported state $\boldsymbol{x}$ and occupation j,*

$$\psi_j(\boldsymbol{x}) \equiv EV(\boldsymbol{x}) - \nu_j(\boldsymbol{x}) = -\varrho \log \mathbb{P}_j(\boldsymbol{x}).$$

Moreover,

$$\nu_j(\boldsymbol{x}) = \underbrace{\tilde{w}_j(\boldsymbol{x}) + \beta \Psi_j(\boldsymbol{x})}_{\text{known}} + \frac{\eta_{jg}}{1-\beta} - \kappa_0 \mathbf{1}_{\{j \neq d\}},$$

where

$$\Psi_j(\boldsymbol{x}) \equiv \sum_{s=1}^{\infty} \beta^{s-1} \left[Q_j^s \left(\tilde{w}_j + \psi_j \right) \right] (\boldsymbol{x}).$$

Because $\psi_j = -\varrho \log \mathbb{P}_j$, the term $\Psi_j(\boldsymbol{x})$ depends only on identified wages and observed choice probabilities, and is therefore known.

Proof. Taking logs of the logit CCP gives

$$\varrho \log \mathbb{P}_j(\boldsymbol{x}) = \nu_j(\boldsymbol{x}) - \varrho \log \sum_{m \in \mathcal{J}} \exp\left(\frac{\nu_m(\boldsymbol{x})}{\varrho} \right) = \nu_j(\boldsymbol{x}) - EV(\boldsymbol{x}),$$

so that $\psi_j(\boldsymbol{x}) = EV(\boldsymbol{x}) - \nu_j(\boldsymbol{x}) = -\varrho \log \mathbb{P}_j(\boldsymbol{x})$. This identity holds for every occupation j, whether or not the worker actually chooses j: when the choice probability of j is low, its value lies far below $EV(\boldsymbol{x})$ and the correction is large. Consider a state $\boldsymbol{x}$ and an occupation j. If the worker is repeatedly assigned to j, she never switches occupation and therefore never pays the switching cost. At every state $\boldsymbol{y}$ along this path I substitute $EV(\boldsymbol{y}) = \nu_j(\boldsymbol{y}) + \psi_j(\boldsymbol{y})$ into the choice-specific value index, which gives a recursion for the ex-ante value function through the assignment operator,

$$EV(\boldsymbol{y}) = \tilde{w}_j(\boldsymbol{y}) + \eta_{jg} + \psi_j(\boldsymbol{y}) + \beta (Q_j EV)(\boldsymbol{y}).$$

Applying the assignment operator Q_j at the initial state and iterating this identity S

times yields

$$(Q_j EV)(\boldsymbol{x}) = \sum_{s=1}^{S} \beta^{s-1} \left[Q_j^s \left(\tilde{w}_j + \psi_j \right) \right] (\boldsymbol{x}) + \frac{1-\beta^S}{1-\beta} \eta_{jg} + \beta^S (Q_j^{S+1} EV)(\boldsymbol{x}).$$

When $S \to \infty$, Assumption 7 gives

$$(Q_j EV)(\boldsymbol{x}) = \Psi_j(\boldsymbol{x}) + \frac{\eta_{jg}}{1-\beta}.$$

Substituting this expression into the choice-specific value index and using $1 + \beta/(1-\beta) = 1/(1-\beta)$ gives

$$v_j(\boldsymbol{x}) = \tilde{w}_j(\boldsymbol{x}) + \beta \Psi_j(\boldsymbol{x}) + \frac{\eta_{jg}}{1-\beta} - \kappa_0 \mathbf{1}_{\{j \neq d\}}.$$

□

Repeated assignment to j applies the correction term at every future state. This choice of reference action is convenient: because the worker always stays in j, she pays the switching cost only on the initial move, whereas she receives the amenity in every period, so that it enters as a geometric discounted sum with factor $1/(1-\beta)$.

This representation is a special case of the forward CCP representation of Arcidiacono and Miller (2011). Their general representation can average over several future reference actions, whereas I place all the weight on j. As a result, Q_j^s weights future states only through their transition probabilities, and the choice probabilities enter only through the correction $\psi_j = -\varrho \log \mathbb{P}_j$.

Theorem 2 (Identification of the dynamic choice parameters) *Let Assumptions 6–8 hold and normalize $\eta_{0g} = 0$ for every g. Then*

$$\left(\{\eta_{jg}\}_{j \neq 0,\, g \in \mathcal{G}}, \kappa_0 \right)$$

is identified.

Proof. By Lemma 1, the index $\tilde{w}_j(\boldsymbol{x}) + \beta \Psi_j(\boldsymbol{x})$ is known, and the lemma implies

$$\varrho \log \frac{\mathbb{P}_j(\boldsymbol{x})}{\mathbb{P}_m(\boldsymbol{x})} - \left[\left(\tilde{w}_j(\boldsymbol{x}) + \beta \Psi_j(\boldsymbol{x}) \right) - \left(\tilde{w}_m(\boldsymbol{x}) + \beta \Psi_m(\boldsymbol{x}) \right) \right] = \frac{\eta_{jg} - \eta_{mg}}{1-\beta} - \kappa_0 \left(\mathbf{1}_{\{j \neq d\}} - \mathbf{1}_{\{m \neq d\}} \right). \tag{58}$$

Amenities. Fix g and $j \neq 0$, and evaluate equation (58) at a supported state $\boldsymbol{x}_{dg}$ with $d \notin \{j, 0\}$, as guaranteed by Assumption 8. Because both j and 0 are then moves, their switching costs cancel. Comparing occupation j with the baseline normalized

occupation where $\eta_{0g} = 0$ yields

$$\eta_{jg} = (1-\beta)\left\{\varrho \log \frac{\mathbb{P}_j(\boldsymbol{x}_{dg})}{\mathbb{P}_0(\boldsymbol{x}_{dg})} - \left[\left(\tilde{w}_j(\boldsymbol{x}_{dg}) + \beta\Psi_j(\boldsymbol{x}_{dg})\right) - \left(\tilde{w}_0(\boldsymbol{x}_{dg}) + \beta\Psi_0(\boldsymbol{x}_{dg})\right)\right]\right\}.$$

Thus common-origin comparisons between two moves identify every amenity relative to the base occupation.

Switching cost. At a supported state $\boldsymbol{x}_{dg}$ whose occupation d admits a destination $j \neq d$, compare moving to j with staying in d. Equation (58) gives

$$\kappa_0 = \frac{\eta_{jg} - \eta_{dg}}{1-\beta} - \varrho \log \frac{\mathbb{P}_j(\boldsymbol{x}_{dg})}{\mathbb{P}_d(\boldsymbol{x}_{dg})} + \left[\left(\tilde{w}_j(\boldsymbol{x}_{dg}) + \beta\Psi_j(\boldsymbol{x}_{dg})\right) - \left(\tilde{w}_d(\boldsymbol{x}_{dg}) + \beta\Psi_d(\boldsymbol{x}_{dg})\right)\right].$$

All terms on the right-hand side are now known, so κ_0 is identified. Intuitively, wages, amenities and continuation values imply a certain amount of switching; if workers move less than that, the gap is attributed to the switching cost. □

Remark. A just-identifying subset contains

$$|\mathcal{G}|(|\mathcal{J}| - 1) + 1$$

log-odds equations: one move-versus-move comparison for each (j, g) with $j \neq 0$, and one move-versus-stay comparison for the common switching cost. In the application this is $3 \times 7 + 1 = 22$. Other supported states and destination comparisons provide overidentifying restrictions and are used in the estimation procedure.

This comparison argument requires that the switching cost be the same across destinations and be paid only when the worker moves. If switching costs varied with the destination, the direction of the move, or experience, they would no longer cancel in the first step, and I would have to identify them jointly with the amenities.

C.3 Bellman Fixed Point

The worker's dynamic problem has a unique solution. The identification proof for the dynamic choice parameters relies on this property: given amenities and switching costs, the model pins down continuation values.

Throughout, $\boldsymbol{x}_t$ is the state defined in equation (23), dropping the worker index. The price path is part of the state, so the future evolution of prices is known from the worker's point of view.

Let $f(\boldsymbol{x}' \mid \boldsymbol{x}, j)$ denote the density of next period's state when the worker chooses occupation j today; it includes the update of beliefs after the wage signal. The deter-

ministic value of choosing occupation j is

$$v_{jt}(\boldsymbol{x}_t; EV) = \mathbb{E}\left[\log W_{jt} \mid \boldsymbol{x}_t\right] + \eta_{jg} - \mathcal{C}(j, d_{t-1}) + \beta \int EV(\boldsymbol{x}')\, f(\boldsymbol{x}' \mid \boldsymbol{x}_t, j)\, \mathrm{d}\boldsymbol{x}',$$

where the current taste shocks have been integrated out, and the Bellman operator is

$$(\mathcal{T}EV)(\boldsymbol{x}_t) = \varrho \log \left[\sum_{j \in \mathcal{J}} \exp\left(\frac{v_{jt}(\boldsymbol{x}_t; EV)}{\varrho} \right) \right].$$

I normalize the taste shocks to have mean zero. Under the other common normalization, the value function would only shift by the constant $\varrho\gamma_E/(1-\beta)$, where γ_E is Euler's constant, and the CCPs would not change.

The difficulty in this setting is that beliefs are not bounded. The worker's beliefs about her skills, $\tilde{\boldsymbol{\theta}}$, are normal random variables that enter the expected log wage linearly, and hence her value function. The standard contraction argument for bounded value functions therefore does not apply. I use instead the weighted contraction mapping theorem of Boyd (1990) and show that $\mathcal{T}$ is a contraction once value functions are compared relative to the size of beliefs. This delivers existence, uniqueness, and convergence of value iteration from any starting value (Lemma 2).

Lemma 2 (Bellman contraction in a weighted norm) *Suppose $\beta \in (0,1)$, $\varrho > 0$, $\varsigma_{\varepsilon j}^2 > 0$ for every j, and prices, amenities, switching costs and returns to experience are bounded. Then the Bellman equation has a unique solution EV among value functions that are continuous in beliefs and grow at most linearly in beliefs, and iterating the Bellman operator from any such starting value converges to it at rate β.*

Proof.

Since the value function grows at most as fast as beliefs, I measure the distance between two value functions relative to a weight function w that captures the size of beliefs:

$$w(\boldsymbol{x}) = \left(1 + \|\tilde{\boldsymbol{\theta}}\|^2 + \operatorname{tr}\tilde{\boldsymbol{\Sigma}}\right)^{1/2}, \qquad \|EV - \widetilde{EV}\|_w = \sup_{\boldsymbol{x}} \frac{|EV(\boldsymbol{x}) - \widetilde{EV}(\boldsymbol{x})|}{w(\boldsymbol{x})}.$$

The constant bounds the parts of the payoff that do not depend on beliefs, such as prices, amenities and switching costs, and keeps w away from zero. The term $\|\tilde{\boldsymbol{\theta}}\|^2$ captures the level of beliefs, which drives the expected log wage. The term $\operatorname{tr}\tilde{\boldsymbol{\Sigma}}$ captures the residual uncertainty about skills. The last two terms sum to the expected squared size of the true skill vector, $\mathbb{E}[\|\boldsymbol{\theta}\|^2 \mid \boldsymbol{x}] = \|\tilde{\boldsymbol{\theta}}\|^2 + \operatorname{tr}\tilde{\boldsymbol{\Sigma}}$. A value function has finite w-norm exactly when it grows at most linearly in beliefs. I restrict attention to value functions that are continuous in beliefs and grow at most linearly in beliefs. The Bellman operator preserves continuity, because beliefs are updated by a normally distributed wage signal, so the expected continuation value is continuous in beliefs, and the log-sum of finitely many continuous choice-specific values is continuous.

By Boyd (1990), $\mathcal{T}$ is a contraction of modulus β on this set of value functions if three conditions hold:

1. *Monotonicity:* $EV \leq \widetilde{EV}$ implies $\mathcal{T}EV \leq \mathcal{T}\widetilde{EV}$.
2. *Linear growth:* $\mathcal{T}0$, the value of a worker who cares only about the current payoff, grows at most linearly in beliefs.
3. *Discounting:* $\mathcal{T}(EV + c\,w) \leq \mathcal{T}EV + \beta c\,w$ for every constant $c \geq 0$.

Condition 1. Only the continuation term of $v_{jt}(\boldsymbol{x}_t; EV)$ depends on EV. If $EV \leq \widetilde{EV}$ at every state, the expected continuation value is weakly higher under $\widetilde{EV}$, so $v_{jt}(\boldsymbol{x}_t; EV) \leq v_{jt}(\boldsymbol{x}_t; \widetilde{EV})$ for every occupation j. The log-sum is increasing in each choice-specific value, since its derivatives are the choice probabilities, which are positive, and therefore

$$(\mathcal{T}EV)(\boldsymbol{x}_t) = \varrho \log\Big[\sum_{j\in\mathcal{J}} \exp\Big(\frac{v_{jt}(\boldsymbol{x}_t; EV)}{\varrho}\Big)\Big] \leq \varrho \log\Big[\sum_{j\in\mathcal{J}} \exp\Big(\frac{v_{jt}(\boldsymbol{x}_t; \widetilde{EV})}{\varrho}\Big)\Big] = (\mathcal{T}\widetilde{EV})(\boldsymbol{x}_t).$$

Condition 2. With no continuation value, $\mathcal{T}0$ is the log-sum of the flow payoffs. It lies between the highest flow payoff and the highest flow payoff plus $\varrho \log|\mathcal{J}|$, the option value of the taste shocks with $|\mathcal{J}|$ choices. The only unbounded term of the flow payoff is the expected skill component $\boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}}$, which is bounded in absolute value by a constant times $\|\tilde{\boldsymbol{\theta}}\| \leq w(\boldsymbol{x})$; prices, amenities, switching costs and returns to experience are bounded by assumption. Hence $\mathcal{T}0$ is bounded by a constant times w, which is where the constant in w is needed.

Condition 3. The key property of w is that it does not grow on average from one period to the next. A wage signal moves beliefs in either direction, but not systematically: the expected belief next period equals the current belief. Learning does, however, shift mass between the two terms of w. As the worker learns, her beliefs move away from the prior mean, so $\|\tilde{\boldsymbol{\theta}}\|^2$ rises on average, while $\operatorname{tr}\tilde{\boldsymbol{\Sigma}}$ falls by exactly the same amount. Because priors are rational, the sum is the worker's own forecast of the squared size of her skill vector, $w(\boldsymbol{x})^2 = \mathbb{E}[1 + \|\boldsymbol{\theta}\|^2 \mid \boldsymbol{x}]$, and since skills are constant this forecast is a martingale. Next period's w^2 therefore equals this period's on average and, since the square root is concave,

$$\int w(\boldsymbol{x}')\, f(\boldsymbol{x}' \mid \boldsymbol{x}, j)\, \mathrm{d}\boldsymbol{x}' \leq w(\boldsymbol{x}).$$

The uncertainty term is essential here: with the level of beliefs alone, w would grow on average as the worker learns. Raising the continuation value from EV to $EV + c\,w$ raises each choice-specific value by $\beta c \int w(\boldsymbol{x}')\, f(\boldsymbol{x}' \mid \boldsymbol{x}, j)\, \mathrm{d}\boldsymbol{x}' \leq \beta c\, w(\boldsymbol{x})$. The log-sum of values that each rise by at most δ rises by at most δ, because its derivatives are the choice probabilities, which are nonnegative and sum to one. Hence $\mathcal{T}(EV + c\,w) \leq \mathcal{T}EV + \beta c\,w$.

All three conditions hold, so $\mathcal{T}$ is a contraction of modulus β, and the contraction mapping theorem gives existence, uniqueness and convergence of the iterations from

any starting value that is continuous in beliefs and grows at most linearly in beliefs. This fixed point is the integrated value function of the occupational choice problem. □

In the computation, the state space is a bounded grid, the expectation is replaced by Gauss–Hermite weights, which are positive and sum to one, and points outside the grid are projected back onto it. Beliefs are then bounded, the same argument applies in the ordinary sup norm, and the computed Bellman operator is also a contraction. The Smolyak coefficient iteration additionally interpolates between grid points, so this result does not cover it; I check its convergence numerically (Appendix D.5.1).

D Appendix: Estimation and Numerical Methods

D.1 Externally Set and Calibrated Parameters

Table 3: Externally Set and Calibrated Parameters

Parameter	Description	Empirical Moment	Value
Preferences & Demographics			
β	Discount factor		0.900
χ	Per-period exit hazard	Expected career length (12.5 yrs)	0.080
Education Distribution			
Low education (L)	Population share	Empirical shares	67.9%
Medium education (M)	Population share	Empirical shares	20.1%
High education (H)	Population share	Empirical shares	12.0%
Human Capital Accumulation			
$\boldsymbol{\omega}_j$	Task importance weights	Mihaylov and Tijdens (2019)	
CES Final-Good Aggregator			
σ	Elasticity of substitution	Althoff and Reichardt (2026)	1.57
α_j	CES aggregator weights	Stage-1 prices & outputs	Calibrated[a]
P	Aggregate price level	Final-good numeraire	$\equiv 1$

Notes: Parameters are either set externally, to standard values or to match aggregate statistics, or calibrated in closed form. The per-period exit hazard χ is exogenous and not internalized by workers; it only generates the stationary age-experience distribution, and is set externally to imply an expected career length of 12.5 years, close to the average career span in the large sample. [a]The CES aggregator weights α_j are calibrated in closed form as $\alpha_j = P_j Y_j^{1/\sigma} / (\sum_m P_m Y_m)^{1/\sigma}$ (equivalently $\alpha_j^\sigma = s_j P_j^{\sigma-1}$, with wage-bill share s_j), where $P_j = \exp(p_j)$ is the Stage-1 occupation price in levels and Y_j the occupation output (wage bill deflated by P_j). This makes the dual CES price index $P = (\sum_j \alpha_j^\sigma P_j^{1-\sigma})^{1/(1-\sigma)} = 1$ at the Stage-1 prices. The scale of $\{\alpha_j\}$ is the final-good numeraire, so the estimated price vector is itself the stationary equilibrium.

Table 3 lists the parameters that are set externally or calibrated outside the likelihood (Section 4). The discount factor $\beta = 0.9$ is a standard value in the dynamic discrete choice literature. The exit hazard $\chi = 0.08$ is set externally and implies an expected career length of 12.5 years, close to the average career span in the large sample; it is exogenous, not internalized by workers, and only serves to generate the stationary age-experience distribution (Section 3.6). Education shares π_g match the large sample: 67.9% low-educated, 20.1% medium-educated, and 12.0% high-educated workers. The task importance weights $\boldsymbol{\omega}_j$ are from Mihaylov and Tijdens (2019); ω_{jk} governs the time

a worker in occupation j spends on task k and therefore how task-specific experience accumulates. The CES elasticity of substitution is $\sigma = 1.57$, and the aggregator weights $\{\alpha_j\}$ are calibrated in closed form by equation (40), so that the CES inverse demand holds exactly at the Stage-1 prices and the final-good price index equals one; the scale of $\{\alpha_j\}$ is the final-good numeraire.

D.2 Stage 1 Estimation Results: Wage Technology and Skill Distributions

Table 4: Stage 1: Expectation-Maximization Estimates of Wage Equation Parameters

Panel A: Loadings, Shock Variances, and Occupational Prices

Occupation	$\lambda_{j,C}$	$\lambda_{j,M}$	$\varsigma^2_{\varepsilon j}$	p_j
Cluster 0	1.000	0.000	0.0235	1.538
Cluster 1	0.000	1.000	0.0200	1.443
Cluster 2	0.202	0.826	0.0197	1.442
Cluster 3	0.875	0.000	0.0271	1.520
Cluster 4	0.527	0.567	0.0210	1.481
Cluster 5	0.350	0.737	0.0211	1.496
Cluster 6	0.827	0.375	0.0264	1.522
Cluster 7	0.541	0.491	0.0218	1.455

Panel B: Task Importance Weights by Occupation

Occupation	$\omega_{j,C}$ (Cognitive)	$\omega_{j,M}$ (Manual)
Cluster 0	0.988	0.012
Cluster 1	0.009	0.991
Cluster 2	0.162	0.838
Cluster 3	0.327	0.673
Cluster 4	0.448	0.552
Cluster 5	0.692	0.308
Cluster 6	0.946	0.054
Cluster 7	0.082	0.918

Panel C: Skills Distribution by Education Group

	Low (L)	Medium (M)	High (H)
$\boldsymbol{\mu}_g = \begin{pmatrix} \mu_C \\ \mu_M \end{pmatrix}$	$\begin{pmatrix} -0.183 \\ -0.136 \end{pmatrix}$	$\begin{pmatrix} -0.075 \\ -0.030 \end{pmatrix}$	$\begin{pmatrix} 0.302 \\ 0.196 \end{pmatrix}$
$\boldsymbol{\Sigma}_g = \begin{pmatrix} \sigma_C^2 & \rho\sigma_C\sigma_M \\ \rho\sigma_M\sigma_C & \sigma_M^2 \end{pmatrix}$	$\begin{pmatrix} 0.082 & 0.007 \\ 0.007 & 0.069 \end{pmatrix}$	$\begin{pmatrix} 0.109 & 0.012 \\ 0.012 & 0.112 \end{pmatrix}$	$\begin{pmatrix} 0.097 & 0.008 \\ 0.008 & 0.227 \end{pmatrix}$
Implied correlation ρ_{CM}	0.099	0.109	0.055

Panel A reports the skill loadings λ_{jk}, the idiosyncratic wage–shock variance $\varsigma^2_{\varepsilon j}$, and the log occupation prices p_j (the occupation intercept in the wage equation). By normalization *Cluster 0* is the cognitive anchor ($\lambda_{0,C} = 1$, $\lambda_{0,M} = 0$) and *Cluster 1* the manual anchor ($\lambda_{1,C} = 0$, $\lambda_{1,M} = 1$). Returns to task-specific experience are estimated as a flexible step function of tenure with annual knots; the estimated profiles are shown in Figure 7. Panel B reports task importance weights ω_{jk} from the Mihaylov and Tijdens (2019) classification under the $|\mathcal{J}| = 8$ cluster partition; these govern task aggregation and time allocation within occupations. Panel C reports EM estimates of prior means $\boldsymbol{\mu}_g$ and covariance matrices $\boldsymbol{\Sigma}_g$ of cognitive (θ_C) and manual (θ_M) latent skills by education group $g \in \{L, M, H\}$ (genders pooled), with implied correlation ρ_{CM} below. Skill means are identified up to location and anchored by the sample-share-weighted normalization $\sum_g \pi_g \boldsymbol{\mu}_g = \mathbf{0}$, with π_g the estimation-sample group shares.

Table 4 reports the full estimation results. Panel A shows the skill loadings λ_{jk}, the idiosyncratic wage variances $\varsigma^2_{\varepsilon j}$, and the log occupation prices p_j. For identification, Cluster 0 is the cognitive anchor ($\lambda_0 = (1, 0)$) and Cluster 1 the manual anchor ($\lambda_1 = (0, 1)$). Occupations differ sharply in their loadings: some load mainly on cognitive skills (Clusters 0, 3, and 6), others mainly on manual skills (Clusters 1, 2, and 5), and Clusters 4 and 7 load comparably on both. I estimate the returns to task-specific experience as a stepwise function of tenure bins; Figure 7 reports the estimated profiles, which are aligned with the skill loadings: occupations that value a given skill dimension also reward experience in it. Panel B reports the task importance weights ω_{jk} from Mihaylov and Tijdens (2019) under the $|\mathcal{J}| = 8$ cluster partition, which govern time allocation within occupations and the accumulation of task-specific experience. Panel C shows the estimated skill distributions by education group (genders pooled), normalized so that the sample-share-weighted mean of skills across groups is zero, $\sum_g \pi_g \boldsymbol{\mu}_g = \mathbf{0}$. Mean skills rise with education and the cognitive comparative advantage widens.

D.3 Stage 2 Estimation Results: Dynamic Choice Parameters

Table 5: Stage 2: NFXP Estimates of Dynamic Choice Parameters

Switching cost and taste-shock scale	
κ_0 (switching cost)	1.103
ϱ (matched)	0.271

Occupation–education amenities η_{jg}

Occupation	Low (L)	Medium (M)	High (H)
Cluster 0 (base)	0.000	0.000	0.000
Cluster 1	0.169	0.088	−0.219
Cluster 2	0.188	0.119	−0.034
Cluster 3	0.176	0.123	0.116
Cluster 4	0.242	0.211	0.198
Cluster 5	0.191	0.143	0.133
Cluster 6	0.087	0.047	0.017
Cluster 7	0.204	0.134	0.061

Notes: The utility switching cost is a single constant κ_0, paid whenever the worker changes occupation and independent of the origin, the destination, and accumulated experience. Amenities η_{jg} are occupation- and education-specific, normalized so that the baseline occupation, Cluster 0, has $\eta = 0$ within each education group (21 free amenities). The taste-shock scale ϱ is weakly identified from continuation values without exogenous payoff shifters and is therefore set to match the average occupational mobility rate in the large sample.

D.4 EM Algorithm: Computational Details

The EM algorithm follows the derivations of James (2012) and Hincapié (2020), adapted to the structure of my model.

For worker i employed in occupation j at date t,

$$w_{ijt} \;=\; p_j \;+\; \boldsymbol{\lambda}_j' \boldsymbol{\theta}_i \;+\; h_j(\boldsymbol{\tau}_{it}) \;+\; \varepsilon_{ijt}, \qquad \varepsilon_{ijt} \sim \mathcal{N}\big(0, \varsigma^2_{\varepsilon j}\big), \tag{59}$$

where p_j is the occupation log price, $\boldsymbol{\theta}_i \in \mathbb{R}^K$ collects latent skills, $\boldsymbol{\lambda}_j \in \mathbb{R}^K_+$ are skill loadings, and $\boldsymbol{\tau}_{it} \in \mathbb{R}^K_+$ are raw cumulative task tenures. The term $h_j(\boldsymbol{\tau}_{it})$ is parameterized as a stepwise function, mapping the continuous tenure $\boldsymbol{\tau}_{it}$ into tenure bins with corresponding dummy coefficients to capture the non-linear wage progression flexibly. Workers belong to observed groups $g(i) \in \mathcal{G}$ defined by education. The skill prior is conditionally Gaussian within group,

$$\boldsymbol{\theta}_i \mid g(i) = g \;\sim\; \mathcal{N}\big(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g\big). \tag{60}$$

Because $g(i)$ is observed in the data, the model is a conditional Gaussian factor model. The group weights π_g are fixed at empirical sample shares and are not updated by EM.

The parameter vector iterated by the algorithm is

$$\boldsymbol{\Gamma}^q = \Big\{ (\boldsymbol{\mu}_g^q, \boldsymbol{\Sigma}_g^q)_{g\in\mathcal{G}},\; (p_j^q, \boldsymbol{\lambda}_j^q, \boldsymbol{\delta}_j^q, \varsigma_{\varepsilon j}^{2,q})_{j\in\mathcal{J}} \Big\},$$

with q indexing the EM iteration.

D.4.1 E-step

Conditional on $\boldsymbol{\Gamma}^q$ and on $g(i)$, the posterior of $\boldsymbol{\theta}_i$ is multivariate normal. Define the residual after netting out occupation prices and the tenure profile,

$$r_{it}^q \;:=\; w_{ijt} \;-\; p^q_{j(i,t)} \;-\; h^q_{j(i,t)}(\boldsymbol{\tau}_{it}). \tag{61}$$

Completing the square in $\boldsymbol{\theta}_i$ in the product of the wage likelihood implied by (59) and the group prior (60) delivers

$$\boldsymbol{V}_i^q = \left((\boldsymbol{\Sigma}^q_{g(i)})^{-1} + \sum_{t=t_0(i)}^{T(i)} \frac{1}{\varsigma^{2,q}_{\varepsilon j(i,t)}} \boldsymbol{\lambda}^q_{j(i,t)} \big(\boldsymbol{\lambda}^q_{j(i,t)}\big)' \right)^{-1}, \tag{62}$$

$$\boldsymbol{\zeta}_i^q = \boldsymbol{V}_i^q \left((\boldsymbol{\Sigma}^q_{g(i)})^{-1} \boldsymbol{\mu}^q_{g(i)} + \sum_{t=t_0(i)}^{T(i)} \frac{\boldsymbol{\lambda}^q_{j(i,t)}\, r^q_{it}}{\varsigma^{2,q}_{\varepsilon j(i,t)}} \right), \tag{63}$$

$$\boldsymbol{\theta}_i \mid \text{data}, \boldsymbol{\Gamma}^q, g(i) \;\sim\; \mathcal{N}\big(\boldsymbol{\zeta}_i^q,\; \boldsymbol{V}_i^q\big). \tag{64}$$

Equations (62)–(63) give the posterior mean and variance of each worker's skills in closed form, and I keep them for the M-step.

D.4.2 M-step

In the M-step I maximize the expected complete-data log-likelihood over $\boldsymbol{\Gamma}$, given the E-step posteriors. The complete-data log-likelihood is the sum over workers of the log prior density of $\boldsymbol{\theta}_i$, equation (60), and of the log density of each wage, equation (59). Its expectation under the E-step posterior $\boldsymbol{\theta}_i \mid w_i, d_i, \boldsymbol{\Gamma}^q \sim \mathcal{N}(\boldsymbol{\zeta}_i^q, \boldsymbol{V}_i^q)$ is

$$\begin{aligned} Q(\boldsymbol{\Gamma} \mid \boldsymbol{\Gamma}^q) = & -\tfrac{1}{2}\sum_i \Big[\log\big|2\pi\boldsymbol{\Sigma}_{g(i)}\big| + \operatorname{tr}\big(\boldsymbol{\Sigma}_{g(i)}^{-1}\boldsymbol{V}_i^q\big) + \big(\boldsymbol{\zeta}_i^q - \boldsymbol{\mu}_{g(i)}\big)'\boldsymbol{\Sigma}_{g(i)}^{-1}\big(\boldsymbol{\zeta}_i^q - \boldsymbol{\mu}_{g(i)}\big)\Big] \\ & -\tfrac{1}{2}\sum_i\sum_t \Big[\log\big(2\pi\varsigma_{\varepsilon j}^2\big) + \frac{\big(w_{ijt} - p_j - \boldsymbol{\lambda}_j'\boldsymbol{\zeta}_i^q - h_j(\boldsymbol{\tau}_{it})\big)^2 + \boldsymbol{\lambda}_j'\boldsymbol{V}_i^q\boldsymbol{\lambda}_j}{\varsigma_{\varepsilon j}^2}\Big], \end{aligned} \tag{65}$$

with $j = j(i,t)$ being the occupation chosen by individual i in period t. The Q-function is the sum of a prior part, which separates across groups, and a wage part, which separates across occupations, so I can maximize each part in closed form. I first derive the first-order conditions for the prior moments and then those for the wage parameters of each occupation.

Prior moments optimization. Maximizing the prior block of (65) over $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ gives the standard updates

$$\boldsymbol{\mu}_g^{q+1} = \frac{1}{N_g}\sum_{i:\,g(i)=g} \boldsymbol{\zeta}_i^q, \tag{66}$$

$$\boldsymbol{\Sigma}_g^{q+1} = \frac{1}{N_g}\sum_{i:\,g(i)=g} \Big[\boldsymbol{V}_i^q + (\boldsymbol{\zeta}_i^q - \boldsymbol{\mu}_g^{q+1})(\boldsymbol{\zeta}_i^q - \boldsymbol{\mu}_g^{q+1})'\Big]. \tag{67}$$

To pin down the location of the latent factor, I normalize the group-weighted population mean of skills to zero, $\sum_g \pi_g \boldsymbol{\mu}_g = \mathbf{0}$. I impose this normalization at each iteration by subtracting $\boldsymbol{a}^{q+1} := \sum_g \pi_g \boldsymbol{\mu}_g^{q+1}$ from every $\boldsymbol{\mu}_g^{q+1}$ and adding it back to the occupation fixed effect (price).

Wage equation parameters optimization. For each occupation j, I stack the N_j worker–year observations in j into a regressor matrix whose rows contain a constant, the E-step posterior mean of the worker's skills, and the task-tenure bin indicators:

$$\boldsymbol{W}_j \;=\; \big[\,\mathbf{1},\;\; \boldsymbol{\zeta}^q,\;\; \boldsymbol{D}(\boldsymbol{\tau})\,\big] \;\in\; \mathbb{R}^{N_j \times (1+K+|\mathcal{B}|)}, \tag{68}$$

where $\boldsymbol{\zeta}^q$ is the posterior mean for each worker repeated vertically each time the same worker is observed, and $\boldsymbol{D}(\boldsymbol{\tau})$ collects the tenure bin indicators for task-specific experience with total dimension $|\mathcal{B}|$. The corresponding coefficient vector is

$$\boldsymbol{\vartheta}_j \;=\; \big[p_j,\; \boldsymbol{\lambda}_j',\; \boldsymbol{\delta}_j'\big]'.$$

The wage block of (65) is quadratic in $\boldsymbol{\vartheta}_j$, so updating the wage parameters is just a per-occupation least-squares regression of the stacked log wages $\boldsymbol{y}_j$ on $\boldsymbol{W}_j$. Its first-order

conditions are given by the following closed-form equations

$$\boldsymbol{\vartheta}_j^{q+1} \;=\; \boldsymbol{S}_j^{-1}\boldsymbol{b}_j, \qquad \boldsymbol{S}_j \;=\; \boldsymbol{W}_j'\boldsymbol{W}_j \;+\; \boldsymbol{\Delta}_j^{q}, \qquad \boldsymbol{b}_j \;=\; \boldsymbol{W}_j'\boldsymbol{y}_j. \tag{69}$$

The only departure from textbook OLS is the correction $\boldsymbol{\Delta}_j^{q} = \sum_{t\in j} \boldsymbol{V}_{i(t)}^{q}$, which adds the summed posterior variances $\sum_{t\in j} \boldsymbol{V}_i^{q}$ to the $K \times K$ loading block of $\boldsymbol{S}_j$ and is zero elsewhere. This term is the curvature of the $\boldsymbol{\lambda}_j'\boldsymbol{V}_i^{q}\boldsymbol{\lambda}_j$ correction in (65); it corrects the loadings $\boldsymbol{\lambda}_j$ for the residual posterior uncertainty about $\boldsymbol{\theta}_i$. Intuitively, residual wages are noisy measurements of multidimensional skills, because of the iid shocks and because skills enter wages only through the loadings. If I had observed every worker for many periods moving across different occupations, then this uncertainty would be close to zero for everyone. This system of equations has dimension $1 + K + |\mathcal{B}|$ and is solved separately for each occupation, delivering $(p_j^{q+1}, \boldsymbol{\lambda}_j^{q+1}, \boldsymbol{\delta}_j^{q+1})$ jointly.

Including the E-step posterior $\boldsymbol{\zeta}^{q}$ as a regressor in $\boldsymbol{W}_j$ implements the correction for selection on beliefs of Section 4.1: regressing log wages on $(\mathbf{1}, \boldsymbol{\zeta}^{q}, \boldsymbol{D}(\boldsymbol{\tau}))$ per occupation absorbs endogenous occupational choice through the observed history, and so controls for belief-based selection into occupation, while the augmentation $\boldsymbol{\Delta}_j^{q}$ corrects the loading block for residual posterior uncertainty.

Anchored normalization. The factor model is identified up to a $K \times K$ invertible transformation of $(\boldsymbol{\theta}, \boldsymbol{\lambda})$. To eliminate this rotation and scale indeterminacy, I pin the full loading vector at two anchor occupations: a cognitive anchor a_C and a manual anchor a_M. In the default implementation of the EM algorithm I set the anchor loadings $\boldsymbol{\lambda}_{a_C} = (1, 0)'$ and $\boldsymbol{\lambda}_{a_M} = (0, 1)'$. This choice imposes four scalar restrictions (two scale, two rotation), which is exactly what is required to identify the orientation of the latent skill space. At the two anchor occupations I substitute $\boldsymbol{\lambda}_j$ out of (69) and solve the reduced system for $(p_j, \boldsymbol{\delta}_j)$ only.

I also constrain the skill loadings to be non-negative, $\boldsymbol{\lambda}_j \geq \mathbf{0}$, so that the parameters have economic content and skills have a meaningful ordinal ranking: if $\lambda_{jk} < 0$ for any occupation j in task k then a very low skill worker in dimension k might have an unreasonably high productivity in occupation j.

Residual variance. Given the updated $(p_j^{q+1}, \boldsymbol{\lambda}_j^{q+1}, \boldsymbol{\delta}_j^{q+1})$, the variance update is

$$\varsigma_{\varepsilon j}^{2,q+1} \;=\; \frac{1}{N_j} \sum_{(i,t):\, j(i,t)=j} \Big[\big(r_{it}^{q+1} - (\boldsymbol{\lambda}_j^{q+1})'\boldsymbol{\zeta}_i^{q}\big)^2 + (\boldsymbol{\lambda}_j^{q+1})'\boldsymbol{V}_i^{q}\boldsymbol{\lambda}_j^{q+1} \Big], \tag{70}$$

where r_{it}^{q+1} is the residual from (61) evaluated at the new wage parameters. The second term inside the bracket is the variance-of-the-projection correction $\boldsymbol{\lambda}_j'\boldsymbol{V}_i^{q}\boldsymbol{\lambda}_j$ from (65) and ensures that posterior uncertainty about $\boldsymbol{\theta}_i$ is correctly transmitted into $\varsigma_{\varepsilon j}^2$.

The algorithm iterates (62)–(64) and (66)–(70) until convergence. I use two different stopping rules. The EM algorithm stops when the mean squared change in the vector of parameters across consecutive iterations falls below 10^{-9} or when the absolute change in the Q-function is lower than 10^{-4}.

D.4.3 Likelihood Evaluation

Because the EM algorithm is subject to local maxima, I use different initial guesses for the parameters and then evaluate the observed-data wage log-likelihood by integrating out $\boldsymbol{\theta}_i$ under the group prior. For a candidate parameter vector $\boldsymbol{\Gamma}$,

$$\tilde{\mathcal{L}}_i^w(\boldsymbol{\Gamma}) \;=\; \int_{\mathbb{R}^K} \left\{ \prod_{t=t_0(i)}^{T(i)} \phi\Big(w_{ij(i,t)t};\; p_{j(i,t)} + \boldsymbol{\lambda}'_{j(i,t)}\boldsymbol{\theta}_i + h_{j(i,t)}(\boldsymbol{\tau}_{it}),\; \varsigma^2_{\varepsilon j(i,t)}\Big) \right\} \phi\big(\boldsymbol{\theta}_i; \boldsymbol{\mu}_{g(i)}, \boldsymbol{\Sigma}_{g(i)}\big)\, d\boldsymbol{\theta}_i, \tag{71}$$

where $\phi(\cdot\,; \mu, \varsigma^2)$ is the univariate normal density and $\phi(\cdot\,; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ the multivariate normal density.

I approximate (71) by Monte Carlo. For each worker i I fix a sequence of standard-normal draws $\boldsymbol{Z}_i^{(b)} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and map them through the current prior,

$$\boldsymbol{\theta}_i^{(b)} = \boldsymbol{\mu}_{g(i)} + \boldsymbol{L}_{g(i)}\, \boldsymbol{Z}_i^{(b)}, \qquad \boldsymbol{L}_g \boldsymbol{L}_g' = \boldsymbol{\Sigma}_g,$$

and form the worker-level time-summed log contribution

$$S_i^{(b)}(\boldsymbol{\Gamma}) = \sum_{t=t_0(i)}^{T(i)} \log \phi\Big(w_{ij(i,t)t};\; p_{j(i,t)} + \boldsymbol{\lambda}'_{j(i,t)}\boldsymbol{\theta}_i^{(b)} + h_{j(i,t)}(\boldsymbol{\tau}_{it}),\; \varsigma^2_{\varepsilon j(i,t)}\Big).$$

The log-likelihood is then evaluated by the log-sum-exp identity,

$$\log \tilde{\mathcal{L}}_i^w(\boldsymbol{\Gamma}) \;\approx\; \operatorname{logsumexp}\big(\{S_i^{(b)}(\boldsymbol{\Gamma})\}_{b=1}^B\big) - \log B, \tag{72}$$

which is numerically stable (it avoids underflow) at extreme parameter values.[34]

D.5 NFXP Algorithm: Computational Details

D.5.1 Group–Occupation–Specific Integrated Value Functions and Fixed Point

In the dynamic model, the worker state at time t is

$$\boldsymbol{x}_t \;\equiv\; \big(g,\, \tilde{\boldsymbol{\theta}}_t,\, \tilde{\boldsymbol{\Sigma}}_t,\, \boldsymbol{\tau}_t,\, d_{t-1}\big) \in \mathcal{X},$$

the decision-relevant part of the state (23): the education group $g \in \mathcal{G}$, the posterior mean $\tilde{\boldsymbol{\theta}}_t \in \mathbb{R}^K$ and covariance $\tilde{\boldsymbol{\Sigma}}_t \in \mathbb{R}^{K\times K}$ over latent skills, task tenures $\boldsymbol{\tau}_t \in \mathbb{R}^K_+$, and last period's occupation $d_{t-1} \in \mathcal{J}$ (occupation prices are constant in the stationary solve and are suppressed). I index the current occupational choice by $j \in \mathcal{J}$ and reserve m for the lagged occupation $d_{t-1} = m$. Conditional on choosing j at t, the realized log

[34] Fixing the common random numbers across iterations ensures that differences in $\sum_i \log \tilde{\mathcal{L}}_i^w(\boldsymbol{\Gamma})$ across starts reflect an improvement in the fit of the wage model rather than simulation noise.

wage in (14) delivers the scalar signal, mirroring (17),

$$s_{jt} \;\equiv\; \log W_{jt} - p_{jt} - h_j(\boldsymbol{\tau}_t) = \boldsymbol{\lambda}_j'\boldsymbol{\theta} + \varepsilon_{jt}, \qquad \varepsilon_{jt} \sim \mathcal{N}(0, \varsigma_{\varepsilon j}^2), \tag{73}$$

and the Kalman recursion (the filter form of Bayes' rule (18)) reads

$$\tilde{\boldsymbol{\theta}}_{t+1} = \tilde{\boldsymbol{\theta}}_t + \boldsymbol{K}_{jt}\, z_{jt}, \qquad z_{jt} \equiv s_{jt} - \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}}_t \sim \mathcal{N}\big(0,\, s_{zz}^{(j)}\big), \tag{74}$$

$$\tilde{\boldsymbol{\Sigma}}_{t+1} = \big(\boldsymbol{I}_K - \boldsymbol{K}_{jt}\boldsymbol{\lambda}_j'\big)\tilde{\boldsymbol{\Sigma}}_t, \qquad \boldsymbol{K}_{jt} = \frac{\tilde{\boldsymbol{\Sigma}}_t\boldsymbol{\lambda}_j}{s_{zz}^{(j)}}, \tag{75}$$

$$s_{zz}^{(j)} \equiv \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\Sigma}}_t\boldsymbol{\lambda}_j + \varsigma_{\varepsilon j}^2, \qquad \boldsymbol{\tau}_{t+1} = \boldsymbol{\tau}_t + \boldsymbol{\omega}_j. \tag{76}$$

The next-period covariance $\tilde{\boldsymbol{\Sigma}}_{t+1}$ and the next-period tenures $\boldsymbol{\tau}_{t+1}$ are both deterministic transitions given $(\boldsymbol{x}_t, j)$; the only random variable is the scalar innovation z_{jt}, with predictive variance $s_{zz}^{(j)}$. The next posterior mean therefore moves only along the Kalman gain $\boldsymbol{K}_{jt}$, and the expectation of the continuation value is a one-dimensional integral over z_{jt} whatever the number of skills K.

Univariate Gauss–Hermite quadrature. I use the change of variables $u = z/\sqrt{2\, s_{zz}^{(j)}}$ to map the Gaussian density into the Hermite weight e^{-u^2}:

$$\mathbb{E}\big[G(\tilde{\boldsymbol{\theta}}_{t+1}, \tilde{\boldsymbol{\Sigma}}_{t+1}, \boldsymbol{\tau}_{t+1}, j) \,\big|\, \boldsymbol{x}_t, j\big] \;=\; \pi^{-1/2}\int_{\mathbb{R}} G\Big(\tilde{\boldsymbol{\theta}}_t + \sqrt{2\, s_{zz}^{(j)}}\, \boldsymbol{K}_{jt}\, u,\; \tilde{\boldsymbol{\Sigma}}_{t+1},\; \boldsymbol{\tau}_t + \boldsymbol{\omega}_j,\; j\Big)\, e^{-u^2}\, du. \tag{77}$$

Letting $\{(u_r, \varpi_r)\}_{r=1}^R$ denote the R-node univariate Gauss–Hermite nodes and weights satisfying $\int_{\mathbb{R}} g(u) e^{-u^2} du \;\approx\; \sum_{r=1}^R \varpi_r\, g(u_r)$, the continuation expectation is approximated by

$$\mathbb{E}\big[G(\boldsymbol{x}_{t+1}) \,\big|\, \boldsymbol{x}_t, j\big] \;\approx\; \pi^{-1/2}\sum_{r=1}^{R} \varpi_r\, G\Big(\tilde{\boldsymbol{\theta}}_t + \sqrt{2\, s_{zz}^{(j)}}\, \boldsymbol{K}_{jt}\, u_r,\; \tilde{\boldsymbol{\Sigma}}_{t+1},\; \boldsymbol{\tau}_t + \boldsymbol{\omega}_j,\; j\Big). \tag{78}$$

The cost is therefore R function evaluations per $(\boldsymbol{x}_t, j)$ rather than R^K, which would have been required by a tensor-product rule applied to the full posterior covariance.

Deterministic index and integrated value. In practice, rather than including the discrete states in the state space, I define group- and occupation-specific value functions, with only the continuous variables remaining as arguments; from here on $\boldsymbol{x} \equiv (\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\Sigma}}, \boldsymbol{\tau})$ denotes this continuous part of the state rather than the full state $\boldsymbol{x}_t$ above. Let

$$EV_{mg}(\boldsymbol{x}) \;\equiv\; EV\big(\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\Sigma}}, \boldsymbol{\tau}, d_{-1} = m,\, g\big), \qquad m \in \mathcal{J},\; g \in \mathcal{G},$$

denote the integrated value function conditional on being in occupation m at the start of the period and belonging to group g. For a current choice $j \in \mathcal{J}$, the choice-j specific deterministic index combines the expected log wage, the amenity, the switching cost,

and the discounted continuation value:

$$v_{j|m,g}(\boldsymbol{x}) \;=\; \underbrace{p_j + \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}} + h_j(\boldsymbol{\tau})}_{\text{expected log wage}} + \eta_{jg} - \mathcal{C}(j,m) \\ + \beta\, \mathbb{E}\big[\, EV_{jg}(\boldsymbol{x}') \;\big|\; \boldsymbol{x}, j \,\big], \tag{79}$$

where group g is fixed over the worker's life and is therefore carried unchanged into the continuation value EV_{jg}, whose incumbent occupation is the current choice j.

With Type-I extreme value taste shocks of scale ϱ normalized to mean zero (Appendix C.3), the occupation-specific integrated values satisfy a log-sum-exp fixed point. Substituting the univariate Gauss–Hermite approximation (78) for the continuation expectation in (79) and writing the deterministic flow utility as $u_{jg}(\boldsymbol{x}) \equiv p_j + \boldsymbol{\lambda}_j'\tilde{\boldsymbol{\theta}} + h_j(\boldsymbol{\tau}) + \eta_{jg}$, the integrated Bellman equation reads, for each $m \in \mathcal{J}$ and $g \in \mathcal{G}$,

$$EV_{mg}(\boldsymbol{x}) \;=\; \varrho \,\log \sum_{j\in\mathcal{J}} \exp\left(\frac{1}{\varrho}\Big[\, u_{jg}(\boldsymbol{x}) - \mathcal{C}(j,m) \right. \\ \left. + \beta\, \pi^{-1/2} \sum_{r=1}^{R} \varpi_r\, EV_{jg}\Big(\tilde{\boldsymbol{\theta}} + \sqrt{2\, s_{zz}^{(j)}}\, \boldsymbol{K}_j\, u_r,\; \tilde{\boldsymbol{\Sigma}}',\; \boldsymbol{\tau} + \boldsymbol{\omega}_j\Big) \,\Big] \right), \qquad \boldsymbol{x} \in \mathcal{X}. \tag{80}$$

For each group g, collect the integrated values across incumbent occupations into the vector $\boldsymbol{EV}_g(\boldsymbol{x}) = \big(EV_{mg}(\boldsymbol{x})\big)_{m\in\mathcal{J}}$. For a given incumbent m, the right-hand side of (80) depends on the continuation values EV_{jg} of every possible choice j, so the $|\mathcal{J}|$ equations form a single vector fixed point,

$$\boldsymbol{EV}_g \;=\; \mathcal{T}(\boldsymbol{EV}_g), \qquad \big[\mathcal{T}(\boldsymbol{EV}_g)\big]_m(\boldsymbol{x}) = \varrho \,\log \sum_{j\in\mathcal{J}} \exp\Big(\tfrac{1}{\varrho}\big[\, u_{jg}(\boldsymbol{x}) - \mathcal{C}(j,m) + \beta\, \mathbb{E}\big(EV_{jg}(\boldsymbol{x}') \mid \boldsymbol{x}, j\big) \,\big] \Big), \tag{81}$$

which must be solved jointly for all $|\mathcal{J}|$ components rather than one occupation at a time. By Lemma 2, $\mathcal{T}$ is a contraction of modulus at most β, so (81) has a unique fixed point $\boldsymbol{EV}_g^{\star}$. The fixed point is solved separately for each $g \in \mathcal{G}$, and from here on I drop the index g.

Smolyak approximation of the integrated value function on a sparse grid. The continuous state $\boldsymbol{x} = (\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\Sigma}}, \boldsymbol{\tau})$, which collects the posterior means, the distinct elements of the posterior variance-covariance matrix, and task experience, can be high-dimensional. I approximate each value function $EV_{mg}(\cdot)$ on a sparse grid using the Smolyak method (Malin et al. 2011; Judd et al. 2014) with Chebyshev polynomials. Because the construction is identical for every incumbent occupation m, I describe it for a generic value function $EV(\boldsymbol{x})$ and suppress the index m.

Fix lower and upper bounds for each component of $\boldsymbol{x}$. Let $D = 2K + K(K+1)/2$ be the continuous-state dimension ($D = 7$ with two skills) and $\bar{\ell}$ the Smolyak level. The

sparse grid is

$$\mathcal{H}^{D,\bar{\ell}} = \bigcup_{\mathbf{i}:\,|\mathbf{i}|=D+\bar{\ell}} \left(\mathcal{S}^{n(i_1)} \times \cdots \times \mathcal{S}^{n(i_D)}\right), \qquad |\mathbf{i}| := i_1 + \cdots + i_D,$$

where $\mathcal{S}^{n(\cdot)}$ are nested univariate nodes (Chebyshev extrema) and $n(\cdot)$ is the node-count function. For a given multi-index $\mathbf{i}$, the tensor-product interpolant for EV is

$$EV^{\mathbf{i}}(\boldsymbol{x}; \boldsymbol{\phi}^{(\mathbf{i})}) = \sum_{l_1=1}^{n(i_1)} \cdots \sum_{l_D=1}^{n(i_D)} \phi^{(\mathbf{i})}_{l_1 \cdots l_D}\, T_{l_1}(x_1) \cdots T_{l_D}(x_D) = \boldsymbol{\phi}^{(\mathbf{i})\prime} \mathbf{T}(\boldsymbol{x}),$$

with $\mathbf{T}(\boldsymbol{x})$ the basis vector of multivariate Chebyshev polynomials. The Smolyak combination gives the global approximation

$$\widehat{EV}^{D,\bar{\ell}}(\boldsymbol{x}) = \sum_{D \le |\mathbf{i}| \le D+\bar{\ell}} (-1)^{D+\bar{\ell}-|\mathbf{i}|} \binom{D-1}{D+\bar{\ell}-|\mathbf{i}|}\, EV^{\mathbf{i}}(\boldsymbol{x}; \boldsymbol{\phi}^{(\mathbf{i})}). \tag{82}$$

Fixed point in coefficient space. Substituting the Smolyak interpolant $\widehat{EV}_j(\boldsymbol{x}) = \boldsymbol{\phi}^{(j)\prime}\mathbf{T}(\boldsymbol{x})$ for the unknown continuation value EV_j inside (80) recasts the integrated Bellman equation as a fixed point in the finite-dimensional Smolyak coefficients $\boldsymbol{\Phi} \equiv [\boldsymbol{\phi}^{(m)}]_{m \in \mathcal{J}} \in \mathbb{R}^{M \times |\mathcal{J}|}$, whose columns index the incumbent occupation and where M is the number of Smolyak basis functions. For each incumbent $m \in \mathcal{J}$,

$$\begin{aligned} \boldsymbol{\phi}^{(m)\prime}\mathbf{T}(\boldsymbol{x}) \;=\; \varrho \,\log \sum_{j \in \mathcal{J}} \exp\Bigg(\frac{1}{\varrho} \Bigg[& u_j(\boldsymbol{x}) - \mathcal{C}(j,m) \\ & + \beta\, \pi^{-1/2} \sum_{r=1}^{R} \varpi_r\, \boldsymbol{\phi}^{(j)\prime} \mathbf{T}\Big(\tilde{\boldsymbol{\theta}} + \sqrt{2\, s_{zz}^{(j)}}\, \boldsymbol{K}_j\, u_r, \\ & \tilde{\boldsymbol{\Sigma}}',\; \boldsymbol{\tau} + \boldsymbol{\omega}_j \Big) \Bigg] \Bigg), \qquad \forall\, \boldsymbol{x} \in \mathcal{X},\; m \in \mathcal{J}. \end{aligned} \tag{83}$$

To obtain the matrix-stacked map solved in practice, evaluate (83) at every node of the sparse grid. Let $\{\boldsymbol{x}^{(n)}\}_{n=1}^{N_G}$ denote the N_G grid points in $\mathcal{H}^{D,\bar{\ell}}$, and let $\boldsymbol{y}^{(n)} \in [-1,1]^D$ be $\boldsymbol{x}^{(n)}$ rescaled to the Chebyshev domain. Define the basis matrix $\mathbf{T} \in \mathbb{R}^{M \times N_G}$ by $\mathbf{T}_{\ell,n} = T_\ell(\boldsymbol{y}^{(n)})$ (basis function $\ell = 1, \ldots, M$, grid point n) and the collocation pseudo-inverse

$$\mathbf{T}^{+} \;\equiv\; (\mathbf{T}\mathbf{T}^{\top})^{-1}\mathbf{T} \;\in\; \mathbb{R}^{M \times N_G};$$

in the unisolvent case this is the same as $\mathbf{T}^{-1}$; I use the pseudo-inverse rather than direct inversion for numerical stability. Stacking the right-hand sides of (83) into the matrix $\widetilde{\mathbf{EV}}(\boldsymbol{\Phi}) \in \mathbb{R}^{|\mathcal{J}| \times N_G}$ with entries $\widetilde{\mathbf{EV}}_{mn}(\boldsymbol{\Phi}) = \widehat{EV}_m(\boldsymbol{x}^{(n)}; \boldsymbol{\Phi})$, the fixed-point map in coefficient space is

$$\boldsymbol{\Phi} \;=\; \mathcal{F}(\boldsymbol{\Phi}) \;\equiv\; \mathbf{T}^{+}\, \widetilde{\mathbf{EV}}(\boldsymbol{\Phi})'. \tag{84}$$

I compute the fixed point $\mathbf{\Phi}^\star$ of (84) by successive substitution, compiled into a single JAX while-loop (Section D.5.6).

D.5.2 Fixed-Point Algorithm for Integrated Value Function Smolyak Approximation

The inner loop of the NFXP solves (84) by iterating on the Smolyak coefficients $\mathbf{\Phi}$: at each iteration I evaluate the right-hand side of (83) at every grid point, integrate over next-period beliefs with Gauss–Hermite quadrature, and update the coefficients. The algorithm box below summarizes the steps.

Algorithm: Fixed-Point Iteration for the Integrated Value Function Smolyak Approximation

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Inputs.

- The parameters $(\hat{\mathbf{\Gamma}}, \mathbf{\Delta})$.
- Sparse-grid nodes $\{\boldsymbol{x}^{(n)}\}_{n=1}^{N_G} \subset \mathcal{H}^{D,\bar{\ell}}$.
- Chebyshev basis $\mathbf{T} \in \mathbb{R}^{M \times N_G}$ and pseudo-inverse $\mathbf{T}^+ = (\mathbf{T}\mathbf{T}^\top)^{-1}\mathbf{T}$.
- Univariate Gauss–Hermite nodes and weights $\{(u_r, \varpi_r)\}_{r=1}^{R}$.
- Tolerance $\epsilon_{\text{tol}} > 0$; maximum iterations $n_{\max}$.

Output. Fixed-point Smolyak coefficient matrix $\mathbf{\Phi}^\star \in \mathbb{R}^{M \times |\mathcal{J}|}$ solving (84).

Initialization. Set $\mathbf{\Phi}^{(0)} = \mathbf{0} \in \mathbb{R}^{M \times |\mathcal{J}|}$.

Repeat for $s = 0, 1, \ldots, n_{\max} - 1$ until convergence. For each grid point n and each choice $j \in \mathcal{J}$:

- Compute the next-period belief covariance $\tilde{\mathbf{\Sigma}}'(\boldsymbol{x}^{(n)}, j)$ from the Kalman update.
- Form the R next-period states $\boldsymbol{x}'_r$ from the Gauss–Hermite nodes (78).
- Evaluate the interpolant $\widehat{EV}_j(\boldsymbol{x}'_r; \mathbf{\Phi}^{(s)}) = \boldsymbol{\phi}^{(s,j)\prime}\mathbf{T}(\boldsymbol{x}'_r)$.
- Compute the choice-specific values $v_{j|m}(\boldsymbol{x}^{(n)})$ for every incumbent $m \in \mathcal{J}$, equation (79).

Compute the integrated value $\widetilde{EV}_m(\boldsymbol{x}^{(n)}) = \varrho \log \sum_j \exp\big(v_{j|m}(\boldsymbol{x}^{(n)})/\varrho\big)$.

Update the coefficients $\mathbf{\Phi}^{(s+1)} = \mathbf{T}^+ \, \widetilde{\mathbf{EV}}(\mathbf{\Phi}^{(s)})'$, equation (84).

Stop when $\big\|\mathbf{\Phi}^{(s+1)} - \mathbf{\Phi}^{(s)}\big\|_F < \epsilon_{\text{tol}}$.

Return $\mathbf{\Phi}^\star = \mathbf{\Phi}^{(s+1)}$.

Numerical stability. For small ϱ, I compute the log-sum-exp in the shifted form, with $v_{\max} = \max_j v_j$,

$$\varrho \log\Big(\sum_j e^{v_j/\varrho}\Big) \;=\; v_{\max} + \varrho \log\Big(\sum_j e^{(v_j - v_{\max})/\varrho}\Big),$$

to avoid overflow when one alternative dominates.

D.5.3 Policy Functions, CCPs, and Simulation

Given $\{\widehat{EV}_m\}_{m \in \mathcal{J}}$, the CCPs at $(\boldsymbol{x}, m)$ follow from (85). To simulate a worker's path:

1. Given $(\tilde{\boldsymbol{\theta}}_t, \tilde{\mathbf{\Sigma}}_t, \boldsymbol{\tau}_t, m)$, compute CCPs $\{\mathbb{P}(d_t = j \mid \boldsymbol{x}_t, m)\}_{j \in \mathcal{J}}$ and draw d_t (or use

the argmax for policy).

2. Draw a wage residual $\varepsilon_{d_t,t} \sim \mathcal{N}(0, \varsigma^2_{\varepsilon d_t})$ and set

$$\log W_{d_t t} = p_{d_t} + \boldsymbol{\lambda}'_{d_t}\boldsymbol{\theta} + h_{d_t}(\boldsymbol{\tau}_t) + \varepsilon_{d_t,t}.$$

   Equivalently, the signal in (73) is $s_{d_t t} = \log W_{d_t t} - p_{d_t} - h_{d_t}(\boldsymbol{\tau}_t)$.

3. Update beliefs via (74)–(76) using $j = d_t$.

4. Update tenures: $\boldsymbol{\tau}_{t+1} = \boldsymbol{\tau}_t + \boldsymbol{\omega}_{d_t}$ (cf. (10)).

5. Update the discrete state: $m \leftarrow d_t$ (next period's lagged occupation).

D.5.4 Conditional Choice Probabilities and Likelihood

Once the fixed-point coefficients $\boldsymbol{\Phi}^\star$ are recovered, conditional choice probabilities and the choice likelihood follow in closed form. Given the integrated value functions $\{\widehat{EV}_j(\boldsymbol{x}; \boldsymbol{\Phi}^\star)\}_{j\in\mathcal{J}}$ and a worker arriving at state $\boldsymbol{x}$ with lagged occupation m, the multinomial-logit CCPs implied by the Type-I extreme value taste shocks are

$$\mathbb{P}\big(d_t = j \mid \boldsymbol{x}, m; \hat{\boldsymbol{\Gamma}}, \boldsymbol{\Delta}\big) \;=\; \frac{\exp\big(v_{j|m}(\boldsymbol{x})/\varrho\big)}{\sum_{\ell\in\mathcal{J}} \exp\big(v_{\ell|m}(\boldsymbol{x})/\varrho\big)}, \qquad j \in \mathcal{J}, \tag{85}$$

where $v_{j|m}(\boldsymbol{x}) \;=\; u_j(\boldsymbol{x}) - \mathcal{C}(j,m) + \beta\,\mathbb{E}\big[EV_j(\boldsymbol{x}') \mid \boldsymbol{x}, j\big]$ is the choice-specific value of moving to occupation j from m.

Evaluation of the choice likelihood on the estimation sample. Using the parameters estimated in the first stage and imposing the rationality assumption, I can compute the exact sequence of beliefs for every worker in the estimation sample. Workers belonging to group g enter the labor market with state

$$\boldsymbol{x}_{i0} = \big(g(i),\, \tilde{\boldsymbol{\theta}}_{i0},\, \tilde{\boldsymbol{\Sigma}}_{i0},\, \boldsymbol{\tau}_{i0},\, d_{i,-1}\big), \qquad \tilde{\boldsymbol{\theta}}_{i0} = \hat{\boldsymbol{\mu}}_{g(i)}, \quad \tilde{\boldsymbol{\Sigma}}_{i0} = \hat{\boldsymbol{\Sigma}}_{g(i)}, \quad \boldsymbol{\tau}_{i0} = \mathbf{0}, \quad d_{i,-1} = \varnothing,$$

that is, rational initial beliefs equal to the group prior, no accumulated task experience, and no incumbent occupation ($d_{i,-1} = \varnothing$, coded as -1, so the first choice bears no switching cost). After entry, the wage net of prices and tenure returns is the signal the worker uses to update her beliefs,

$$s_{it} \;\equiv\; w_{i d_{it} t} - p_{d_{it}} - h_{d_{it}}(\boldsymbol{\tau}_{it}),$$

where d_{it} is her observed occupation at t. Therefore, for each subsequent period I can compute the belief sequence for the mean and the variance–covariance matrix by the

Kalman recursion (74)–(76) evaluated at the observed choice d_{it},

$$\tilde{\boldsymbol{\theta}}_{i,t+1} = \tilde{\boldsymbol{\theta}}_{it} + \boldsymbol{K}_{it}\big(s_{it} - \boldsymbol{\lambda}'_{d_{it}}\tilde{\boldsymbol{\theta}}_{it}\big), \quad \tilde{\boldsymbol{\Sigma}}_{i,t+1} = \big(\boldsymbol{I}_K - \boldsymbol{K}_{it}\boldsymbol{\lambda}'_{d_{it}}\big)\tilde{\boldsymbol{\Sigma}}_{it},$$
$$\boldsymbol{K}_{it} = \frac{\tilde{\boldsymbol{\Sigma}}_{it}\boldsymbol{\lambda}_{d_{it}}}{\boldsymbol{\lambda}'_{d_{it}}\tilde{\boldsymbol{\Sigma}}_{it}\boldsymbol{\lambda}_{d_{it}} + \varsigma^2_{\varepsilon d_{it}}}.$$

These beliefs are the states at which I evaluate the choice likelihood.

The fixed-point algorithm of Section D.5.2 returns the Smolyak coefficient matrix $\boldsymbol{\Phi}^\star$ that approximates the value function at the current parameter estimates. To evaluate the choice likelihood, I must compute CCPs for each observation in the estimation sample. Because the value function is a Chebyshev polynomial, I evaluate it, and hence the CCPs, directly at every observed state.

Stacking (85) across workers and periods yields

$$\log L^c\big(\boldsymbol{\Delta}; \hat{\boldsymbol{\Gamma}}\big) \;=\; \sum_{i=1}^{N}\sum_{t=t_0(i)}^{T(i)} \log \mathbb{P}\big(d_{it} \mid \boldsymbol{x}_{it}, d_{i,t-1}; \boldsymbol{\Phi}^\star(\boldsymbol{\Delta}), \hat{\boldsymbol{\Gamma}}\big). \tag{86}$$

The Stage-1 estimates $\hat{\boldsymbol{\Gamma}}$ enter (86) only through the choice-specific values inside $\boldsymbol{\Phi}^\star$ and are held fixed at their Stage 1 EM estimates.

D.5.5 NFXP Procedure

Stage 1, the EM algorithm, delivers the estimates $\hat{\boldsymbol{\Gamma}}$. Taking these parameters as given, and using them to build the path of beliefs for each worker as described above, the NFXP estimator searches for the dynamic-choice parameters $\boldsymbol{\Delta} = \big(\{\eta_{jg}\}_{j\neq 0,g}, \kappa_0\big)$ that maximize the choice log-likelihood (86), holding the discount factor β and the taste-shock scale ϱ fixed. For each candidate $\boldsymbol{\Delta}^{(k)}$ proposed by the outer optimizer, the procedure executes two steps in sequence.

Step 1 — Inner fixed point. Solve the integrated-value Bellman recursion

$$\boldsymbol{\Phi} \;=\; \mathcal{F}(\boldsymbol{\Phi}) \;\equiv\; \mathbf{T}^{+}\,\widetilde{\mathbf{EV}}(\boldsymbol{\Phi})',$$

i.e. (84), by successive substitution on the Smolyak coefficient matrix $\boldsymbol{\Phi}$, returning the converged coefficients $\boldsymbol{\Phi}^\star\big(\boldsymbol{\Delta}^{(k)}\big)$ via the algorithm of Section D.5.2.

Step 2 — Choice likelihood. Given $\boldsymbol{\Phi}^\star\big(\boldsymbol{\Delta}^{(k)}\big)$, evaluate the CCPs (85) at the observed worker states $\{\boldsymbol{x}_{it}\}$ using the approximated value function, and compute the choice log-likelihood $\log L^c\big(\boldsymbol{\Delta}^{(k)}; \hat{\boldsymbol{\Gamma}}\big)$ in (86).

In the practical implementation, L-BFGS-B (with JAX automatic differentiation; see Section D.5.6) uses $\log L^c$ and its gradient to propose the next candidate $\boldsymbol{\Delta}^{(k+1)}$, and Steps 1–2 are repeated. Iteration continues until $\boldsymbol{\Delta}^{(k)}$ converges.

D.5.6 Differentiable NFXP Implementation (JAX)

I implement the estimator in JAX and run it on a GPU. For a given $\boldsymbol{\Delta}$, I solve the inner fixed point $\boldsymbol{\Phi}^\star$ by iterating the Bellman map $\mathcal{F}$ until the coefficients change by less than 10^{-8}. I obtain the gradient of the likelihood with respect to $\boldsymbol{\Delta}$ by implicit differentiation of the fixed-point condition $\boldsymbol{\Phi}^\star = \mathcal{F}(\boldsymbol{\Phi}^\star)$, which is exact and does not require storing the iterations, and I update $\boldsymbol{\Delta}$ with L-BFGS-B. One evaluation of the likelihood and its gradient, with $N \simeq 2.4$ million observations, takes about 31 minutes on a single NVIDIA H200 GPU.

D.5.7 Two-Step Estimation Summary

The algorithm box below summarizes the two stages of the estimation: the EM algorithm of Appendix D.4 for $\boldsymbol{\Gamma}$ and the NFXP of this appendix for $\boldsymbol{\Delta}$.

Algorithm: Two-Step Estimation (EM + NFXP)

Inputs.
- The estimation panel $\{(w_{it}, d_{it}, \boldsymbol{\tau}_{it}, d_{i,t-1}, g(i))\}$.
- Anchor occupations a_C, a_M; task weights $\{\boldsymbol{\omega}_j\}_{j\in\mathcal{J}}$.
- Starting values and tolerances for both stages.

Outputs. $\hat{\boldsymbol{\Gamma}}$, $\hat{\boldsymbol{\Delta}}$.

Stage 1 — EM [Appendix D.4].

1. Initialize $\boldsymbol{\Gamma}^{(0)}$.
2. Repeat for $q = 0, 1, \ldots$ until $\mathrm{MSD}\big(\boldsymbol{\Gamma}^{(q)}\big) < 10^{-9}$:
   - **E-step** [(62), (63)]: posterior $(\hat{\boldsymbol{\zeta}}_i^{(q)}, \hat{\boldsymbol{V}}_i^{(q)})$ of each worker.
   - **M-step** [(69), (70), (66), (67)]: update the wage parameters of each occupation and the prior moments of each group in closed form.
3. **Output:** $\hat{\boldsymbol{\Gamma}}$, $\{(\hat{\boldsymbol{\zeta}}_i, \hat{\boldsymbol{V}}_i)\}_{i=1}^N$.

Stage 2 — NFXP on $\boldsymbol{\Delta} = (\{\eta_{jg}\}, \kappa_0)$, with β and ϱ fixed [Sections D.5.2, D.5.4].

1. Compute the state path $\{\boldsymbol{x}_{it}\}$ of each worker: beliefs start at $(\hat{\boldsymbol{\mu}}_{g(i)}, \hat{\boldsymbol{\Sigma}}_{g(i)})$ and are updated by the Kalman recursion on the wage residuals at the Stage-1 estimates; tenures $\boldsymbol{\tau}_{it}$ are observed.
2. Initialize $\boldsymbol{\Delta}^{(0)}$, with $\eta_{0g} = 0$ for each $g \in \mathcal{G}$.
3. Repeat for $k = 0, 1, \ldots$ until L-BFGS-B convergence (ftol $< 10^{-6}$, gtol $< 10^{-5}$):
   - **Inner solve**: fixed point $\boldsymbol{\Phi}^\star\big(\boldsymbol{\Delta}^{(k)}\big)$ via the algorithm in Section D.5.2.
   - **CCPs**: evaluate (85) at the observed states $\{\boldsymbol{x}_{it}\}$.
   - **Likelihood** [(86)]: $\log L^c\big(\boldsymbol{\Delta}^{(k)}\big)$.
   - **Outer step**: L-BFGS-B update with the implicit-differentiation gradient.
4. **Output:** $\hat{\boldsymbol{\Delta}}$.

Return $\big(\hat{\boldsymbol{\Gamma}}, \hat{\boldsymbol{\Delta}}\big)$.

D.6 Counterfactual Choice Rules: Learning and Experience Off

The decomposition of occupational mobility in Figures 9 and 10 compares the estimated model with two counterfactual choice rules. Both counterfactuals keep the simulated baseline economy fixed and change only the rule that workers use to choose

their occupation. For each counterfactual $c \in \{\text{lo}, \text{eo}\}$, I first re-solve the fixed point in coefficient space (84) under a modified decision problem, obtaining the Smolyak coefficients $\mathbf{\Phi}^{\text{lo}}$ (learning off) and $\mathbf{\Phi}^{\text{eo}}$ (experience off), while occupation prices, amenities, the switching cost, and the taste-shock scale are held at their baseline values. Let $\boldsymbol{x}_{it} = (\tilde{\boldsymbol{\theta}}_{it}, \tilde{\boldsymbol{\Sigma}}_{it}, \boldsymbol{\tau}_{it})$ and $m = d_{i,t-1}$ denote the realized state and incumbent occupation of worker i of group g in period t of the simulated baseline economy. In the estimated model, the choice-specific value of occupation j is computed from the baseline coefficients $\mathbf{\Phi}^{\star}$ as in (83),

$$\begin{aligned} \nu^{\star}_{j|m}(\boldsymbol{x}_{it}) &= u_{jg}(\boldsymbol{x}_{it}) - \mathcal{C}(j,m) \\ &\quad + \beta\, \pi^{-1/2} \sum_{r=1}^{R} \varpi_r\, \boldsymbol{\phi}^{\star(j)\prime} \mathbf{T}\Big(\tilde{\boldsymbol{\theta}}_{it} + \sqrt{2\, s^{(j)}_{zz}}\, \boldsymbol{K}_j\, u_r,\ \tilde{\boldsymbol{\Sigma}}'_{it},\ \boldsymbol{\tau}_{it} + \boldsymbol{\omega}_j\Big). \end{aligned}$$

Learning off ($\mathbf{\Phi}^{\text{lo}}$). The belief law of motion is replaced by the identity: workers never update their beliefs, so the continuation value of choosing occupation j is evaluated at unchanged beliefs and task experience $\boldsymbol{\tau} + \boldsymbol{\omega}_j$, with no expectation over the wage signal. When the choice rule is evaluated, each worker's beliefs are set to the prior of her group, $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$, while her task experience and incumbent occupation are the realized ones:

$$\nu^{\text{lo}}_{j|m}(\boldsymbol{x}_{it}) = p_j + \boldsymbol{\lambda}_j' \boldsymbol{\mu}_g + h_j(\boldsymbol{\tau}_{it}) + \eta_{jg} - \mathcal{C}(j,m) + \beta\, \boldsymbol{\phi}^{\text{lo}(j)\prime} \mathbf{T}\big(\boldsymbol{\mu}_g,\ \boldsymbol{\Sigma}_g,\ \boldsymbol{\tau}_{it} + \boldsymbol{\omega}_j\big).$$

Experience off ($\mathbf{\Phi}^{\text{eo}}$). The returns to task-specific experience are set to zero, $h_j(\boldsymbol{\tau}) = 0$, and experience is not accumulated, both in the current expected wage and in the continuation value. Learning is unchanged, and the choice rule is evaluated at the worker's realized posterior beliefs:

$$\begin{aligned} \nu^{\text{eo}}_{j|m}(\boldsymbol{x}_{it}) &= p_j + \boldsymbol{\lambda}_j' \tilde{\boldsymbol{\theta}}_{it} + \eta_{jg} - \mathcal{C}(j,m) \\ &\quad + \beta\, \pi^{-1/2} \sum_{r=1}^{R} \varpi_r\, \boldsymbol{\phi}^{\text{eo}(j)\prime} \mathbf{T}\Big(\tilde{\boldsymbol{\theta}}_{it} + \sqrt{2\, s^{(j)}_{zz}}\, \boldsymbol{K}_j\, u_r,\ \tilde{\boldsymbol{\Sigma}}'_{it},\ \boldsymbol{\tau}_{it}\Big). \end{aligned}$$

Counterfactual choice probabilities. For each rule $c \in \{\star, \text{lo}, \text{eo}\}$, the conditional choice probabilities are the multinomial-logit probabilities (85) evaluated at the choice-specific values of that rule,

$$\mathbb{P}^{c}\big(d_t = j \mid \boldsymbol{x}_{it}, m\big) = \frac{\exp\big(\nu^{c}_{j|m}(\boldsymbol{x}_{it})/\varrho\big)}{\sum_{\ell \in \mathcal{J}} \exp\big(\nu^{c}_{\ell|m}(\boldsymbol{x}_{it})/\varrho\big)}, \qquad j \in \mathcal{J}.$$

The counterfactual therefore answers the question: given everything that happened to the worker up to period t, which occupation would she have chosen if her beliefs were fixed at her prior, or if task-specific human capital neither paid off nor accumulated?

The probability of switching is

$$\mathbb{P}^c(\text{switch}_{it}) = 1 - \mathbb{P}^c\big(d_t = m \mid \boldsymbol{x}_{it}, m\big),$$

and the characteristics of the move, such as the task distance or the change in the cognitive task share between m and the destination j', are averaged with weights $\mathbb{P}^c(d_t = j' \mid \boldsymbol{x}_{it}, m)/\mathbb{P}^c(\text{switch}_{it})$ for $j' \neq m$. Using choice probabilities rather than realized choices removes the noise generated by the taste shocks, and evaluating all rules at the same realized states ensures that the differences across lines reflect only differences in the choice rule. Model match quality is the within-(occupation, year) percentile of the latent match $\boldsymbol{\lambda}_j'\boldsymbol{\theta}_i$. As a robustness check, for the learning-off counterfactual I also fully re-simulate the synthetic panel under $\boldsymbol{\Phi}^{\text{lo}}$, so that the changed choices feed back into later states; the resulting mobility dynamics are nearly identical.

E Appendix: Model Fit

E.1 Entry into the Labor Market and Additional Model Fit

I first show how workers enter the labor market in the model and then compare the model with moments of the large sample that the estimation does not target: the distribution of wages at labor-market entry, the distribution of wages within and across occupations, the wage–tenure profile of occupational stayers, and the decline of mobility with occupational tenure. In all the comparisons the data are the log real hourly wage in the large sample, and the model is the baseline stationary simulation of Section 6, with the large-sample education shares.

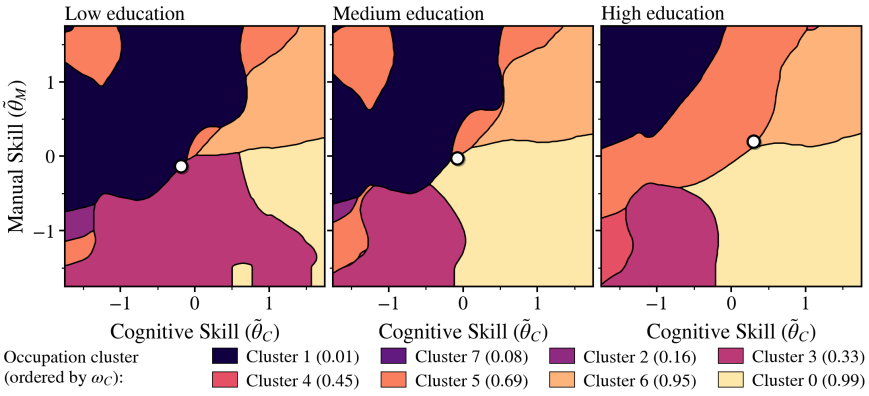
Taste shocks and the entry occupational choice. At the beginning of the career, every worker in education group g shares the same prior beliefs $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ and has no experience, so the systematic part of the choice value is identical within the group. In the deterministic limit $\varrho \to 0$ the entry choice is therefore degenerate: every worker of a specific group would enter the same occupation and workers' paths would diverge only later, through the wage signals they happen to draw and the experience they accumulate. Panel (a) of Figure 27 maps this deterministic assignment as a function of entry beliefs $(\tilde{\theta}_C, \tilde{\theta}_M)$.

The idiosyncratic taste shocks ϵ_{ijt} break this degenerate choice at labor-market entry. With taste shocks, workers from the same group, who hold identical beliefs at entry, spread across the occupations neighboring the one they would choose in the deterministic limit. Panel (b) of Figure 27 plots the expected cognitive task intensity $\mathbb{E}[\omega_C]$ of the entry occupation, in expectation over the taste shocks, as a function of entry beliefs. The dispersion in the occupational choice still depends on the initial prior: Panel (c) reports the probability of the most likely entry occupation, $\max_j \mathbb{P}(\text{entry} = j \mid \tilde{\boldsymbol{\theta}})$, which falls along the lines where a worker is nearly indifferent between occupations, and

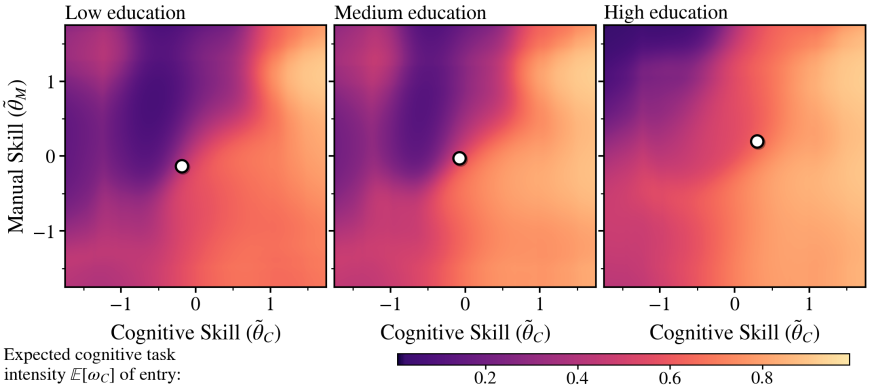
rises toward one where comparative advantage is strong. For example, with the estimated priors, low-educated workers would always enter Cluster 1, which has a very high manual content. However, with taste shocks, they can enter any occupation but are most likely to enter Cluster 3 with $\omega_M = 0.67$ or Cluster 1 with $\omega_M = 0.99$. Over the life cycle, taste shocks also allow the model to rationalize occupational choices and moves that its systematic component alone would not predict.

Wages at labor-market entry. The entry wage distribution is an informative test of the model because, as shown above, the systematic part of the entry choice is identical within an education group. Entrants are assigned to occupations by the prior and by the taste shocks alone, as illustrated above, and their wages reveal skills that neither they nor the econometrician observe at that point. The dispersion of entry wages within an education group therefore reflects only the estimated skill distribution of the group, the loadings of the occupations entrants select into, and the wage noise. Figure 28 compares the quantiles of the entry wage distribution in the model with those in the data, by education group. The model matches the mean and the median of each group (mean log wage 1.31 against 1.32 for low-educated, 1.43 against 1.44 for medium-educated, and 1.84 against 1.77 for high-educated workers) and the dispersion of the two larger groups (standard deviation 0.29 against 0.29 and 0.33 against 0.34), while it under-disperses the entry wages of high-educated workers (0.45 against 0.37). The quantile–quantile plots show where the differences lie: the model compresses both tails of the low- and medium-education distributions slightly (the gap at the 10th percentile is -0.05 to -0.08 log points and at the 90th $+0.04$) and misses the upper tail of the high-education distribution (the gap at the 90th percentile is -0.13). Within education $\times$ entry-occupation cells, the correlation between the mean entry wage in the data and in the model is 0.85.

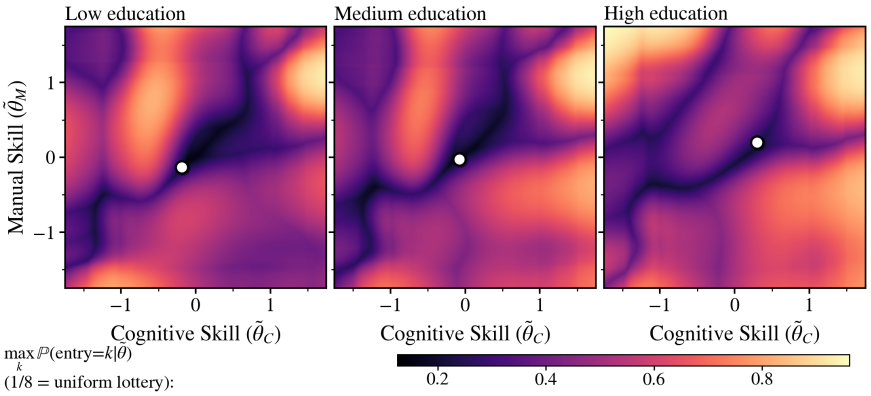
Wage distributions across occupations. Figure 29 compares the distribution of log wages in each of the eight occupations over all worker-years. The quantile–quantile plots in Panel (a) show that the model reproduces the within-occupation dispersion of Clusters 1, 3, 4 and 5 (the standard deviation of the log wage is 0.34 against 0.34 in Cluster 1, 0.36 against 0.33 in Cluster 3, 0.35 against 0.34 in Cluster 4, and 0.42 against 0.41 in Cluster 5), with the model distribution shifted up by 0.1 to 0.2 log points at the median in these occupations, and that it under-disperses the two most cognitive-intensive occupations, Clusters 0 and 6: their medians are matched within 0.08 log points, but the long right tails of high earners are missing in the model (the gap at the 90th percentile is -0.41 and -0.49 log points), so that the standard deviation is 0.60 in the data against 0.44 and 0.40 in the model. Panel (b) overlays the eight distributions: in both the data and the model the wage distributions of different occupations overlap substantially, so a worker's occupation is far from determining her wage, in contrast with models of occupational sorting in which the wage distributions of different occupations are disjoint. The model generates, if anything, more overlap than the



(a) Deterministic entry assignment ($\varrho \to 0$)



(b) Expected cognitive task intensity of the entry occupation, $\mathbb{E}[\omega_C]$



(c) Probability of the most likely entry occupation, $\max_j \mathbb{P}(\text{entry} = j \mid \tilde{\boldsymbol{\theta}})$

Figure 27: The entry occupational choice over the belief space. In every panel the three sub-panels correspond to the three education groups, and each sub-panel maps the entry choice as a function of entry beliefs about cognitive skill ($\tilde{\theta}_C$, horizontal axis) and manual skill ($\tilde{\theta}_M$, vertical axis), holding experience at zero. The white marker locates the group's prior mean $\boldsymbol{\mu}_g$, the point at which all members of the group actually enter. *Panel (a)* shows the occupation cluster that maximizes the systematic choice value, i.e. the choice in the deterministic limit $\varrho \to 0$; clusters are ordered by cognitive task intensity ω_C (legend). *Panel (b)* shows the task intensity ω_C of the entry occupation in expectation over the taste shocks, $\mathbb{E}[\omega_C] = \sum_j \mathbb{P}(\text{entry} = j \mid \tilde{\boldsymbol{\theta}})\, \omega_{jC}$. *Panel (c)* shows the probability mass placed on the single most likely entry occupation; the value $1/8$ corresponds to a uniform lottery over the eight clusters. Panels (b) and (c) are evaluated at the calibrated taste-shock scale $\varrho = 0.271$.

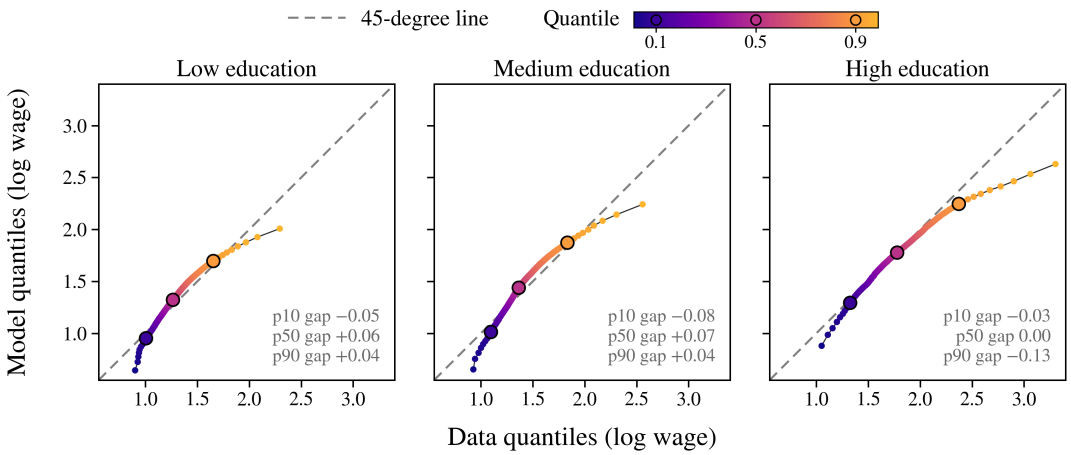


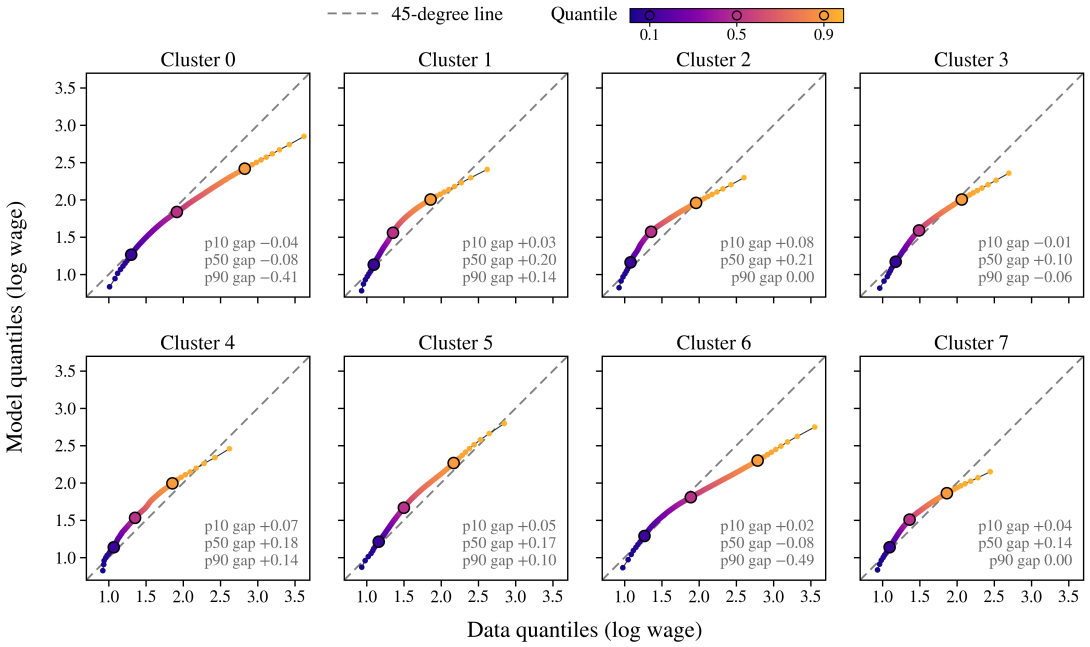
Figure 28: Quantile–quantile plots of the log real hourly wage at labor-market entry, model against data, by education group. Entrants are workers in their first year in the labor market. Each point is a quantile of the entry wage distribution, from the 1st to the 99th percentile; the dashed line is the 45° line, and the gaps between the model and the data at the 10th, 50th, and 90th percentiles are reported in each panel.

data: occupation dummies explain 22% of the variance of log wages in the data and 9% in the model, because the estimated occupation prices and loadings are closer to each other than the raw occupation means, which also reflect the selection of workers across occupations.

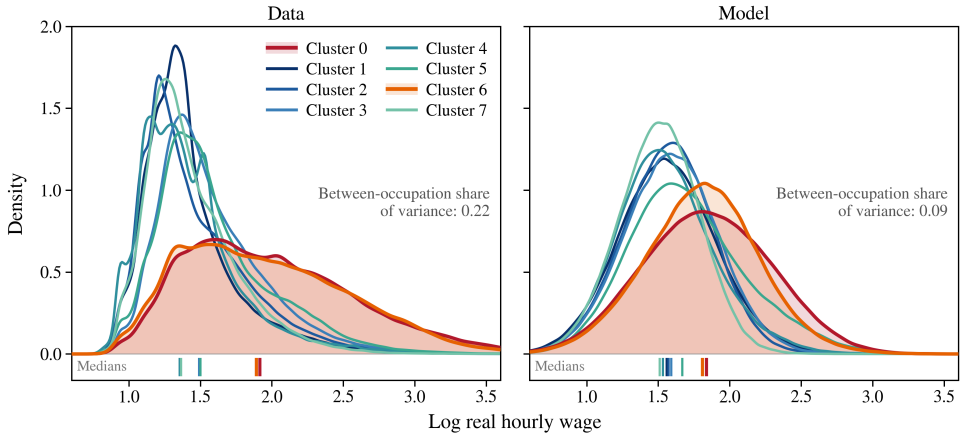
Wage–tenure profiles of stayers and mobility by tenure. Because wages reflect realized rather than expected output, within-occupation wage growth in the model comes from the accumulation of task-specific experience and from the selection of workers who stay. Panel (a) of Figure 30 follows workers observed from entry who remain without interruption in their entry occupation, and plots their mean log wage relative to the entry mean of all entrants. The model tracks the data closely over the first ten years and overshoots thereafter (+0.49 against +0.43 log points after fifteen years). The dotted line isolates the structural return to task experience of the same workers in the model, +0.31 log points after fifteen years; the difference from the model's profile is the selection of stayers, who are increasingly those whose wage signals confirmed a good match. The within-worker wage growth of stayers, which nets out the entry wage, is +0.29 log points after fifteen years in both the data and the model. Panel (b) reports the probability of switching occupation by years of occupational tenure, which complements the hazard by experience of Figure 9. The model reproduces the decline of mobility with tenure, but the decline is flatter than in the data: the model underpredicts switching in the first year of a spell (13% against 19%) and over-predicts it after ten years of tenure (5% against 3%), since the single switching cost κ_0 does not depend on tenure.

E.2 Additional Model-versus-Data Figures

Figures 31 and 32 report additional model-versus-data moments of occupational mobility by within-occupation match quality, and Figures 33 and 34 decompose the effect

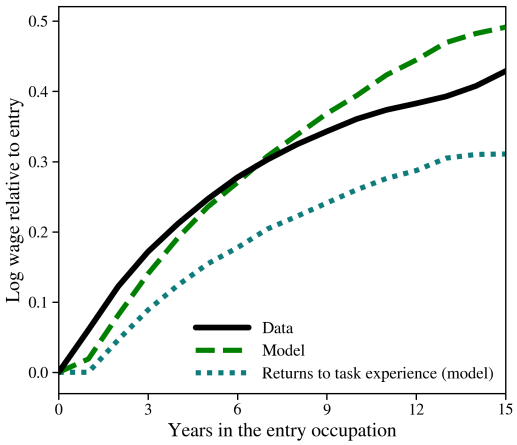


(a) Quantile–quantile plots by occupation

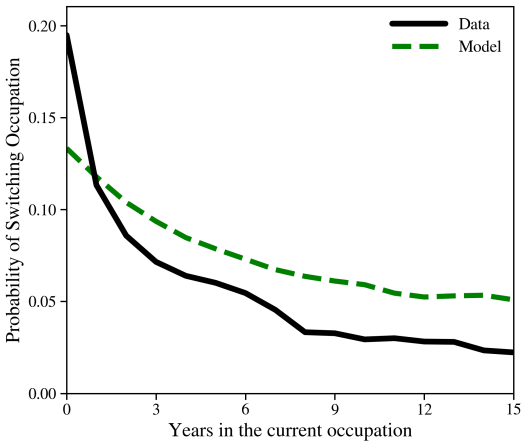


(b) Overlap of the occupation wage distributions

Figure 29: Wage distributions by occupation over all worker-years, data and model. *Panel (a)*: model quantiles against data quantiles of the log real hourly wage in each occupation cluster, from the 1st to the 99th percentile; the dashed line is the 45° line, and the gaps between the model and the data at the 10th, 50th, and 90th percentiles are reported in each panel. *Panel (b)*: densities of the log real hourly wage of the eight occupation clusters overlaid, in the data (left) and in the model (right); the two most cognitive-intensive occupations, Clusters 0 and 6, are shaded, the ticks below the horizontal axis mark the median of each occupation, and each panel reports the share of the variance of log wages explained by occupation dummies.



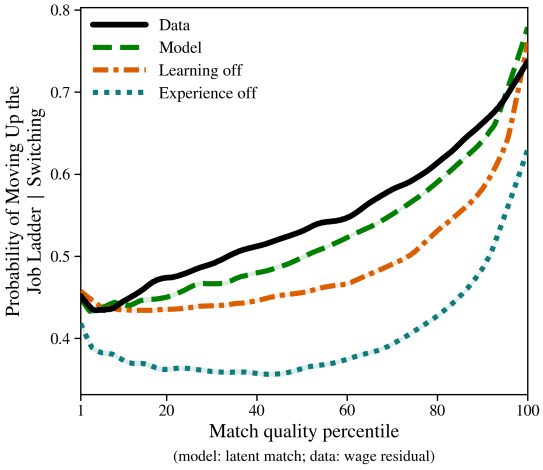
(a) Wage–tenure profile of occupational stayers



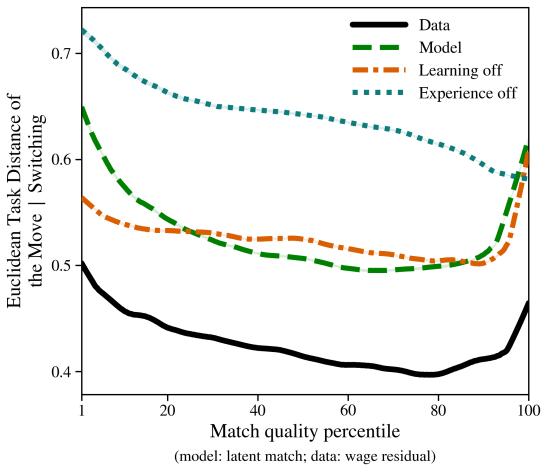
(b) Probability of switching by occupational tenure

Figure 30: Wage growth of stayers and mobility by tenure, data (black, solid) and model (green, dashed). *Panel (a)*: mean log wage, relative to the mean entry wage of all entrants, of workers observed from entry who are still in their entry occupation after the number of years on the horizontal axis; the dotted line is the structural return to task experience of the same workers in the model. The model's stayers are reweighted to the education mix of the entrants in the data. *Panel (b)*: probability of switching occupation between t and $t+1$ by years of tenure in the current occupation.

of learning on mobility into experimentation and passive learning.

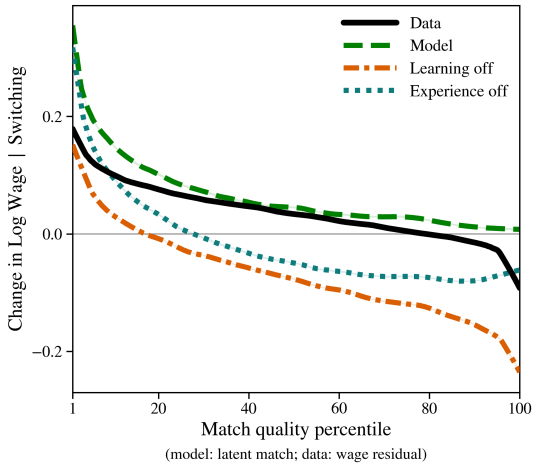


(a) Probability of moving up the job ladder

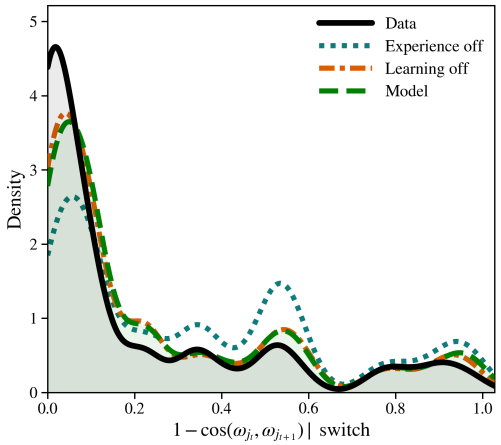


(b) Euclidean task distance of the switch

Figure 31: Additional model-versus-data moments of occupational mobility by within-occupation match-quality percentile (model: latent match quality; data: worker–occupation fixed effect), in the data (black), estimated model (green), *learning off* (orange), and *experience off* (teal). *Panel (a)*: probability that a switch is to an occupation with a higher mean wage rank (moving up the job ladder). *Panel (b)*: Euclidean task distance of the switch, $\|\boldsymbol{\omega}_{j_{t+1}} - \boldsymbol{\omega}_{j_t}\|_2$.

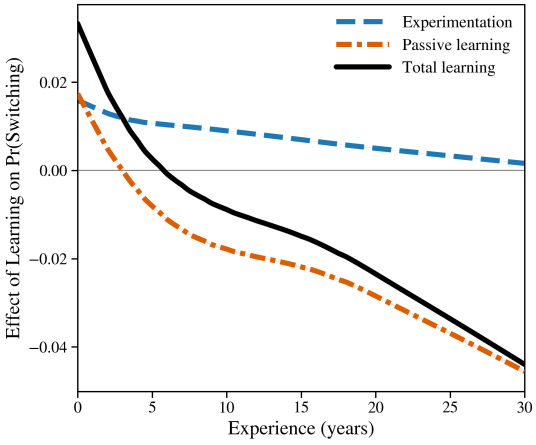


(a) Change in log wage upon switching

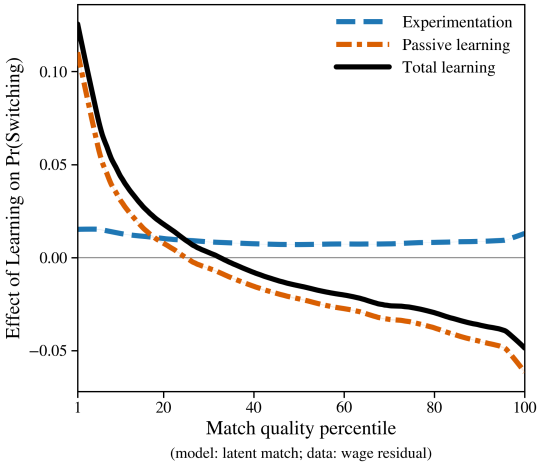


(b) Distribution of the task-content distance of switches

Figure 32: *Panel (a)*: change in log wage upon switching occupation, by within-occupation match-quality percentile (model: latent match quality; data: worker–occupation fixed effect). *Panel (b)*: distribution of the task-content distance of switches, $1-\cos(\omega_{j_t},\omega_{j_{t+1}})$, computed on the task weights of the occupation clusters for both the data and the model, conditional on switching. In both, the data (black), estimated model (green), *learning off* (orange), and *experience off* (teal).

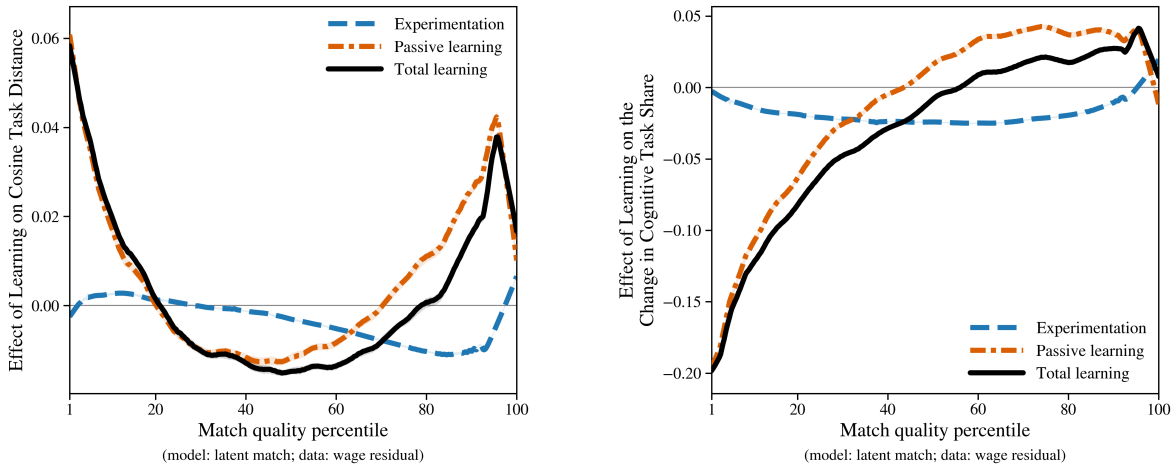


(a) Probability of switching by experience



(b) Probability of switching by match quality

Figure 33: Decomposition of the total effect of learning into experimentation and passive learning. In each panel the total learning effect (black, the gap between the estimated model and the *learning off* counterfactual) is split into the contribution of *experimentation* (blue, the option value of trying occupations to acquire information about comparative advantage) and *passive learning* (orange, the updating of beliefs from realized wage signals). *Panel (a)*: probability of switching by years of experience. *Panel (b)*: probability of switching by within-occupation match-quality percentile.



(a) Task-content distance of the switch

(b) Change in cognitive task content

Figure 34: Decomposition of the total effect of learning into experimentation and passive learning, for the task content of switches by within-occupation match-quality percentile, as in Figure 33: total learning effect (black), *experimentation* (blue), and *passive learning* (orange). *Panel (a)*: task-content distance of the switch, $1 - \cos(\boldsymbol{\omega}_{j_t}, \boldsymbol{\omega}_{j_{t+1}})$, computed on the task weights of the occupation clusters for both the data and the model. *Panel (b)*: change in the cognitive task share, $\Delta\omega_C$.

F Appendix: Additional Results and Counterfactuals

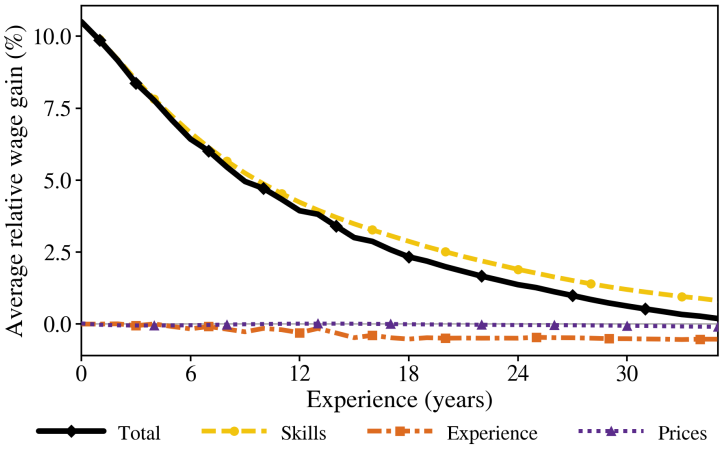


Figure 35: Decomposition of the life-cycle wage gain from removing information frictions (Figure 11, Panel (a)) into channels: Total (black), Skills (yellow, dashed), Experience (orange, dash-dotted), and Prices (purple, dotted). Horizontal axis: years of potential experience; vertical axis: average relative wage gain (percent).

F.1 Learning, Human Capital Accumulation and Wage Growth

I use the model to measure how much learning and human capital accumulation each contribute to life-cycle wage growth. With the counterfactual choice rules of Appendix D.6, I simulate the model in partial equilibrium with learning, human capital accumulation, or both shut down. Because skills are permanent in the model, learning raises wages only by improving sorting, which workers achieve by switching occupations over the

life cycle. Let $G_{h\ell}(e)$ denote cumulative log wage growth from labor-market entry to experience e in the economy in which the human-capital channel is active if $h = 1$ and the learning channel if $\ell = 1$; thus G_{11} is the estimated model, G_{00} the baseline in which both channels are shut down, G_{10} the human-capital-only economy, and G_{01} the learning-only economy. I can decompose cumulative wage growth as

$$G_{11}(e) = \underbrace{G_{00}(e)}_{\text{no HC, no learning}} + \underbrace{\big[G_{10}(e) - G_{00}(e)\big]}_{\text{human capital}} + \underbrace{\big[G_{01}(e) - G_{00}(e)\big]}_{\text{learning}} + \underbrace{\big[G_{11}(e) - G_{10}(e) - G_{01}(e) + G_{00}(e)\big]}_{\text{HC–learning interaction}}. \tag{87}$$

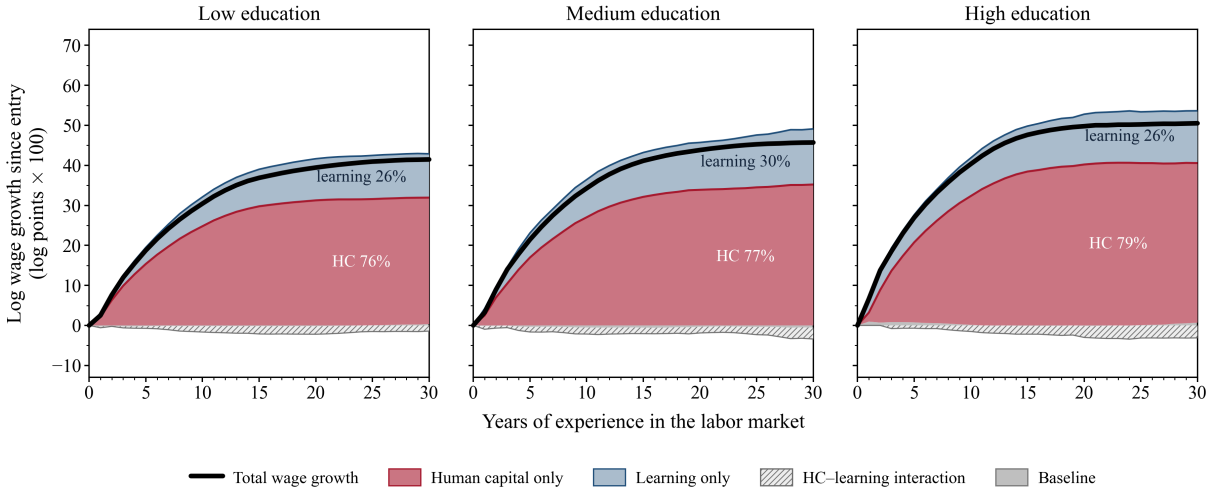


Figure 36: Decomposition of cumulative log wage growth since labor-market entry (vertical axis, log points ×100) by years of experience (horizontal axis), separately for low, medium, and high education. The total (black) is decomposed into the contribution of task-specific human capital (red), of learning about comparative advantage (blue), and their interaction (hatched).

In Figure 36, I plot the wage decomposition separately for each education group. Across groups, learning drives approximately one fourth of total wage growth, and experience accumulation drives the remaining three quarters. Besides starting their careers with higher wage levels, more educated workers see their wages grow more and faster over the life cycle. More educated individuals start their careers in more cognitive-intensive jobs, which have higher and steeper returns to tenure (Deming 2025). This decomposition exercise also shows how learning and human capital accumulation interact in the baseline model. As suggested by the growing interaction term in the decomposition plots, average wage growth in the estimated economy is lower than the sum of the wage growth achieved in the economies with only learning and with only human capital accumulation. This implies that human capital accumulation, while generating wage growth within occupations, in fact prevents some workers from eventually achieving the higher wages that they would achieve in an economy without experience accumulation. This accumulation channel implicitly generates a cost and makes workers less willing to switch occupations later in their careers.

I compute the decomposition in (87) for the economy with the change in tasks induced by AI, and I report the results in Figure 37. AI increases the importance of learning, particularly for high-educated workers, and decreases the importance of human capital accumulation for low- and medium-educated workers: on average, the share of life-cycle wage growth due to human capital accumulation falls from about 79% to about 66%, that is, by about 13 percentage points, while for highly educated workers the negative interaction between learning and human capital widens. In the pre-AI economy, cognitive tasks were an important driver of the returns to experience, both through direct returns and through the accumulation function. In the post-AI economy, cognitive experience is accumulated more slowly and cognitive tasks are less important. At the same time, manual skills become more important and, particularly for highly educated workers, so does learning: given the estimated skill distribution, highly educated workers enter the labor market believing that they have a comparative advantage in performing cognitive tasks. These initial beliefs are particularly penalizing for workers with more mixed skills or higher manual skills, who benefit substantially from learning their comparative advantage and reallocating accordingly.

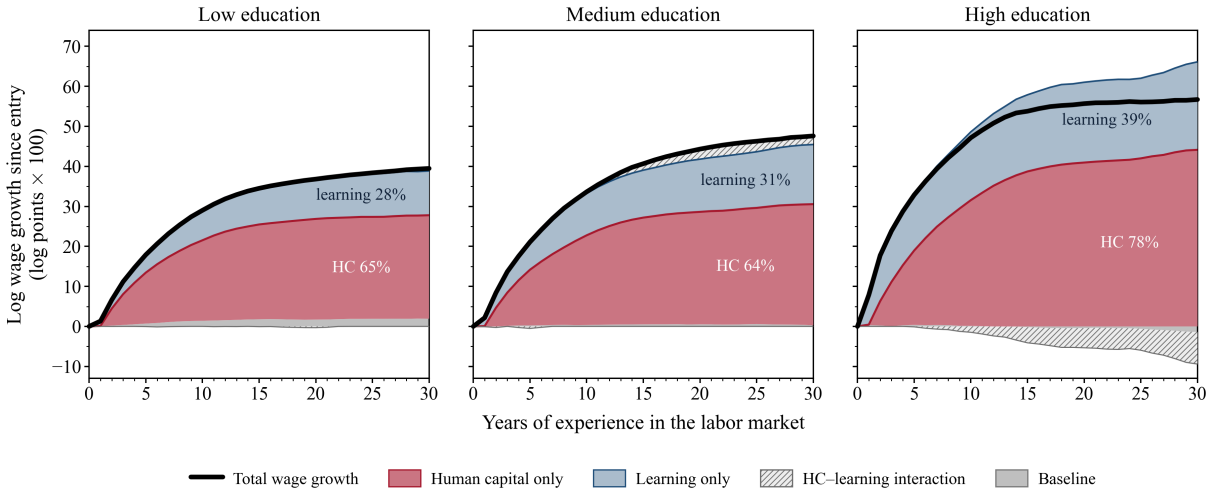
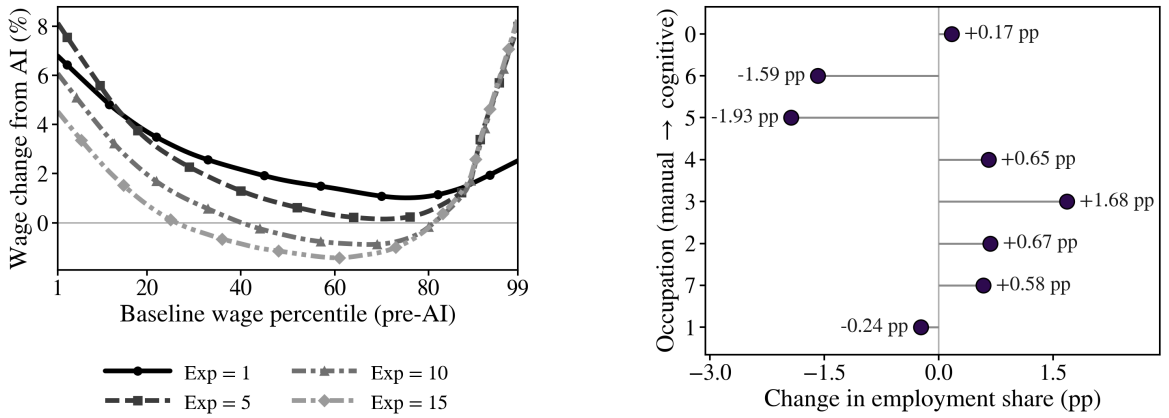


Figure 37: Decomposition of cumulative log wage growth since labor-market entry in the post-AI stationary equilibrium (vertical axis, log points $\times 100$) by years of experience (horizontal axis), separately for low, medium, and high education. The total (black) is decomposed into the contribution of task-specific human capital (red), of learning about comparative advantage (blue), and their interaction (hatched).

F.2 Welfare Measure

I measure welfare with the ex-ante value $EV(\boldsymbol{x}_{it})$ of equation (28), the expected discounted sum of the worker's future flow utilities, log wages plus amenities minus switching costs, given her state. Aggregate welfare is the average of this value over the



(a) Wage gains from AI by baseline wage percentile

(b) Reallocation by task content of the occupation

Figure 38: *Panel (a)*: percent wage change from AI by baseline (pre-AI) wage percentile, for workers with different years of experience (separate lines). *Panel (b)*: change in each occupation cluster's employment share between the pre-AI and post-AI stationary equilibria, in percentage points; clusters are ordered from the most manual-intensive (bottom) to the most cognitive-intensive (top). Employment moves out of the cognitive-intensive Clusters 5 and 6 and into the mixed and manual-intensive clusters, mostly Cluster 3.

stationary cross-section of workers,

$$\mathcal{W} = \frac{1}{|\mathcal{S}|} \sum_{(i,t)\in\mathcal{S}} EV(\boldsymbol{x}_{it}), \tag{88}$$

where $\mathcal{S}$ is the set of worker-periods of the stationary simulation after the burn-in. Because the cross-section contains every stage of the career in its stationary proportions, $\mathcal{W}$ is the expected remaining lifetime utility of a randomly drawn worker.

Since utility is logarithmic in the wage, a permanent increase of every wage by a factor $1 + \Delta^{\text{CEV}}$ raises $\mathcal{W}$ by $\log(1 + \Delta^{\text{CEV}})/(1 - \beta)$. The consumption-equivalent variation of a counterfactual economy relative to a baseline is therefore

$$\Delta^{\text{CEV}} = \exp\left((\mathcal{W}^{\text{cf}} - \mathcal{W}^{\text{base}})(1 - \beta)\right) - 1, \tag{89}$$

the permanent wage increase that makes the average worker in the baseline as well off as the average worker in the counterfactual. This measure includes everything that enters utility: wages, amenities, the switching cost and the option value of the taste shocks.

Table 6 reports the output and welfare gains of each counterfactual. In every counterfactual that changes workers' information set, welfare moves in the same direction as output, and the welfare gain is between a quarter and a half of the output gain. Welfare gains are likely smaller because some workers were sorting on non-wage factors before receiving the information. Once informed, they may choose an occupation that pays a higher wage and produces more output, but give up part of the non-wage components of their utility, the amenities and the idiosyncratic taste shocks. Figures 40

and 41 report the welfare counterparts of the information-provision, life-cycle and transition figures of the main text. Along the AI transition, welfare rises by 0.8% on impact and by 1.1% in the long run; the skill signal delivered to every new cohort raises the long-run gain to 1.5% and closes 17% of the long-run welfare gap with the perfect-information economy, against 20% of the output gap. Welfare and output give a different reading in the switching-cost experiment (Figure 39). The non-monotonicity of output is not replicated in welfare: reducing the switching cost has a direct effect, since the cost itself is a utility loss that the counterfactual removes (Section 6.3), and an indirect effect through re-sorting, as workers move to the occupations in which they draw large taste shocks. Together these two effects outweigh the wage losses, so welfare rises monotonically as the switching cost falls. In the forced-exploration exercise of Appendix F.4, output rises but the entering cohort bears the cost of the forced assignment.

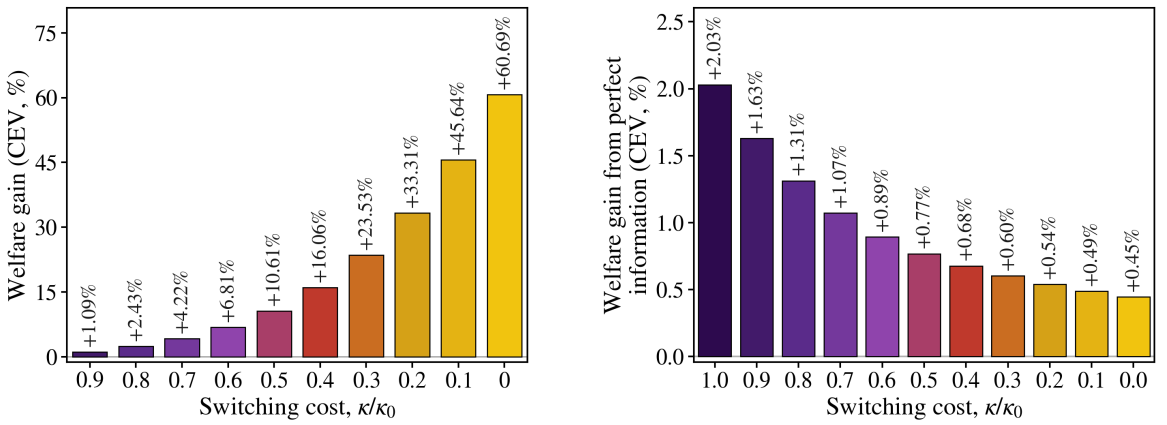
Table 6: Output and Welfare Effects of the Counterfactuals

Counterfactual	Comparison	Output gain (%)	Welfare gain (%)
Perfect information	vs. baseline	5.10	2.03
Skill signal at entry	vs. baseline	0.97	0.32
Skill signal after 2 years	vs. baseline	0.52	0.22
Skill signal after 6 years	vs. baseline	0.23	0.11
Skill signal after 10 years	vs. baseline	0.12	0.06
Perfect information at $\kappa/\kappa_0 = 0.7$	vs. same κ	3.32	1.07
Perfect information at $\kappa/\kappa_0 = 0.5$	vs. same κ	2.52	0.77
Perfect information at $\kappa/\kappa_0 = 0$	vs. same κ	1.62	0.45
Switching cost $\kappa/\kappa_0 = 0.7$	vs. baseline	1.22	4.22
Switching cost $\kappa/\kappa_0 = 0.5$	vs. baseline	0.82	10.61
Switching cost $\kappa/\kappa_0 = 0.3$	vs. baseline	−0.48	23.53
Switching cost $\kappa/\kappa_0 = 0$	vs. baseline	−2.39	60.69
Forced exploration	vs. baseline	1.11	−0.03
AI shock	vs. pre-AI baseline	2.08	1.05
Perfect information, post-AI	vs. post-AI baseline	5.84	2.48
Skill signal at entry, post-AI	vs. post-AI baseline	1.11	0.40

Notes: Output gain is the percent change in CES aggregate output between the two stationary equilibria. Welfare gain is the consumption-equivalent variation of the counterfactual relative to the comparison economy, in percent. All economies are simulated with $N = 200{,}000$ workers for 400 periods, discarding the first 200 periods.

F.3 Switching Costs: Life-Cycle Wage Profiles

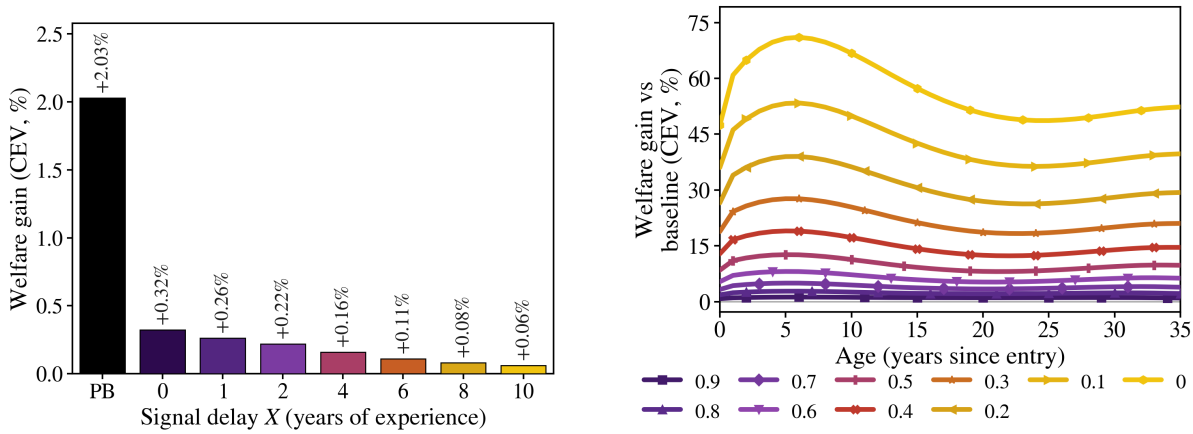
This appendix reports the life-cycle wage profiles behind the switching-cost counterfactuals of Section 6.3, which the main text summarizes by match-quality percentile. Figure 42 reports, for each switching-cost level κ/κ_0, the life-cycle counterparts of Figures 15 (Panel (a), the gain from moving to perfect information) and 16 (Panel (b), the gain relative to the estimated baseline economy). In Panel (b), the wage gain of



(a) Welfare gain from reducing the switching cost

(b) Welfare gain from perfect information, by κ/κ_0

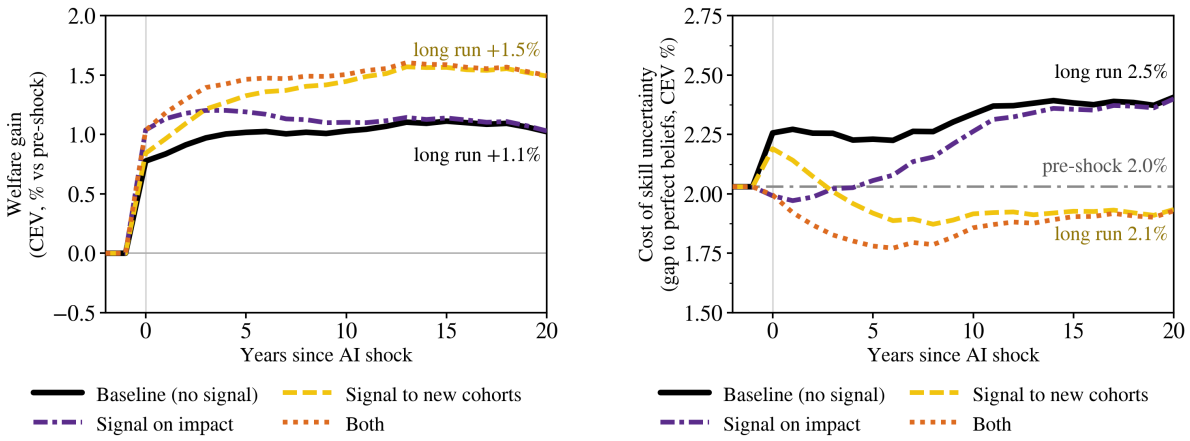
Figure 39: Welfare effects of switching costs and information frictions, the welfare counterparts of Panel (a) of Figures 16 and 15. *Panel (a)*: consumption-equivalent welfare gain of each imperfect-information economy relative to the baseline, by the switching cost as a fraction of its estimated value. *Panel (b)*: consumption-equivalent welfare gain from moving to perfect information at each switching-cost level, both economies solved at the same κ/κ_0.



(a) Welfare gain from the skill signal, by delay

(b) Welfare gain from reducing the switching cost, by age

Figure 40: Welfare counterparts of Figure 13 and of Panel (b) of Figure 42. *Panel (a)*: consumption-equivalent welfare gain relative to the baseline economy when the calibrated skill signal is delivered X years after labor-market entry, and under perfect information. *Panel (b)*: consumption-equivalent welfare gain relative to the baseline economy by years since entry, for each switching cost κ/κ_0; the value at age zero is the welfare gain of the entering cohort.

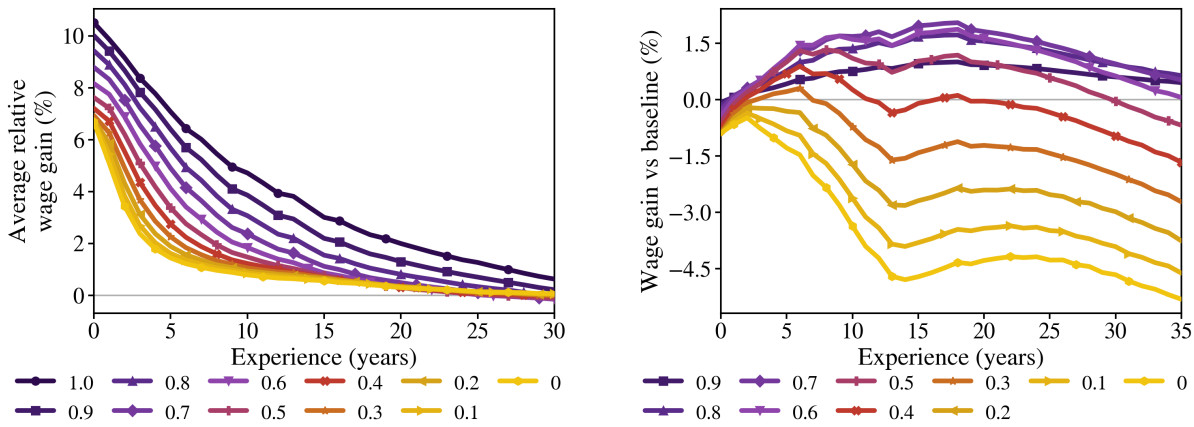


(a) Welfare along the transition

(b) Welfare gap to the perfect-information economy

Figure 41: Welfare counterparts of Figure 19, by years since the AI shock (horizontal axis). Baseline imperfect-information economy with no signal (black, solid); skill signal delivered to all workers on impact (purple, dash-dotted); skill signal delivered to every new cohort at entry (yellow, dashed); both (orange, dotted). *Panel (a)*: consumption-equivalent welfare gain of the cross-section alive in each period relative to the pre-shock stationary economy. *Panel (b)*: consumption-equivalent variation that the cross-section alive in each period would need to be as well off as in the perfect-information economy in the same period; the dash-dotted gray line marks its pre-shock level. Labels report the pre-shock and long-run values.

reducing switching costs gradually becomes more negative over the life cycle: with lower switching costs, taste-shock-driven churning makes workers repeatedly restart their task-specific experience, and these losses compound over the career; when the reduction is extreme, wage losses materialize almost immediately.



(a) Gain from perfect information, by κ/κ_0

(b) Gain over the baseline economy, by κ/κ_0

Figure 42: Life-cycle wage profiles for the switching-cost counterfactuals. Horizontal axis: years of potential experience; vertical axis: average relative wage gain (percent). *Panel (a)*: wage gain from moving to perfect information at each switching cost κ/κ_0. *Panel (b)*: wage gain relative to the estimated baseline economy at each κ/κ_0.

F.4 Forced Exploration at the Beginning of the Career

The switching-cost counterfactual (Section 6.3) shows that frictions to mobility are a key driver of persistent mismatch, but even with low switching costs the baseline information structure limits how quickly workers can sort optimally. A complementary policy question is whether a targeted intervention at the very start of the career can accelerate learning and improve lifetime sorting and wages. I study a one-period "forced exploration" policy: in their first period, workers are assigned to the occupation that maximizes the information gain about their latent skills. I measure the learning value of occupation j for entrants in group g as the mutual information between skills and the first wage signal,

$$IG_{jg}^{\text{entry}} \equiv I(\boldsymbol{\theta}_i; s_{ij1} \mid g) = \frac{1}{2} \log\left(1 + \frac{\boldsymbol{\lambda}_j' \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j}{\varsigma_{\varepsilon j}^2}\right), \tag{90}$$

where $\boldsymbol{\lambda}_j$ are the skill loadings of occupation j, $\boldsymbol{\Sigma}_g$ is the prior skill covariance of group g, and $\varsigma_{\varepsilon j}^2$ is the occupation-specific wage noise variance. This has a natural signal-to-noise interpretation: an occupation is more informative when it loads strongly on dimensions about which entrants are ex ante uncertain and when wages are a precise signal of skills[35]. Each worker is assigned to $\arg\max_j IG_{jg}^{\text{entry}}$ in her first period; from period 2 onward she chooses freely under her updated beliefs. The assignment involves no transfers: workers receive the going market wage in the assigned occupation; the only constraint is on occupational choice in the first period.

The forced occupations are those that best reveal a worker's comparative advantage, and they turn out to be the two *anchor* occupations: the cognitive anchor (Cluster 0) and, for a smaller share of workers, the manual anchor (Cluster 1). This is intuitive: the anchors load on a single skill dimension, so their wage is a noisy signal of a single skill and therefore more informative than the wage of an occupation that mixes skills. Figure 44 shows the occupational shares at entry, after two years, and after ten years, under forced exploration and in the baseline.

At entry, workers are concentrated in the two anchor occupations, unlike the baseline, where entrants are spread across all eight occupations (Figure 44). This forced bunching triggers a strong general-equilibrium price response: excess labor supply in

[35] Mutual information is the reduction in uncertainty in the signal due to knowledge of latent skills,

$$I(\boldsymbol{\theta}_i; s_{ij1} \mid g) = h(s_{ij1} \mid g) - h(s_{ij1} \mid \boldsymbol{\theta}_i, g),$$

where $h(Y) = \mathbb{E}[-\log f_Y(Y)]$ denotes differential entropy. Under the model's linear-Gaussian signal structure, $s_{ij1} = \boldsymbol{\lambda}_j' \boldsymbol{\theta}_i + \varepsilon_{ij1}$, $\boldsymbol{\theta}_i \mid g \sim N(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$, $\varepsilon_{ij1} \sim N(0, \varsigma_{\varepsilon j}^2)$, the signal is scalar Gaussian with unconditional variance $\boldsymbol{\lambda}_j' \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j + \varsigma_{\varepsilon j}^2$ and conditional variance $\varsigma_{\varepsilon j}^2$. Since the entropy of a scalar Gaussian $N(m, v)$ is $h(Y) = \frac{1}{2}\log(2\pi e\, v)$, it follows that

$$I(\boldsymbol{\theta}_i; s_{ij1} \mid g) = \frac{1}{2}\log\Big(2\pi e\left[\boldsymbol{\lambda}_j' \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j + \varsigma_{\varepsilon j}^2\right]\Big) - \frac{1}{2}\log\Big(2\pi e\, \varsigma_{\varepsilon j}^2\Big) = \frac{1}{2}\log\left(1 + \frac{\boldsymbol{\lambda}_j' \boldsymbol{\Sigma}_g \boldsymbol{\lambda}_j}{\varsigma_{\varepsilon j}^2}\right).$$

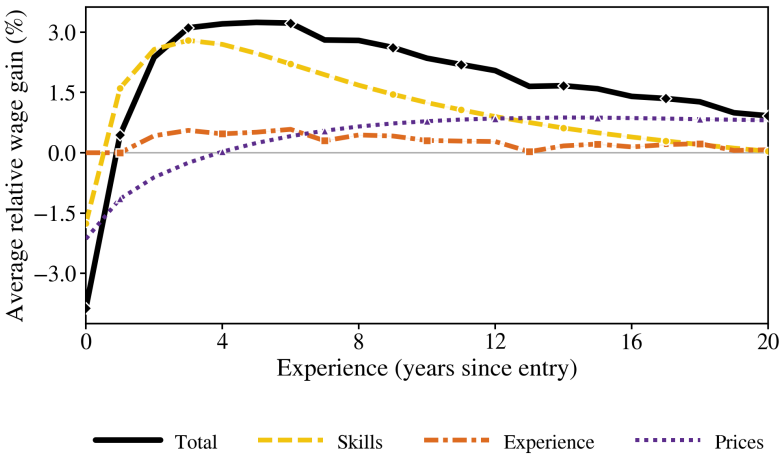


Figure 43: Life-cycle wage gain from forced exploration relative to the baseline economy, decomposed into channels: Total (black), Skills (yellow, dashed), Experience (orange, dash-dotted), and Prices (purple, dotted). Horizontal axis: years of potential experience since entry; vertical axis: average relative wage gain (percent).

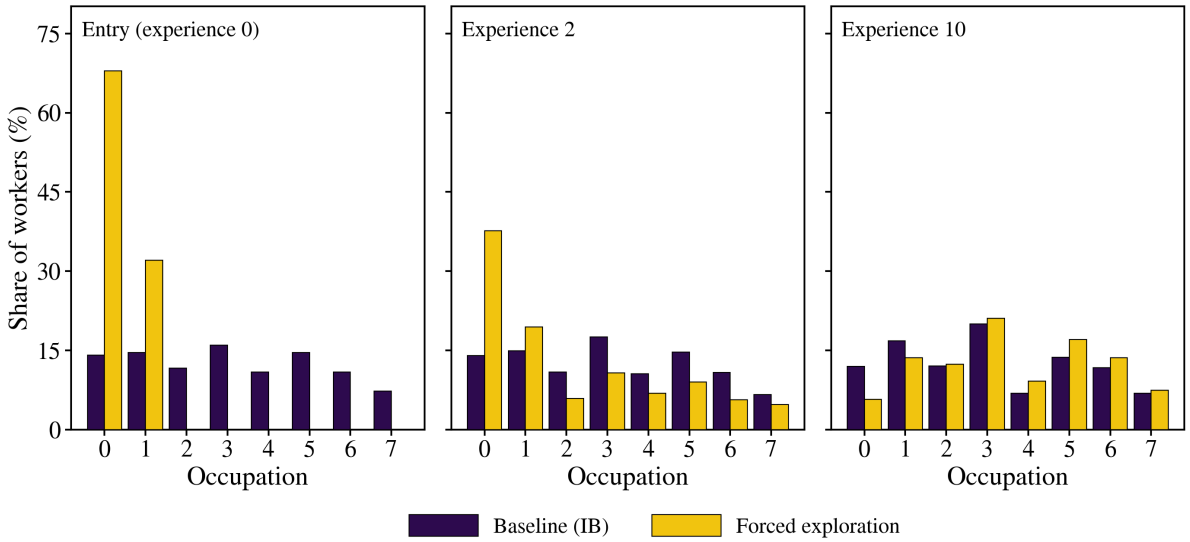


Figure 44: Share of workers in each occupation cluster (vertical axis) under forced exploration (yellow) and in the baseline imperfect-information economy (dark purple), at entry, after two years, and after ten years of experience.

the informative occupations drives their prices down while prices in the other occupations rise. This drop in prices produces a sharp wage loss on impact, about 3.9% at entry (Figure 43). From the second period workers choose freely, and they reallocate quickly to occupations that better match their comparative advantage in tasks. The Skills channel jumps to about +2.6% within two years and stays above the baseline for most of the career, suggesting that in this counterfactual workers are indeed better matched on skills. The total wage gain turns positive after the first year, reaching about +3.2% by the third year and staying close to that level until about the sixth year. As shown in Figure 44, the occupational distribution converges back toward the baseline distribution but workers are now better matched within each occupation. The net re-

sult is that the later gains more than offset the short-run wage loss: aggregate output in the exploration economy is 1.11% higher than in the baseline. The welfare effect is instead negligible: the population consumption-equivalent gain is -0.03%, and the entering cohort, which bears the forced assignment, loses 6.4%, because the assignment overrides the occupation these workers would have chosen (Appendix F.2).

A policy that temporarily redirects entrants toward their most informative occupation can therefore raise productivity later in the career, at the cost of short-run wage losses.

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